



US012385051B2

(12) **United States Patent**
Venkatesan et al.

(10) **Patent No.: US 12,385,051 B2**

(45) **Date of Patent: Aug. 12, 2025**

(54) **SHIGELLA-BASED MINIMAL PLASMID
RECOMBINANT INVASION CONSTRUCT**

(71) Applicant: **THE GOVERNMENT OF THE
UNITED STATES AS
REPRESENTED BY THE
SECRETARY OF THE ARMY**, Fort
Detrick, MD (US)

(72) Inventors: **Malabi M. Venkatesan**, Bethesda, MD
(US); **Lakshmi Chandrasekaran**,
Gaithersburg, MD (US); **Akamol E.
Suvarnapunya**, Rockville, MD (US)

(73) Assignee: **THE GOVERNMENT OF THE
UNITED STATES AS
REPRESENTED BY THE
SECRETARY OF THE ARMY**, Fort
Detrick, MD (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 471 days.

(21) Appl. No.: **17/923,853**

(22) PCT Filed: **May 5, 2021**

(86) PCT No.: **PCT/US2021/030882**

§ 371 (c)(1),

(2) Date: **Nov. 7, 2022**

(87) PCT Pub. No.: **WO2021/236327**

PCT Pub. Date: **Nov. 25, 2021**

(65) **Prior Publication Data**

US 2023/0183719 A1 Jun. 15, 2023

Related U.S. Application Data

(60) Provisional application No. 63/021,463, filed on May
7, 2020.

(51) **Int. Cl.**

C12Q 1/68 (2018.01)

C12N 15/00 (2006.01)

C12N 15/70 (2006.01)

C12N 15/74 (2006.01)

(52) **U.S. Cl.**

CPC **C12N 15/74** (2013.01); **C12N 15/70**
(2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

9,434,772 B2 9/2016 Maurelli

FOREIGN PATENT DOCUMENTS

WO 20210236327 A2 11/2021

OTHER PUBLICATIONS

PCT/US2021/030882, PCT Search Report & Written Opinion mailed
Feb. 2, 2022, 10 pages.

Coster, Trinka S. et al., "Vaccination against Shigellosis with
Attenuated *Shigella flexneri* 2a Strain SC602", *Infection and Immu-*
nity, Jul. 1999, vol. 67, No. 7, pp. 3437-3443.

GenBank CP050984.1, *Shigella flexneri* strain FDAARGOS_716
plasmid unnamed1, complete sequence, downloaded from Internet
Jan. 7, 2022, 15 pages.

Kotloff, Karen L. et al., "Shigella flexneri 2a Strain CVD 1207, with
Specific Deletions in virG, sen, set, and guaBA, Is Highly Attenu-
ated in Humans", *Infection and Immunity*, Mar. 2000, vol. 68, No.
3, pp. 1034-1039.

Levine, Myron M. et al., Clinical trials of *Shigella* vaccines: two
steps forward and one step back on a long, hard road, *Nature*, July
1007, vol. 5, pp. 540-553.

Maurelli, Anthony T. et al., "Temperature-Dependent Expression of
Virulence Genes in *Shigella* Species", *Infection and Immunity*, Jan.
1984, vol. 43, No. 1, pp. 195-201.

Venkatesan, Malabi M. et al., "Use of *Shigella flexneri* ipaC and
ipaH Gene Sequences for the General Identification of *Shigella* spp.
and Enteroinvasive *Escherichia coli*", *Journal of Clinical Microbi-*
ology, Dec. 1989, vol. 27, No. 12, pp. 2687-2691.

Yang, Fan et al., "Genome dynamics and delivery of *Shigella*
species, the etiologic agents of bacillary dysentery", *Nucleic Acids*
Research, 2005, vol. 33, No. 19, pp. 6445-6458.

EESR mailed Jun. 6, 2024, 8 pages.

Bliven, Kimberly A. "Pathoadaption of the intracellular bacteria
Shigella and *Chlamydia*: Virulence, antivirulence, and tissue trop-
ism", 2015, XP093159307.

Primary Examiner — Jennifer E Graser

(74) *Attorney, Agent, or Firm* — Timothy H. Van Dyke;
Wolter Van Dyke Davis, PLLC

(57) **ABSTRACT**

Recombinant *Shigella* minimal invasion plasmid constructs
pRISM and pRISM-G and a method of inducing an immune
response to *Shigella* in a subject, the method comprising
administering the composition to the subject in an amount
sufficient to induce an immune response to *Shigella* in the
subject.

12 Claims, 10 Drawing Sheets

Specification includes a Sequence Listing.

FIG. 1

Atlas: 11 Sep 2018

Created with SnapGene®

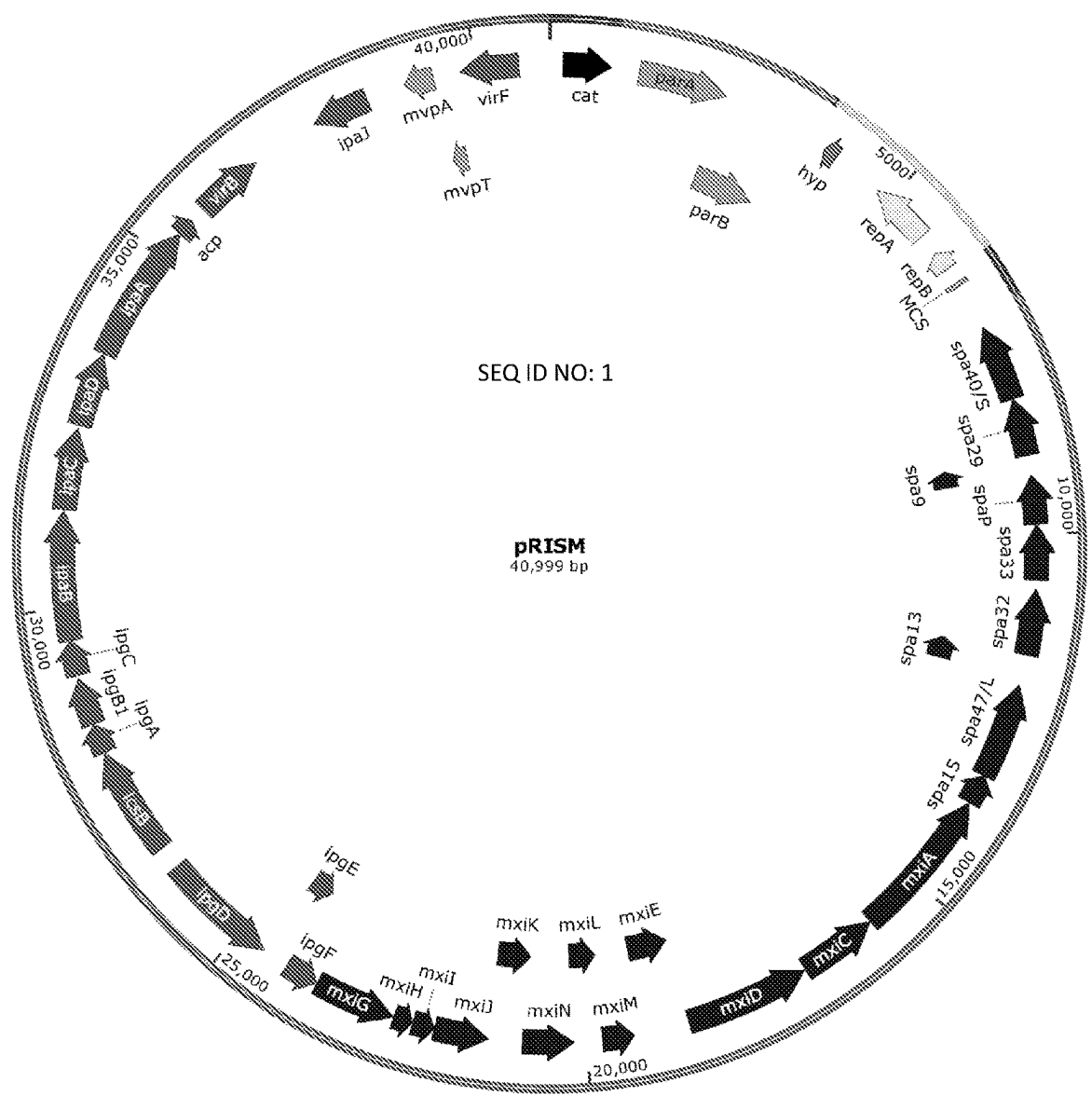


FIG. 2

Alias: 11 Sep 2018

 [CC BY-SA 4.0](https://creativecommons.org/licenses/by-sa/4.0/)

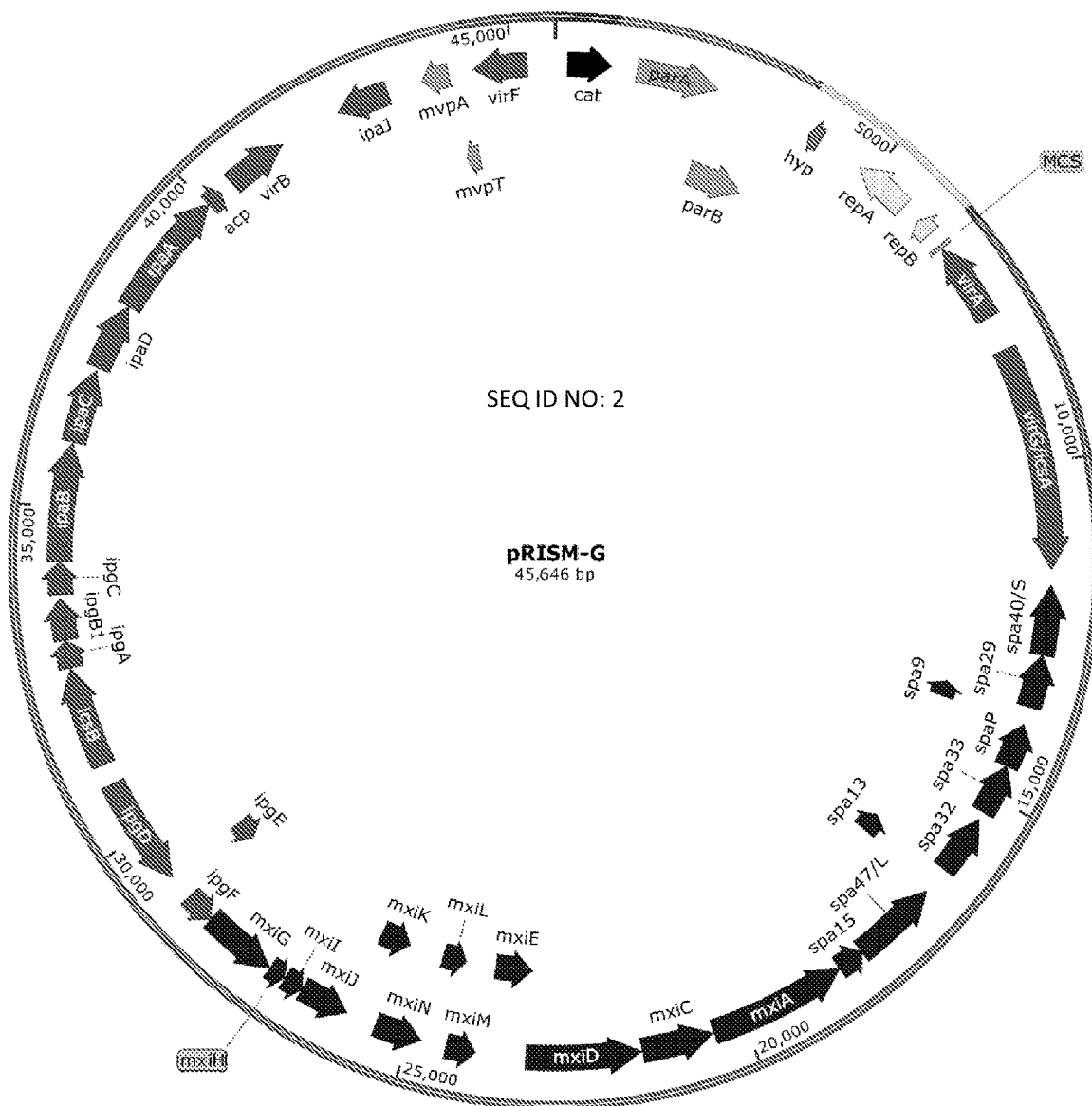


FIG. 3

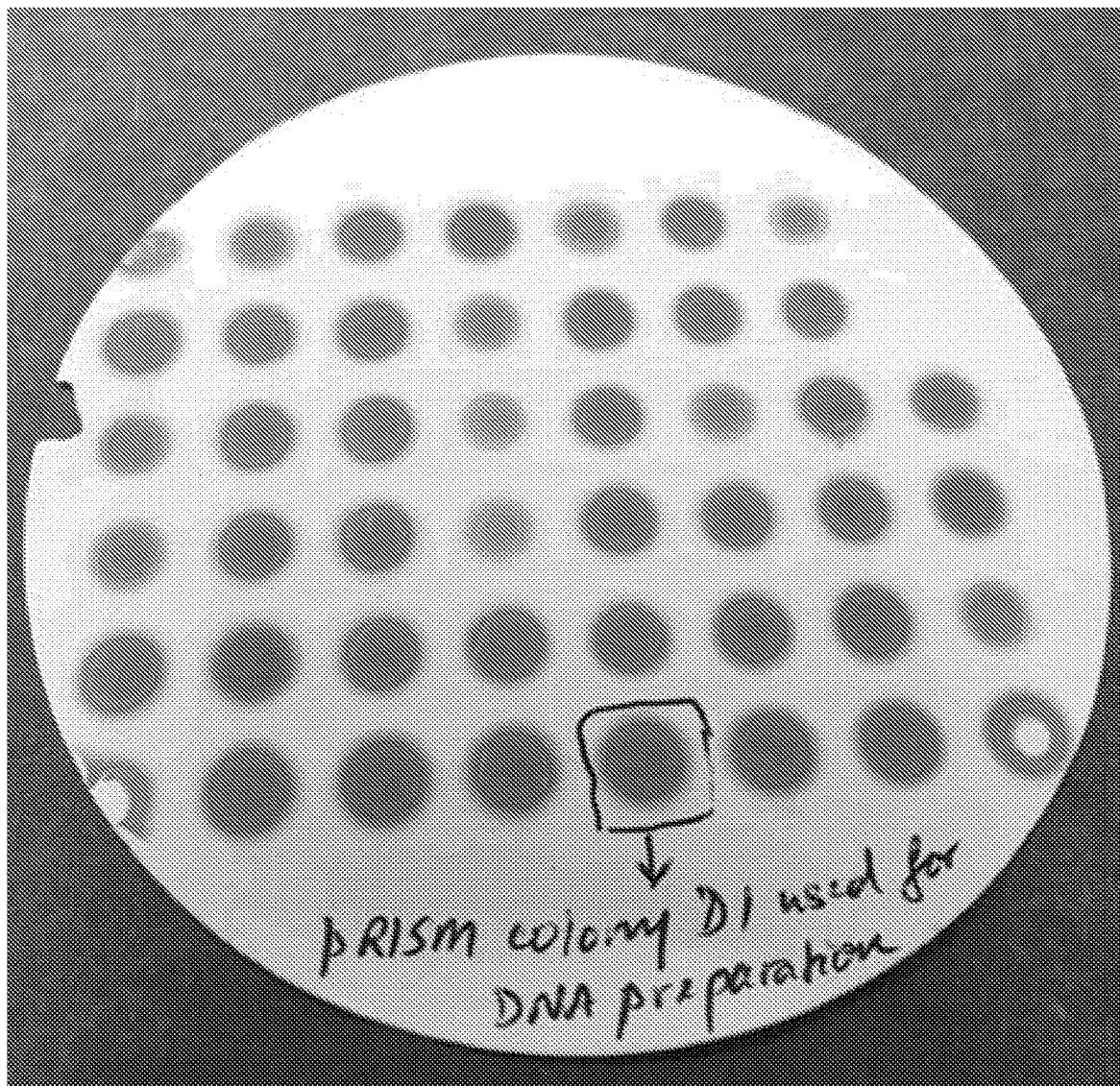


FIG. 4

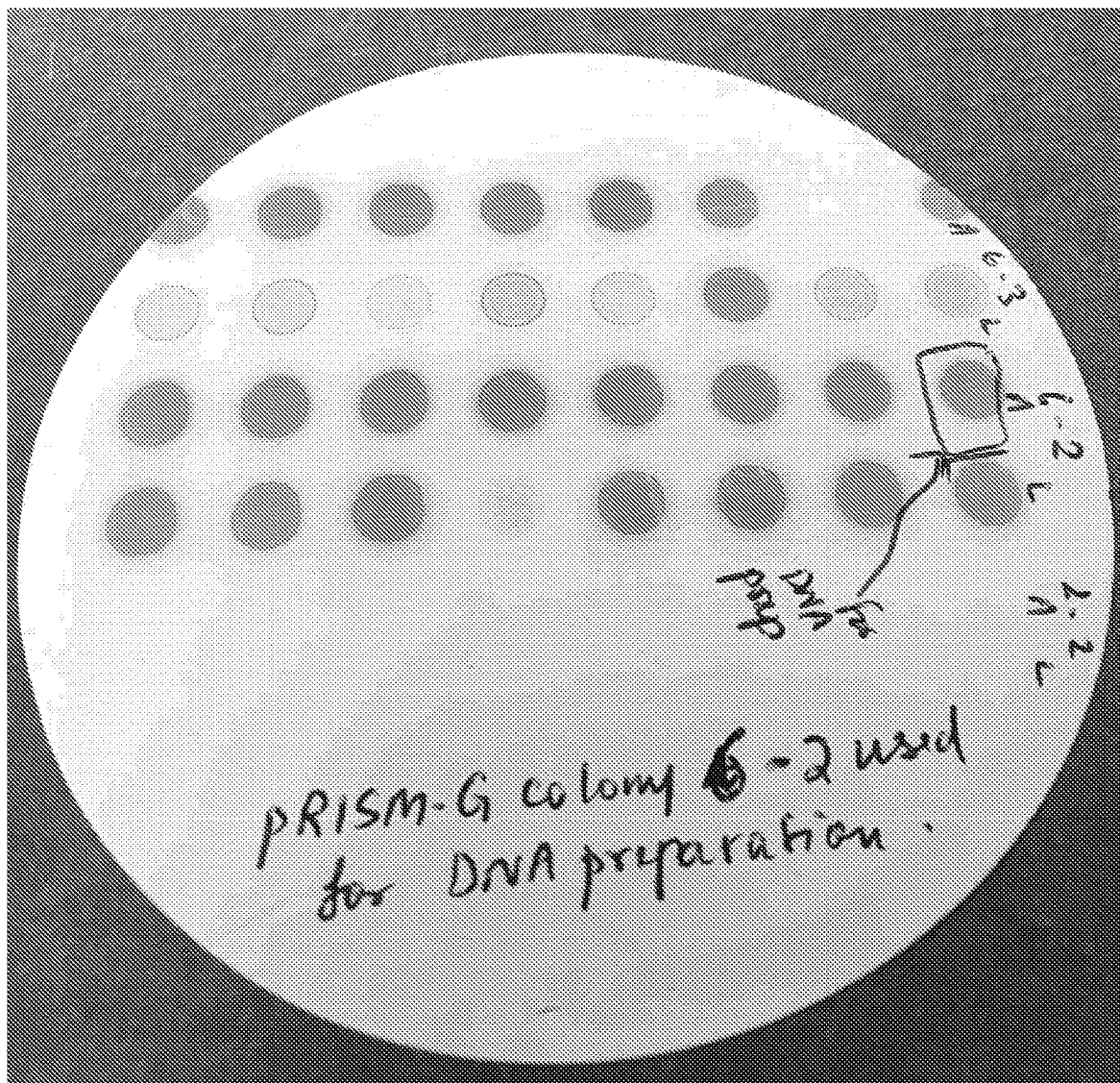


FIG. 5A

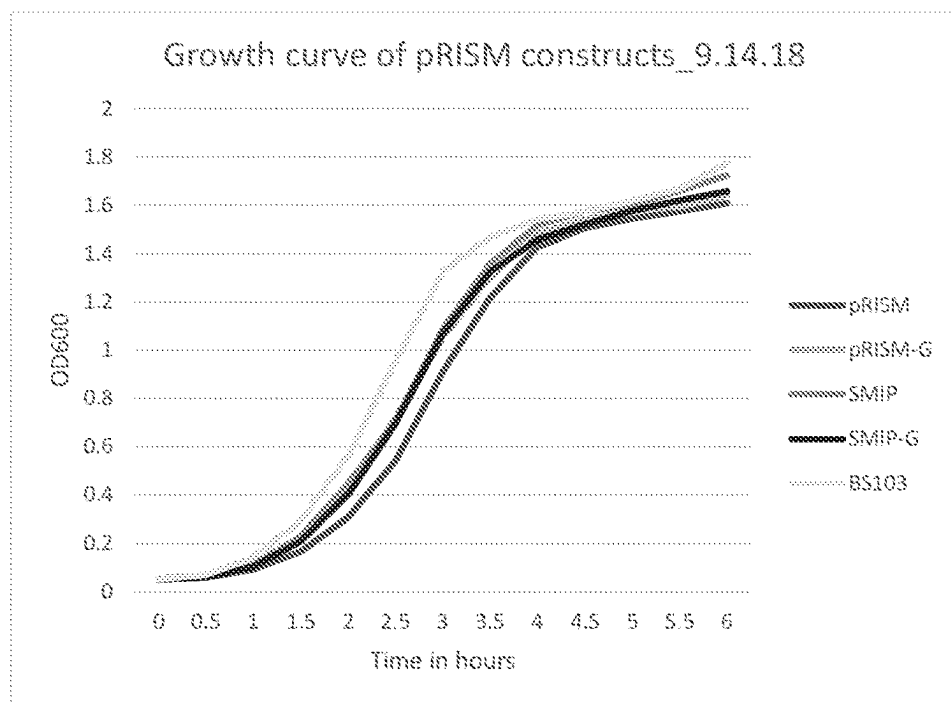


FIG. 5B

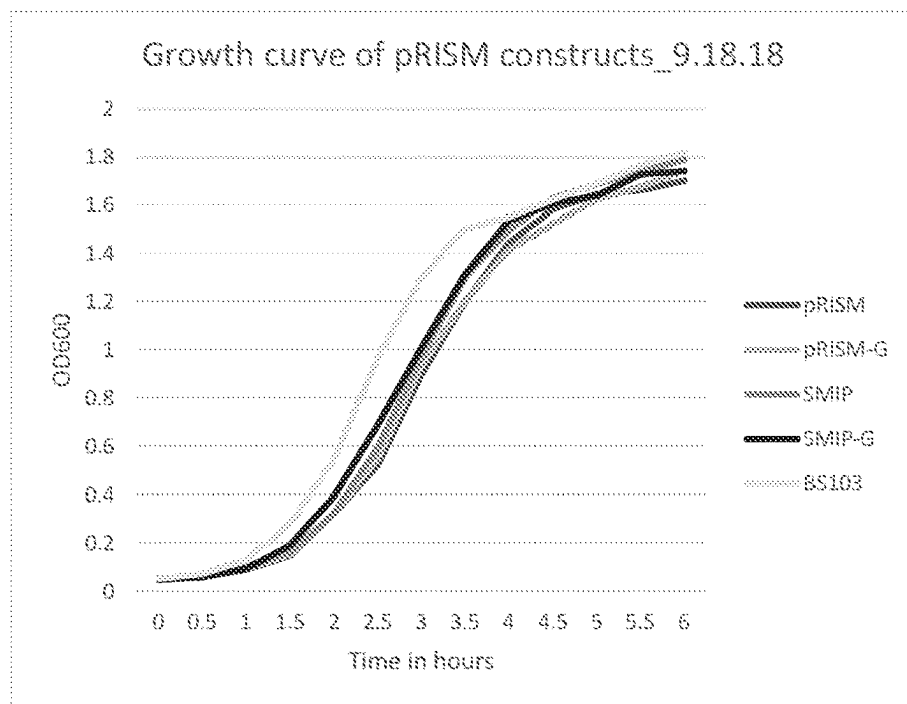


FIG. 6

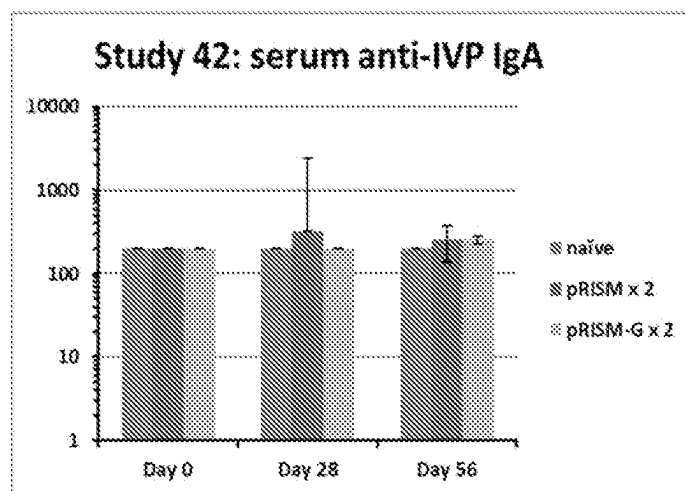
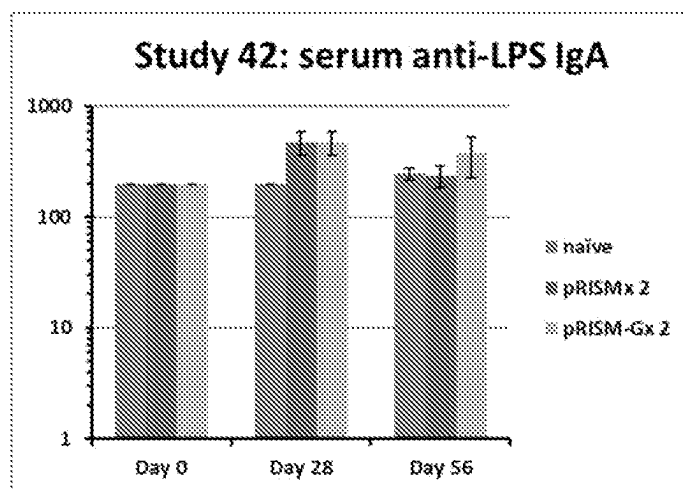


FIG. 7

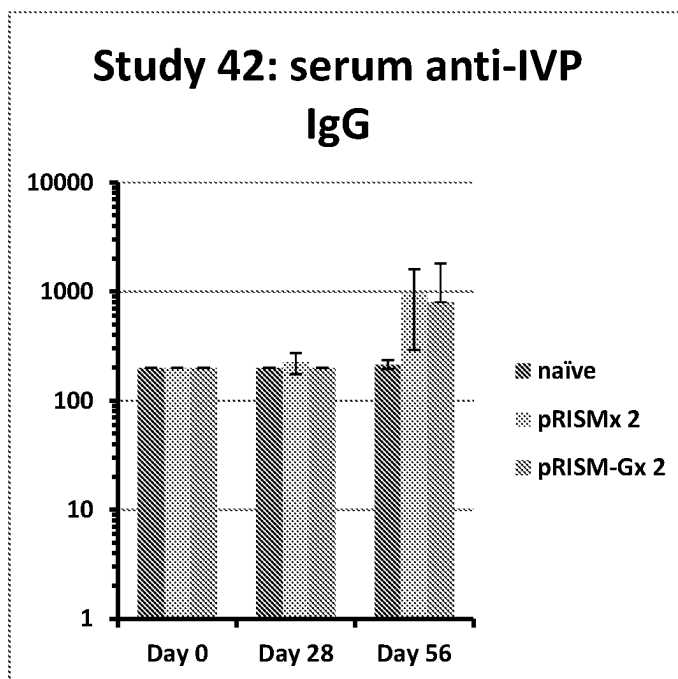
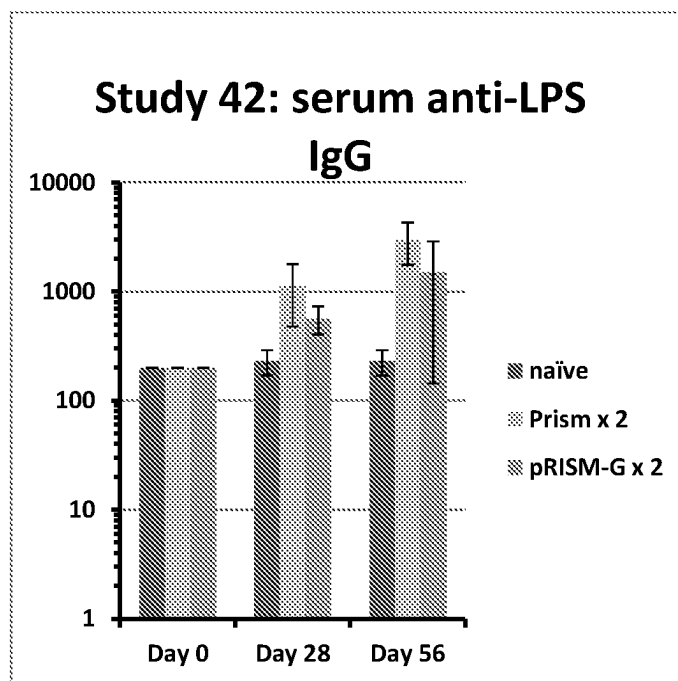
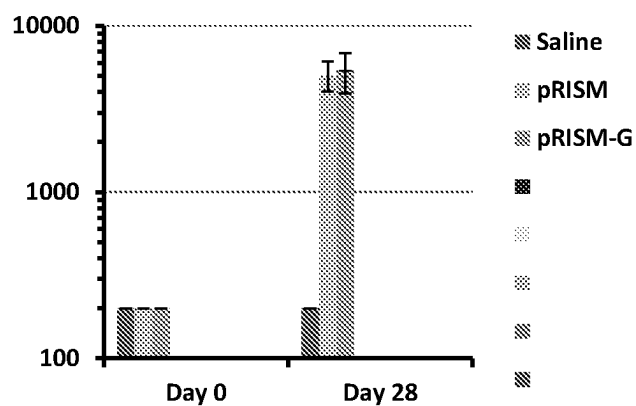


FIG. 8

**Study 42: mucosal anti-S.
flex 2a LPS IgA**



**Study 42: mucosal anti-S.
flex 2a IVP IgA**

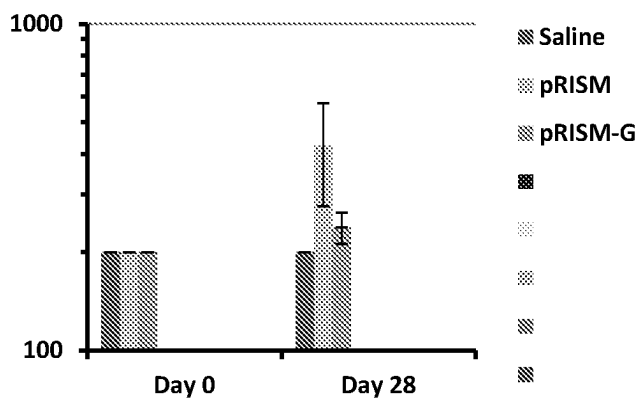


FIG. 9

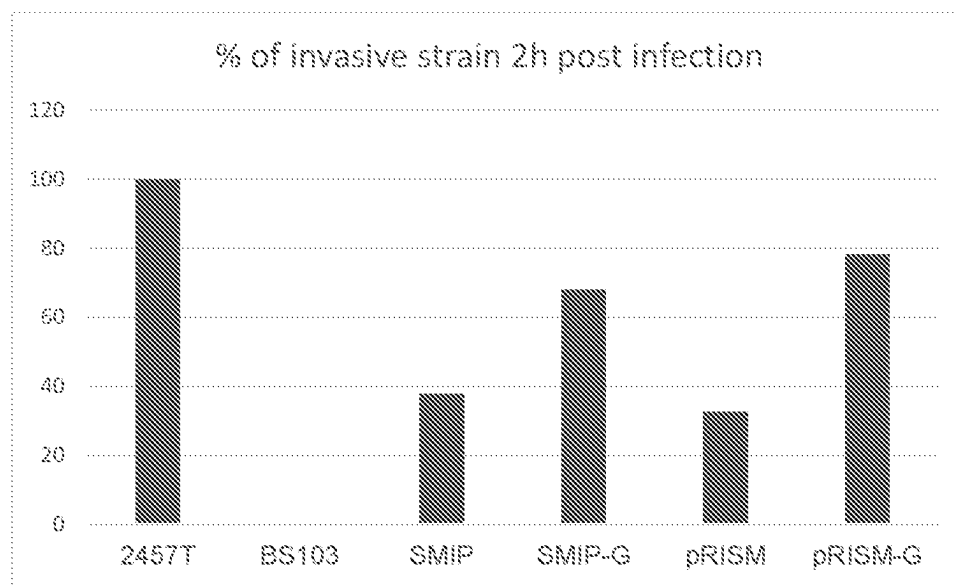
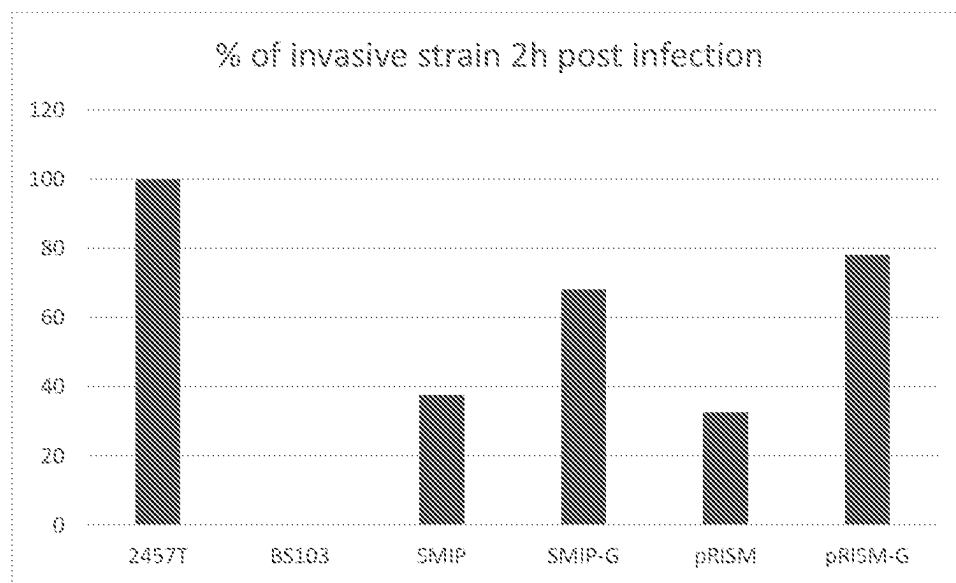


FIG. 10

1

SHIGELLA-BASED MINIMAL PLASMID RECOMBINANT INVASION CONSTRUCT

RELATED APPLICATIONS

This application claims priority to U.S. Provisional Application No. 63/021,463, filed May 7, 2020, which is incorporated by reference in its entirety.

STATEMENT AS TO RIGHTS OR INVENTIONS MADE UNDER FEDERALLY SPONSORED RESEARCH AND DEVELOPMENT

The invention was made with government support from the Walter Reed Army Institute of Research, a subordinate organization of the United States Army Medical Research and Materiel Command. The United States Government has certain rights in the invention.

SEQUENCE LISTING

The instant application contains a Sequence Listing, which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on May 3, 2021, is named "WRAIR_18_28_PCT_ST25.txt" and is 113,272 bytes in size.

BACKGROUND

Live, attenuated strains of bacteria have been successfully used as vaccines to protect humans and animals from disease. The successful vaccine strain is modified to reduce its ability to cause disease to an acceptable level while it retains sufficient pathogenic potential to stimulate a protective immune response in the vaccinated host. Therefore a delicate balance must be achieved as extensive attenuation generally results in reduced immunity.

Shigella is a genus of Gram-negative bacteria belonging to the family Enterobacteriaceae and the etiologic agent of bacillary dysentery or shigellosis, the symptoms of which include abdominal pain, diarrhea, fever, vomiting, and blood or mucus in the stool. *Shigella* is transmitted by a fecal-oral route, typically through contaminated food or water. As a result, *Shigella* is more of a public health threat in developing countries where proper sanitation and hygiene are lacking.

The development of an effective live, attenuated strain of *Shigella* to protect humans against bacillary dysentery (shigellosis) has been hampered by the inability to strike a balance between a need for the strain to invade the intestinal epithelium and reduction of pathogenicity to an acceptable level. Invasive strains that are protective tend to be "reactogenic," i.e., they cause diarrhea and/or fever in the host. The conventional approach to attenuate *Shigella* is to mutate or delete genes on the *Shigella* invasion plasmid in the hopes of reducing the reactogenicity of the vaccine strain while preserving immunogenicity. Levine, M. et al. 2007. Clinical trials of *Shigella* vaccines: two steps forward and one step back on a long, hard road. Nat Rev Microbiol 5:540-553. This strategy is a "top-down" approach starting with a wild type, virulent strain of *Shigella*. Genes are then systematically inactivated until the strain is sufficiently attenuated to be clinically safe to administer yet still stimulate a robust immune response. For example, the attenuating mutation in *S. flexneri* 2a strain SC602 is a deletion of *virG* (*icsA*) which abolishes the ability of the bacterium to spread from cell to

2

cell after invasion. Coster, T. S. et al. 1999. Vaccination against shigellosis with attenuated *Shigella flexneri* 2a strain SC602. Infect. Immun. 67:3437-3443. A gene on the invasion plasmid that encodes an enterotoxin and another enterotoxin gene on the chromosome were targets in construction of *S. flexneri* 2a strain CVD1207 leading to attenuation and reduced reactogenicity. Kotloff, K. L. et al. 2000. *Shigella flexneri* 2a strain CVD 1207, with specific deletions in *virG* (*icsA*), *sen*, *set*, and *guaBA*, is highly attenuated in humans. Infect Immun 68:1034-1039. A major disadvantage of this strategy is that the attenuated strains that are protective have the undesirable side effect of causing diarrhea and/or fever. Further, to achieve the appropriate balance in the approach between preserving immunogenicity and reducing reactogenicity, other genes on the invasion plasmid (also known as the virulence plasmid), and possibly the chromosome as well, that may contribute to reactogenicity must be identified and inactivated.

An alternate strategy is disclosed in U.S. Pat. No. 9,434,772. Vectors described in this application were designed starting with the minimal set of genes from the *Shigella* invasion plasmid that mediate invasion of and multiplication within host epithelial cells. These genes were cloned into a plasmid to produce *Shigella* minimal invasion plasmid (SMIP). The SMIP can be transferred into a plasmid-cured strain of *Shigella* or an enteroinvasive *Escherichia coli* (EIEC), where it can be used as a vaccine strain. Additionally, the SMIP can serve as a platform to which other genes can be added in an attempt to improve immunogenicity of the vaccine strain. This strategy is an improvement over the "top-down" strategy because it allows one to directly determine the contribution of any putative virulence gene by adding it back to the SMIP, i.e., the role of the gene can be measured in the context of the minimal gene complement required for invasion rather than in the larger context of all of the genes normally present on the invasion plasmid.

Inventors have discovered a different approach using PRISM and PRISM-G minimal plasmid constructs described in this application provide an improved platform for developing a live, attenuated vaccine strain of *Shigella* or enteroinvasive *Escherichia coli* (EIEC), that achieves the balance between immunogenicity and reduced reactogenicity in patients. In addition to inserting other *Shigella* or EIEC genes into the recombinant *Shigella* minimal invasion plasmid, it is also possible to insert a gene that encodes an immunogenic protein from a pathogen other than *Shigella* or EIEC or some other non-*Shigella* or non-EIEC protein of interest. In other words, a PRISM and PRISM-G *Shigella* or EIEC strain comprising the recombinant *Shigella* minimal invasion plasmid can be used as a broad antigen delivery system for delivering any antigen of interest. *Shigella* or EIEC invasion into colonic mucosa stimulates a strong immune response in the intestinal mucosa. Furthermore, *Shigella* or EIEC invasion in the colon can also stimulate lymphoid tissues adjacent to the colon, and, thus, it could stimulate a strong mucosal immune response in the reproductive tract. Thus, the recombinant *Shigella* minimal invasion plasmid can be used as a live vaccine platform for delivery or expression of recombinant antigens in host cells. That is, the recombinant *Shigella* minimal invasion plasmid can be used as a vehicle for delivering DNA vaccines to mucosal lymphoid tissues or for expression and delivery of recombinant antigens. As such, the recombinant *Shigella* minimal invasion plasmid can be used in methods of inducing an immune response in a subject. In one embodiment, the immune response is directed to *Shigella*. In other

embodiments, the immune response is directed to a bacterial, viral, fungal, parasitic, or mammalian protein of interest.

SUMMARY

It has been discovered that the recombinant *Shigella* minimal invasion plasmid constructs pRISM and pRISM-G can be used as a vehicle for delivering DNA vaccines to mucosal lymphoid tissues or for expression and delivery of recombinant antigens. As such, the recombinant *Shigella* minimal invasion plasmid can be used in methods of inducing an immune response in a subject.

In one embodiment, the immune response is directed to *Shigella*. In other embodiments, the immune response is directed to a bacterial, viral, fungal, parasitic, or mammalian protein of interest.

A recombinant *Shigella* minimal invasion plasmid construct, pRISM. In some embodiments, PRISM comprises SEQ ID NO: 1. A recombinant *Shigella* minimal invasion plasmid constructs pRISM-G. In some embodiments, pRISM-G comprises SEQ ID NO: 2. In one embodiment A composition comprising the *Shigella* or *E. coli* bacterium comprising a recombinant *Shigella* minimal invasion plasmid constructs PRISM and a pharmaceutically acceptable excipient. A composition comprising the *Shigella* or *E. coli* bacterium comprising a recombinant *Shigella* minimal invasion plasmid constructs pRISM-G and a pharmaceutically acceptable excipient.

A vaccine composition comprising the *Shigella* or *E. coli* bacterium comprising a recombinant *Shigella* minimal invasion plasmid constructs PRISM and at least one pharmaceutically acceptable adjuvant.

A vaccine composition comprising the *Shigella* or *E. coli* bacterium comprising a recombinant *Shigella* minimal invasion plasmid constructs pRISM-G and at least one pharmaceutically acceptable adjuvant.

A method of inducing an immune response to *Shigella* in a subject, the method comprising the step of: administering a composition further comprising a recombinant *Shigella* minimal invasion plasmid construct PRISM or pRISM-G to the subject in an amount sufficient to induce an immune response to *Shigella* in the subject.

BRIEF DESCRIPTION OF THE DRAWINGS

The following drawings form part of the present specification and are included to further demonstrate certain embodiments of the present invention. The invention may be better understood by reference to one or more of these drawings in combination with the detailed description of specific embodiments presented herein.

FIG. 1 shows a plasmid map of a recombinant *Shigella* minimal invasion plasmid construct, pRISM (SEQ ID NO: 1).

FIG. 2 shows a plasmid map of a recombinant *Shigella* minimal invasion plasmid construct, pRISM-G (SEQ ID NO: 2).

FIG. 3 shows PRISM construct colony D1 used for DNA preparation.

FIG. 4 shows pRISM-G construct colony D1 used for DNA preparation.

FIGS. 5A and 5B show growth curves of pRISM constructs compared to the virulence plasmid-cured *Shigella* strain BS103 and SMIP (FIG. 5A) and SMIP-G (FIG. 5B).

FIG. 6 shows mean titers of serum IgA responses in naive guinea pigs or guinea pigs immunized with pRISM or

pRISM-G assayed by ELISA from serum and eyewash samples collected on Day 0, Day 28 and Day 56. Upper and lower error bars together represent one standard deviation from the GMT. A. Serum IgA response to Invaplex. B. Serum IgA response to LPS.

FIG. 7 shows mean titers of serum IgG responses in naive guinea pigs or guinea pigs immunized with PRISM or pRISM-G assayed by ELISA from serum and eyewash samples collected on Day 0, Day 28 and Day 56. Upper and lower error bars together represent one standard deviation from the GMT. A. Serum IgA response to Invaplex. B. Serum IgA response to LPS.

FIG. 8 shows mean titers of mucosal IgA responses in naive guinea pigs or guinea pigs immunized with pRISM or pRISM-G assayed by ELISA from serum and eyewash samples collected on Day 0, Day 28 and Day 56. Upper and lower error bars together represent one standard deviation from the GMT. A. Serum IgA response to Invaplex. B. Serum IgA response to LPS.

FIG. 9 depicts average results of two colonies/*Shigella* minimal plasmid strains for the ability to invade HeLa epithelial cells in a gentamicin protection assay. Invasiveness is plotted as percentages of the invasiveness of a wild-type 2457T Congo red-positive invasion-positive isolate. BS103 is a Congo red-negative invasion negative colony.

FIG. 10 shows that pRISM and pRISM-G constructs are invasive and comparable to SMIP and SMIP-G constructs.

DETAILED DESCRIPTION

Reference will now be made in detail to various exemplary embodiments. It is to be understood that the following detailed description is provided to give the reader a fuller understanding of certain embodiments, features, and details of aspects of the invention, and should not be interpreted as a limitation of the scope of the invention. In order that the present invention may be more readily understood, certain terms are first defined. Additional definitions are set forth throughout the detailed description.

Shigella is a genus of Gram-negative bacteria belonging to the family Enterobacteriaceae and the etiologic agent of bacillary dysentery or shigellosis, the symptoms of which include abdominal pain, diarrhea, fever, vomiting, and blood or mucus in the stool. There are four species of *Shigella*: *Shigella dysenteriae*, *Shigella flexneri*, *Shigella boydii*, and *Shigella sonnei* (also referred to as serogroups A, B, C, and D, respectively). *Shigella* is transmitted by a fecal-oral route, typically through contaminated food or water. *Shigella* is very closely related to enteroinvasive *Escherichia coli* (EIEC), which also cause dysentery, with recent evidence suggesting that *Shigella* and EIEC are derived from multiple origins of *E. coli* and form a single pathovar. Yang et al., (2005) Nucleic Acids Res., 33 (19): 6445-58. Certain O-antigens associated with EIEC are identical to those found in *Shigella* spp. and many plasmid-associated virulence determinants are common to both EIEC and *Shigella*. Id. *Shigella* and EIEC invade colonic epithelial multiply intracellularly, and spread to neighboring cells. The genetic elements responsible for these invasive properties are located on the bacterial chromosome and on a large (about 220 kb) plasmid called the invasion or virulence plasmid that is unique to the virulent strains of *Shigella* and EIEC. *Shigella* variants that have lost the invasion plasmid also lose their invasive properties. In addition to the invasion plasmid, many chromosomal genes also contribute to virulence, including set1A, set1B (enterotoxin), iucA-D (aerobactin production

and transport), iutA (iron uptake), pic (serine protease/mucinase), and sigA (protease).

Virulence of *Shigella* is dependent on temperature. At 37° C. *Shigella* are fully virulent, whereas at 30° C. *Shigella* lose the ability to both invade epithelial cells and provoke keratoconjunctivitis in guinea pigs. Maurelli et al., (1984) Infect. Immun. 43 (1): 195-201. When *Shigella* grows at 37° C., the VirF protein induces the expression of the VirB protein, which in turn activates the ipa, mxi, and spa promoters, causing the activation of the spa and mxi operons. This results in the synthesis of a protein complex called the Mxi-Spa translocon. Contact between the bacterium and the host epithelial cell activates the translocon, leading to the secretion of invasion plasmid antigens (Ipa). IpaA, IpaB, and IpaC form a complex that interacts with the cellular membrane and induces a signal cascade that causes the bacterium to be internalized into the cytoplasm via an endosome. The Ipa proteins also mediate the release of the bacterium from the endosome.

A cell entry region of about 30 kb is generally conserved in invasion plasmids and includes the mxi-spa gene locus, the virB gene, the ipa gene locus, and the ipg gene locus. Yang et al., (2005) Nucleic Acids Res., 33 (19) 6445-58. See FIG. 5. The cell entry region is bracketed by insertion sequence 100 and insertion sequence 600 in all invasion plasmids, suggesting the transmission of a common invasion plasmid in *Shigella* or the transmission of a cell entry region to all invasion plasmids from a common source. Yang et al., (2005) Nucleic Acids Res., 33 (19): 6445-58. The cell entry region has the markings of a typical pathogenicity island (PAI), such as G+C content distinctly different from the surrounding genes, indicating genetic instability. The remaining .about. 190 kb of virulence plasmid DNA sequence has a mosaic character. It contains genes related to IncFIIA-type plasmids. In addition, the plasmid is littered with copies of IS elements and remnants of IS elements that account for 53% of the open reading frames on the plasmid as well as a class of virulence genes known as the osp genes. These virulence genes encode secreted effector molecules that are transported by the type III secretion system (T3SS). Given the unusual mosaic makeup of the virulence plasmid, it is likely that it was formed by sequential gene acquisition and possibly gene duplication and rearrangement. The presence of three separate and distinct ospD genes on the *S. flexneri* 2a plasmid is evidence that argues for gene duplication after gene acquisition. In any event, it is apparent that the virulence plasmid present in contemporary strains of *Shigella* spp. arose in a step-wise fashion.

In addition to invasion, a fully virulent strain of *Shigella* has several other hallmark phenotypes, including intracellular replication, intercellular spread, the ability to induce apoptosis in macrophages, the ability to protect a host mammalian cell from apoptosis, the ability to signal polymorphonuclear leukocyte migration across a model monolayer of polarized epithelial cells, and the ability to generate a strong inflammatory reaction in the guinea pig model for keratoconjunctivitis (Sereny test). Post-invasion phenotypes are important for full virulence and the state of the art for identifying genes responsible for these phenotypes has been to inactivate a gene and test for loss or reduction of virulence phenotypes.

The present disclosure provides a recombinant vector comprising the cell entry region of a *Shigella* or EIEC invasion plasmid and a virF gene from a *Shigella* or EIEC invasion plasmid and optionally a selection gene. In one embodiment, the cell entry region is from a *Shigella flexneri* invasion plasmid. In yet another embodiment, the recombi-

nant vector comprises an ipa gene locus from a *Shigella* or EIEC invasion plasmid, an ipg gene locus from a *Shigella* or EIEC invasion plasmid, a virB gene from a *Shigella* or EIEC invasion plasmid, a mxi-spa gene locus from a *Shigella* or EIEC invasion plasmid, and a virF gene from a *Shigella* or EIEC invasion plasmid. The recombinant vector is capable of directing the expression of genes that have been inserted into the vector. Thus, in certain aspects, the recombinant vector further comprises a nucleic acid encoding one or more foreign proteins, including, for example, one or more bacterial viral, fungal, parasitic, or mammalian proteins, or an immunogenic portion thereof. The recombinant vector can be used to express any protein of interest. In one embodiment, the protein is from an enterohemorrhagic or enterotoxigenic *E. coli*, including, but not limited to the intimin protein, the colonization factor antigen I fimbrial adhesion, or the heat labile toxin (LT) subunit B. In another embodiment, the protein is from *Salmonella*, including, for example, a type III secretion system-1 secreted protein.

The nucleic acid encoding the one or more bacterial, viral, fungal, parasitic, or mammalian proteins, or an immunogenic portion thereof or the one or more *Shigella* or EIEC proteins, or an immunogenic portion thereof, may be inserted into a replicable vector for cloning (amplification of the DNA) or for expression. Various vectors are publicly available. The appropriate nucleic acid sequence may be inserted into the vector by a variety of procedures known in the art. For example, DNA may be inserted into an appropriate restriction endonuclease site(s) in the vector using techniques known in the art. As compared to other mutated *Shigella* invasion plasmids (Coster, T. S. et al., Infect. Immun. 67:3437-3443; Kotloff, K. L. et al. 2000. Infect Immun 68:1034-1039. Levine, M. M. et al., Nat Rev Microbiol 5:540-553), the recombinant vectors provided in this application contain a substantially smaller portion of the invasion plasmid. In one embodiment, the recombinant vector comprises no more than about 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, or 200 kilobase pairs of a *Shigella* or EIEC invasion plasmid. Preferably the recombinant vector comprises no more than about 30-50 kilobase pairs of the invasion plasmid. The recombinant vectors disclosed herein can also be much smaller in overall size than the mutated invasion plasmids generated using a conventional approach. Thus, in certain embodiments, the recombinant vector has no more than about 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, or 200 kilobase pairs. Preferably the recombinant vector has no more than about 40-100 kilobase pairs. More preferably the recombinant vector has no more than about 40-60 kilobase pairs.

Suitable vectors may be chosen or constructed to contain appropriate regulatory sequences, including promoter sequences, terminator sequences, enhancer sequences, a selection gene, and other sequences. Typical selection genes encode proteins that (a) confer resistance to antibiotics or other toxins, e.g., ampicillin, neomycin, kanamycin, chloramphenicol, or tetracycline, (b) complement auxotrophic deficiencies, or (c) supply critical nutrients not available from complex media, e.g., the gene encoding D-alanine racemase. Such selection genes are well known in the art. Construction of suitable vectors containing one or more of these regulatory sequences employs standard ligation techniques which are known to the skilled artisan. See e.g., Sambrook et al., Molecular Cloning: A Laboratory Manual, 3rd ed., Cold Spring Harbor Laboratory Press (2001). Many established techniques used with vectors, including the manipulation, preparation, mutagenesis,

sequencing, and transfection of DNA, are described in Current Protocols in Molecular Biology, Ausubel et al. eds., John Wiley & Sons (2010).

The invention further provides a host cell that comprises at least one recombinant vector described herein. Systems for cloning and expressing polypeptides in a variety of host cells are known in the art. Suitable host cells include mammalian cells, insect cells, plant cells, yeast cells, or prokaryotic cells, e.g., *E. coli*. Preferably the host cell is a bacterium. In some embodiments, the host cell is a *Shigella* (e.g., *Shigella dysenteriae*, *Shigella flexneri*, *Shigella boydii*, and *Shigella sonnei*) or EIEC bacterium comprising the recombinant vector described in this application. In one embodiment, the *Shigella* or EIEC bacterium is a plasmid-cured bacterium that does not contain an invasion plasmid. A further aspect of the disclosure provides a method of introducing the recombinant vector into a host cell. For eukaryotic cells, suitable transfection techniques may include calcium phosphate, DEAE-Dextran, electroporation, or liposome-mediated transfection. For bacterial cells, suitable techniques may include, but are not limited to, calcium chloride transformation, electroporation, and transfection using bacteriophage. These techniques are well known in the art. See e.g., Current Protocols in Molecular Biology, Ausubel et al. eds., John Wiley & Sons (2010). DNA introduction may be followed by a selection method (e.g., antibiotic resistance) to select cells that contain the vector.

The vectors and bacterial strains disclosed in this invention provide an improved platform for developing an attenuated vaccine strain of *Shigella* or EIEC that achieves the balance between immunogenicity and reduced reactogenicity in a subject. In one embodiment, the attenuated strain is a plasmid-cured *Shigella* or EIEC comprising a recombinant vector as described in this application (hereinafter referred to as "vaccine strain"). As noted above, in one embodiment, the recombinant vector of the vaccine strain further comprises a nucleic acid encoding one or more foreign proteins, including, for example, one or more bacterial, viral, fungal, parasitic, or mammalian proteins, or an immunogenic portion thereof. As noted above, in another embodiment, the recombinant vector of the vaccine strain further comprises a nucleic acid encoding one or more *Shigella* or EIEC proteins, or an immunogenic portion thereof. Also disclosed is a composition comprising the vaccine strain, at least one pharmaceutically acceptable excipient, and optionally an adjuvant (hereinafter referred to as "vaccine composition").

The pharmaceutically acceptable excipient can be chosen from, for example, diluents such as starch, microcrystalline cellulose, dicalcium phosphate, lactose, sorbitol, mannitol, sucrose, methyl dextrans; binders such as povidone, hydroxypropyl methylcellulose, dihydroxy propylcellulose, and sodium carboxymethylcellulose; and disintegrants such as croscopovidone, sodium starch glycolate, croscarmellose sodium, and mixtures of any of the foregoing. The pharmaceutically acceptable excipient can further be chosen from lubricants such as magnesium stearate, calcium stearate, stearic acid, glyceryl behenate, hydrogenated vegetable oil, glycerine fumarate and glidants such as colloidal silicon dioxide, and mixtures thereof. In some embodiments of the present invention, the pharmaceutically acceptable excipient is chosen from microcrystalline cellulose, starch, talc, povidone, croscopovidone, magnesium stearate, colloidal silicon dioxide, sodium dodecyl sulfate, and mixtures of any of the foregoing. The excipients of the present invention can be intragranular, intergranular, or mixtures thereof.

The vaccine composition can be formulated as freeze-dried or liquid preparations according to any means suitable

in the art. Non-limiting examples of liquid form preparations include solutions, suspensions, syrups, slurries, and emulsions. Suitable liquid carriers include any suitable organic or inorganic solvent, for example, water, alcohol, saline solution, buffered saline solution, physiological saline solution, dextrose solution, water propylene glycol solutions, and the like, preferably in sterile form. After formulation, the vaccine composition can be incorporated into a sterile container which is then sealed and stored at a low temperature (e.g., 4° C.), or it can be freeze dried. The vaccine composition can be formulated in either neutral or salt forms. Pharmaceutically acceptable salts include the acid addition salts (formed with the free amino groups of the active polypeptides) and which are formed with inorganic acids such as, for example, hydrochloric or phosphoric acids, or organic acids such as acetic, oxalic, tartaric, mandelic, and the like. Salts formed from free carboxyl groups can also be derived from inorganic bases such as, for example, sodium, potassium, ammonium, calcium, or ferric hydroxides, and such organic bases as isopropylamine, trimethylamine, 2-ethylamino ethanol, histidine, procaine, and the like.

The vaccine composition can comprise agents that enhance the protective efficacy of the vaccine, such as adjuvants. Adjuvants include any compound or compounds that act to increase an immune response to *Shigella* or to any antigen delivered by the recombinant vector, thereby reducing the quantity of antigen necessary in the vaccine, and/or the frequency of administration necessary to generate a protective immune response. Adjuvants can include for example, emulsifiers, muramyl dipeptides, pyridine, aqueous adjuvants such as aluminum hydroxide, chitosan-based adjuvants, and any of the various saponins, oils, and other substances known in the art, such as Amphigen, LPS, bacterial cell wall extracts, bacterial DNA, CpG sequences, synthetic oligonucleotides and combinations thereof (Schijns et al. (2000) Curr. Opin. Immunol. 12:456), Mycobacterialphlei (*M. phlei*) cell wall extract (MCWE) (U.S. Pat. No. 4,744,984), *M. phlei* DNA (M-DNA), and *M. phlei* cell wall complex (MCC). Compounds which can serve as emulsifiers include natural and synthetic emulsifying agents, as well as anionic, cationic and nonionic compounds. Among the synthetic compounds, anionic emulsifying agents include, for example, the potassium, sodium and ammonium salts of lauric and oleic acid, the calcium, magnesium and aluminum salts of fatty acids, and organic sulfonates such as sodium lauryl sulfate. Synthetic cationic agents include, for example, cetyltrimethylammonium bromide, while synthetic nonionic agents are exemplified by glycerylestere (e.g., glyceryl monostearate), polyoxyethylene glycol esters and ethers, and the sorbitan fatty acid esters (e.g., sorbitan monopalmitate) and their polyoxyethylene derivatives (e.g., polyoxyethylene sorbitan monopalmitate). Natural emulsifying agents include acacia, gelatin, lecithin and cholesterol.

Other suitable adjuvants can be formed with an oil component, such as a single oil, a mixture of oils, a water-in-oil emulsion, or an oil-in-water emulsion. The oil can be a mineral oil, a vegetable oil, or an animal oil. Mineral oils are liquid hydrocarbons obtained from petrolatum via a distillation technique, and are also referred to in the art as liquid paraffin, liquid petrolatum, or white mineral oil. Suitable animal oils include, for example, cod liver oil, halibut oil, menhaden oil, orange roughy oil and shark liver oil, all of which are available commercially. Suitable vegetable oils, include, for example, canola oil, almond oil, cottonseed oil, corn oil, olive oil, peanut oil, safflower oil, sesame oil, soybean oil, and the like. Freund's Complete

Adjuvant (PCA) and Freund's Incomplete Adjuvant (FIA) are two common adjuvants that are commonly used in vaccine preparations, and are also suitable for use in the present invention. Both FCA and FIA are water-in-mineral oil emulsions; however, FCA also contains a killed *Mycobacterium* sp. Immunomodulatory cytokines can also be used in the vaccine compositions to enhance vaccine efficacy, for example, as an adjuvant. Non-limiting examples of such cytokines include interferon alpha (IFN- α), interleukin-2 (IL-2), and granulocyte macrophage-colony stimulating factor (GM-CSF), or combinations thereof. The vaccine composition can be prepared using techniques well known to those skilled in the art including, but not limited to, mixing, sonication and microfluidation. The adjuvant can comprise from about 10% to about 80% (v/v) of the vaccine composition, more preferably about 20% to about 50% (v/v), and more preferably about 20% to about 30% (v/v), or any integer within these ranges.

The vaccine composition can be administered to any animal, and preferably is a mammal such as a human, mouse, rat, hamster, guinea pig, rabbit, cat, dog, monkey, cow, horse, pig, and the like. Humans are a preferred embodiment. Administration of the vaccine composition can be by infusion or injection (e.g., intravenously, intramuscularly, intracutaneously, subcutaneously, intrathecal, intraduodenally, intraperitoneally, and the like). The vaccine composition can also be administered intranasally, vaginally, rectally, orally, intratonsillar, or transdermally. Additionally, the vaccine composition can be administered by "needle-free" delivery systems. The effective amount of the vaccine composition may be dependent on any number of variables, including without limitation, the species, breed, size, height, weight, age, overall health of the patient, the type of formulation, or the mode or manner or administration. The appropriate effective amount can be routinely determined by those of skill in the art using routine optimization techniques and the skilled and informed judgment of the practitioner and other factors evident to those skilled in the art. Preferably, a therapeutically effective dose of the vaccine composition described herein will provide the therapeutic preventive benefit without causing substantial toxicity to the subject. The vaccine composition can be administered to a patient on any schedule appropriate to induce and/or sustain an immune response against *Shigella* or any other protein of interest. For example, patients can be administered a vaccine composition as a primary immunization as described and exemplified herein, followed by administration of a secondary immunization, or booster, to bolster and/or maintain protective immunity. The vaccine administration schedule, including primary immunization and booster administration, can continue as long as needed for the patient, for example, over the course of several years, to over the lifetime of the patient. The frequency of primary vaccine and booster administration and dose administered can be tailored and/or adjusted to meet the particular needs of individual patients, as determined by the administering physician according to any means suitable in the art. The vaccine composition may be administered prophylactically (before exposure to *Shigella*, EIEC, or other foreign pathogen) or therapeutically (after exposure to *Shigella*, EIEC, or other foreign pathogen).

In another aspect, the vaccine strain or vaccine composition can be used in a method of inducing an immune response. The immune response can be induced in a naive subject who has not previously been exposed to *Shigella* or EIEC (or some other foreign pathogen). Alternatively, the

immune response can be induced in a subject who has been previously exposed to *Shigella* or EIEC (or some other foreign pathogen) and used to enhance an existing immune response. In one embodiment, the method comprises administering to a subject a vaccine strain or a vaccine composition, wherein the administration of the vaccine strain or the vaccine composition induces an immune response against a protein, or immunogenic portion thereof, encoded by the recombinant vector in the vaccine strain or the vaccine composition.

In certain embodiments, the recombinant vector comprises a nucleic acid encoding a foreign protein, or an immunogenic portion thereof, as discussed previously. In other embodiments, the recombinant vector comprises a nucleic acid encoding a *Shigella* or EIEC protein, or an immunogenic portion thereof, as discussed previously. Alternatively, the recombinant vector may comprise a first nucleic acid encoding a foreign protein, or immunogenic portion thereof, and a second nucleic acid encoding a *Shigella* or EIEC protein, or immunogenic portion thereof. One embodiment is directed to a method of inducing an immune response against *Shigella* or EIEC in a subject, the method comprising administering to the subject a vaccine strain or a vaccine composition in an amount sufficient to induce an immune response to *Shigella* or EIEC in the subject. In certain aspects, the vaccine strain or the vaccine composition induces significantly less transepithelial migration of polymorphonuclear neutrophils (PMNs), as compared to the 2457T *Shigella flexneri* strain or, in certain embodiments, does not induce transepithelial migration of PMNs. Transepithelial migration of PMNs can be measured using routine methods, such as those described in this application.

Another embodiment is directed to a method of inducing an immune response to a bacterial, viral, fungal, parasitic, or mammalian protein in a subject, the method comprising administering to the subject a vaccine strain or a vaccine composition in an amount sufficient to induce an immune response to the bacterial, viral, fungal, parasitic, or mammalian protein in the subject. In these methods of inducing an immune response, the immune response can be measured using routine methods in the art, such as those disclosed in this application. These routine methods include, but are not limited to, measuring an antibody response, such as an antibody response directed against the protein encoded by the recombinant vector, and measuring cellular proliferation, including, for example, by measuring tritiated thymidine incorporation or cytokine (e.g., IFN- γ) production.

Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, suitable methods and materials are described below. All publications, patent applications, patents, and other references mentioned herein are incorporated by reference in their entirety. In case of conflict, the present specification, including definitions, will control. In addition, the materials, methods, and examples are illustrative only and not intended to be limiting.

11

EXAMPLES

Example 1 pRISM Construct

Construction of a recombinant *Shigella* minimal invasion plasmid construct, PRISM (SEQ ID NO: 1.)

A core plasmid called SMIP for *Shigella* minimum invasion plasmid was constructed comprising the following features, as summarized in FIG. 1 (SEQ ID NO: 1).

Example 2 pRISM-G Construct

Construction of a recombinant *Shigella* minimal invasion plasmid construct, pRISM-G A core plasmid called SMIP for *Shigella* minimum invasion plasmid was constructed comprising the following features, as summarized in FIG. 2 (SEQ ID NO: 2).

PRISM construct colony D1 used for DNA preparation (FIG. 3). PRISM-G construct colony D1 used for DNA preparation (FIG. 4).

HeLa cell invasion assay: Live attenuated strains of *Shigella* are expected to retain the ability to invade non-phagocytic cells in order to generate protective immune responses in the host. The *Shigella* minimal plasmid constructs that invade and become intracellular will be protected from the effects of gentamicin whereas extracellular bacteria will be killed. PRISM and pRISM-G minimal plasmid constructs in a non-invasive *Shigella* background were evaluated for their invasiveness by the gentamicin protection assay with Hela cells. Alongside, wild type invasive *S. flexneri* 2a strain 2457T, non-invasive strain BS103 (positive and negative controls respectively) and SMIP and SMIP-G were also tested.

Gentamicin protection assay with Hela cells was performed as follows. HeLa monolayers were grown to semi-confluence in a 75 cm² flask in complete medium (cMEM) containing 10% heat-inactivated fetal bovine serum, 2 mM L-glutamine and 2 mM pyruvate. One flask was trypsinized with 0.25% trypsin and concentration of cells was adjusted to 2×10⁵ cells/ml in cMEM. Twenty four well-plates were seeded with 2 ml of Hela cells and grown overnight at 37° C. in 5% CO₂. Log phase cultures of bacteria (grown in LB medium with appropriate antibiotics for strain maintenance if needed) were added at an estimated multiplicity of infection of 10. After addition of bacteria, the plate was centrifuged in a Sorvall swinging bucket rotor at 500×g for 5 minutes at 25° C. and the plate was incubated at 37° C. in 5% CO₂ for 1.5 h. The plates were washed three times with Hank's balanced salt solution and then incubated with cMEM containing 50 µg/ml gentamycin for 2 h at 37° C. Hela cells were then lysed in a 0.1% Triton X-100 solution in PBS for 10 minutes. The bacteria in the lysate were plated on Tryptic Soy Agar plates and bacterial colonies counted after growth at 37° C. for 18 h. The figure below shows that pRISM and pRISM-G constructs are invasive and comparable to SMIP and SMIP-G constructs.

Shigella minimal plasmid strains were tested for the ability to invade HeLa epithelial cells in a gentamicin protection assay. Invasiveness is plotted as percentages of the invasiveness of a wild-type 2457T Congo red-positive invasion-positive isolate. BS103 is a Congo red-negative invasion negative colony. Average for two colonies/strain are shown in FIG. 9.

HeLa cell invasion assay: Live attenuated strains of *Shigella* are expected to retain the ability to invade non-phagocytic cells in order to generate protective immune responses in the host. The *Shigella* minimal plasmid con-

12

structs that invade and become intracellular will be protected from the effects of gentamicin whereas extracellular bacteria will be killed. PRISM and pRISM-G minimal plasmid constructs in a non-invasive *Shigella* background were evaluated for their invasiveness by the gentamicin protection assay with Hela cells. Alongside, wild type invasive *S. flexneri* 2a strain 2457T, non-invasive strain BS103 (positive and negative controls respectively) and SMIP and SMIP-G were also tested.

Gentamicin protection assay with Hela cells was performed as follows. HeLa monolayers were grown to semi-confluence in a 75 cm² flask in complete medium (cMEM) containing 10% heat-inactivated fetal bovine serum, 2 mM L-glutamine and 2 mM pyruvate. One flask was trypsinized with 0.25% trypsin and concentration of cells was adjusted to 2×10⁵ cells/ml in cMEM. Twenty four well-plates were seeded with 2 ml of Hela cells and grown overnight at 37° C. in 5% CO₂. Log phase cultures of bacteria (grown in LB medium with appropriate antibiotics for strain maintenance if needed) were added at an estimated multiplicity of infection of 10. After addition of bacteria, the plate was centrifuged in a Sorvall swinging bucket rotor at 500×g for 5 minutes at 25° C. and the plate was incubated at 37° C. in 5% CO₂ for 1.5 h. The plates were washed three times with Hank's balanced salt solution and then incubated with cMEM containing 50 µg/ml gentamycin for 2 h at 37° C. in 5% CO₂. Hela cells were then lysed in a 0.1% Triton X-100 solution in PBS for 10 minutes. The bacteria in the lysate were plated on Tryptic Soy Agar plates and bacterial colonies counted after growth at 37° C. for 18 h. FIG. 10 shows that pRISM and pRISM-G constructs are invasive and comparable to SMIP and SMIP-G constructs.

Shigella minimal plasmid strains were tested for the ability to invade HeLa epithelial cells in a gentamicin protection assay. Invasiveness is plotted as percentages of the invasiveness of a wild-type 2457T Congo red-positive invasion-positive isolate. BS103 is a Congo red-negative invasion negative colony. Average for two colonies/strain are shown.

Growth curves of pRISM constructs compared to the virulence plasmid-cured *Shigella* strain BS103 and SMIP and SMIP-G. PRISM and SMIP constructs were transformed into BS103 (FIGS. 5A and 5B).

Immunogenicity and Efficacy of pRISM and pRISM-G Constructs:

Guinea pigs were immunized ocularly on Day 0 and day 14 with ~1×10⁸ CFUs of either PRISM or PRISM-G while the control group received only saline. Eyes were observed and scored for disease 4 days post-treatment. Serum and eye wash samples were collected from all the guinea pigs on Day 0 (pre-immune sample) and Day 28 (2 weeks after the second immunization) for immune responses. One month after the second immunization, the three groups were ocularly challenged with ~1×10⁵ CFUs of virulent *Shigella flexneri* strain 2457T to determine vaccine efficacy. The eyes were scored for disease (intensity of keratoconjunctivitis or Sereny reaction) for 4 days post-challenge. The protective efficacy of the pRISM constructs were assessed using a scoring criteria established in the lab. The guinea pig eyes which were immunized with PRISM or PRISM-G showed an overall protective efficacy of 92% (p<0.0557) and 100% (p<0.0096) respectively compared to the control group. The guinea pigs were euthanized at the end of the scoring period on Day 56 when serum samples were collected by heart bleed.

SBA is a measure of antibodies capable of killing *Shigella* in a serotype-specific manner in the presence of exogenous

13

complement. SBA titers are used as a measure of immune response and vaccine efficacy. Serum bactericidal activity (SBA) assay was conducted on serum samples from Day 0, day 28 and Day 56 and eye wash samples of guinea pigs on Day 0 and Day 28. Results of the assays are shown below in Table 1.

TABLE 1

		Serum		Eyewash	
		Avg KI*	Fold increase over baseline	Avg KI*	Fold increase over baseline
PRISM	Day 0	20		20	
	Day 28	869	43	2991	150
	Day 56	4868	234		
PRISM-G	Day 0	20		20	
	Day 28	458	23	2430	122
	Day 56	8956	448		
Control	Day 0	20		20	
	Day 28	23	0	20	
	Day 56	399	20		

*Killing Index when 50% of bacteria are killed by the serotype specific antibody

SBA titers clearly show a serotype-specific killing response after receiving two doses of the vaccine on day 28 in serum and mucosal samples of guinea pigs immunized with pRISM and pRISM-G constructs.

IgA and IgG to LPS and Invaplex (IVP, an aqueous extract of virulent *Shigella* containing a macromolecular complex of LPS, IpaB and IpaC) were also assayed by ELISA from

14

serum and eyewash samples collected on Day 0, Day 28 and Day 56. Results are shown below (FIGS. 6-8).

Serum IgG to LPS: On day 28 PRISM and pRISM-G immunized guinea pig sera had a 5.6 fold and 3 fold increase in mean titer respectively compared to control sera. On day 56 there was an increase in the IgG responses. PRISM sera showed 15 fold increase and pRISM-G sera showed 7.5 fold increase in mean titer compared to control.

Serum IgG to IVP: pRISM sera showed 4.75 fold increase and pRISM-G sera showed 4 fold increase over control on day 56. On day 28 no significant increase was seen.

Serum IgA to IVP: On day 28 PRISM sera had about 2 fold increase in mean titer over control whereas pRISM-G sera IgA titers were comparable to control. On day 56 PRISM and pRISM-G sera did not show any significant difference compared to control sera.

Mucosal IgA to LPS: On day 28 PRISM and pRISM-G samples had ~25 fold increase in mean titer compared to control samples.

Mucosal IgA to IVP: pRISM and pRISM-G samples had a ~2fold increase in mean titer compared to control samples.

While specific aspects of the subject disclosure have been discussed, the above specification is illustrative and not restrictive. Many variations of the disclosure will become apparent to those skilled in the art upon review of this specification and the claims below. The full scope of the disclosure should be determined by reference to the claims, along with their full scope of equivalents, and the specification, along with such variations.

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 2

<210> SEQ ID NO 1

<211> LENGTH: 40999

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Recombinant Plasmid

<400> SEQUENCE: 1

```

gctagcatgc gcaaatTTAA agcgctgata tcgatcgcgc gccctgcagg taccgTtCgt      60
atagcataca ttatacgaag ttattgatcg gcacgtaaga ggTtccaaCT ttcaccataa      120
tgaataaaga tcactaccgg gcgtattttt tgagttgtcg agattttcag gagctaagga      180
agctaaaatg gagaaaaaaa tcactggata taccaccggt gatataTccc aatggcatcg      240
taaagaacat ttgaggcat ttcagtcagt tgctcaatgt acctataacc agaccgttca      300
gctggatatt acggcctttt taaagaccgt aaagaaaaat aagcacaagt tttatccggc      360
ctttattcac attcttgccc gctgatgaa tgctcatccg gaattacgta tggcaatgaa      420
agacgggtgag ctggtgatat gggatagtgt tcacccttgt tacaccgttt tccatgagca      480
aactgaaacg ttttcatcgc tctggagtga ataccacgac gatttccggc agtttctaca      540
catatatctg caagatgtgg cgtgttacgg tgaaaacctg gcctatttcc ctaaagggtt      600
tattgagaat atgttttttcg tctcagccaa tccctgggtg agtttcacca gttttgattt      660
aaacgtggcc aatatggaca actttcttcgc ccccgttttc accatgggca aatattatac      720
gcaaggcgac aaggtgctga tgccgctggc gattcagggt catcatgcgg tttgtgatgg      780
cttccatgtc ggcagatgct taatgaatac aacagtactg cgatgaataa cttcgtataa      840
tgtatgctat acgaacggta cctgcagggt catgatgatc attgcaattg gatccaagca      900

```

-continued

ctacccccgc	ctgttgcgac	agtcacgtca	tcgccgtccg	ttaccggaac	atgtttcccg	960
cgaaatacac	catctggagc	ctgaagaaag	ctgttggtccg	gagtgtgggtg	gtgaactgga	1020
ttattcgggg	gaaatcagcg	tctttctccc	ggaaaggcat	actatcttac	agtaatattc	1080
attccgatga	ttaataaaac	agcaatgcgc	gcatttttaa	tataagcatg	tacaatatgt	1140
acttgcacgc	attaccctgc	taatgcagtt	ttttttgtaa	taagttaaaa	tcctacatag	1200
cacggagggg	gaatgactag	ttttgagcag	ttaagtaagg	ttgcgcagcg	tgagataaaa	1260
atgttgctcg	ccctaacaaa	acagattcag	gagcaaaagc	aggagtttca	ggctgatgtt	1320
ttttatcaag	tctattcaaa	gtctgtctga	gctaagcttc	caaaattaac	gcgcgcaagc	1380
gtggatgggtg	cagtcggtga	gatggaggct	cagggtctac	agtttgaaaa	acgccagca	1440
ggaaccgcga	caaaatatgc	attaaccatc	cagaacatta	tcgatatcta	tgacatcgt	1500
ggcattcccta	aatatcgcga	tcgttatagt	gaggcatatt	ccatattcat	cggtagtcta	1560
aaaggcgggtg	tgtctaaaac	cgtgagtagt	gtttccgtag	ctcacgcctc	gcgtgctcat	1620
cctcacctgc	tgtcagaaga	tttacgtatt	ctgctccttg	atttagatcc	ccagtcatca	1680
gccactatgt	ttctgaacta	cttgcatgcg	gtggggctgg	ttgatacaac	cgaccgcag	1740
gctatgtttgc	agaacgtgag	tcgtgaggaa	ttactggaag	attttatcgt	tcgctcagtt	1800
atccctggcg	tttatgttat	gccggcttct	attgacgacg	cctttattgc	atcaaaactgg	1860
gatactctgt	gcgaggagca	tctattagga	caaaacaaac	acgcgatatt	gcgtgagaat	1920
attattgata	agctgaagca	tgatttcgac	ttcctcctta	ttgatacagg	cccgcatctg	1980
gacgcatttt	tgaaaaatgc	cattgtctgc	gcagatatca	tgtttacacc	ggtgccgcca	2040
gcacaagttg	atttccattc	cacgcttaaa	tatttggctc	gtttacctga	acttgtacag	2100
attattgaac	aggatggatg	ttcatgccgt	ttgcaggcaa	atattggctt	catgtcgaaa	2160
ctggcgaata	agtcagatca	taaatactgc	catagcctga	caaaagagat	tttcggtggg	2220
gacatgcttg	atgtttcaat	gccacgtctt	gatggttttg	agagaagcgg	cgagtcgttt	2280
gataccgtga	tatctgcgaa	tcctgttact	tatgttggca	gtggtgaagc	tttgaaaaat	2340
gcacgtatgg	cagcagaaga	tttcgctaaa	gcagtttttg	atcgtattga	atttataagg	2400
gccaaactact	gatggaaaac	agaaagcacc	gaccgactat	tgcccgtaacg	ctgaatacaa	2460
atatcctgaa	taatactgaa	gaaatatcgg	ctccagttca	tgtatttacc	ctcaataactg	2520
gccgaaaggc	aaaatttact	gagatcaagg	ttgatcatga	caaagtggat	actcaaaactt	2580
ttgtgggtga	agagggtgaat	ggccgcgaac	agacagccct	gacgccggac	tcactaaaag	2640
acatcaccag	aactatacgt	ttacaacaat	tctatccatg	tattgggatt	cggactggcg	2700
atcttattga	gattttggat	ggttcccgtc	gtcgggcagc	agctttgcta	tgcaagggtg	2760
gattacgtgt	acttgtaacg	gatgatgagc	ttactgttcc	tgaagctcag	catctggcga	2820
aagatttgca	gacgtcactt	gaacataata	ttcgtgaaat	tgggttcgct	ttagtacgtc	2880
tcaaagaggc	cggatgaac	cagaagcaaa	ttgcggaaaag	agagggattg	tcggcagcta	2940
aggtgactcg	tgccctccaa	gcagcgtccg	tacctaaaga	ttttgtgtct	ctttttcctg	3000
ttcagtcctga	gctgacatac	gccgattaca	ggcaactggc	tgaattaagt	gaacgactga	3060
ggctgggtga	tatatctatt	gatgaagttg	tgaagaatat	ctcgcagagt	attgagctta	3120
ttactgctga	tgacaatctt	tctgaagacg	agggtaaaaa	cagtattatg	cggcttataa	3180
ccaaggaaat	gtcttcattg	cttgattcag	gagtaaagga	taaagcgggt	gtaactttgt	3240
tatggaaatt	tgatagtaaa	gataaatattg	ctcgaaaacg	agtaaaggga	cgtacgttct	3300

-continued

catatgaatt	cggcagatta	ccgttggaag	ttcaggataa	gtagatcgt	atgattgctc	3360
ttgtactaaa	agataatctg	aattcgctgt	gaaaattcaa	gtttttcgcc	agcaactcgc	3420
cagtttttaa	ttggctgttt	taaaagaaat	tttgctttta	ttagcgtttt	ttttgaacag	3480
tatttcacgg	tgaatcgcc	tgtttcaccc	tatcttcttt	gcttgaaaaa	gatagtttct	3540
cttagccgtt	taacagtaac	cgatgagata	gtcttcccc	tctttcacgg	cagccttcaa	3600
ttgcgccatg	atggataaac	tttaacaaca	aaagtaagcg	ggataccatg	tctgaacaga	3660
aaatcaccgg	cgatccata	tatagggccc	gggttataat	tacctcaggt	cgacgtccca	3720
tggccattcg	aattcgccct	tatgcaatga	atcaatcaga	gccatccagt	ttaatcgtcg	3780
gttcctgtta	tatggagcaa	aagtgggtgt	gaagaaacat	ccgttttggtg	gtgttttttt	3840
aatctttttg	gggttttaat	tctatcgtc	caagtcttaa	cgatatccct	gaaacgacca	3900
cttaatatta	ctcacacttt	tgcaaatgca	ttaaatgctg	cttgatttat	tctgatgccg	3960
caccagcggc	atggaaggcg	gcacgctgtg	gttacatgtg	ataccggagt	aaaaccgccg	4020
aagcccgggc	tcagccggta	ctgattgaca	gatttcacct	tacccatccc	cagccctgcc	4080
aagccatacc	cgttttcagc	catgatagag	cttctgtgcg	cggtcggagt	ggccccgacg	4140
agggttttac	ccgaagtcgg	ggcgatatct	cgcgttagcg	ggccgtgagg	gccgcttacg	4200
agcgtgtatt	cgatacgtcc	agccagaaga	ctgacagcga	tgatgatgta	gttacaacat	4260
tcacaattaa	aagcgactct	gttcccgcgc	tttgggcggg	ggcggcgggc	cttttcagtt	4320
atgagggggg	cgttgtgggt	tcagttttgc	gttggaactg	ggtttttttag	gggttttatgt	4380
ttgcatgttg	taactaaacc	ggcttcgggt	gggggcccgc	cgttcgcgtg	gggaggtgca	4440
tatttgtctg	tccacaggac	aggcagtgaa	caggttttct	ttttaaatga	atgtaattaa	4500
gtagtttaaa	ggagatataa	acagggtgtt	aaaagataca	ttgcaccctg	taggggttgac	4560
gtctggcgct	ttatgacgtg	aacggttgta	accttatggg	gaagtccctt	gcagtttaat	4620
gtggataagc	aaaattcccc	gtctgtgagg	cgtgttttgt	atcaaaaaca	ggggggaccg	4680
gatgcacctg	aagggtgatg	atgaggttgt	ttttgtatg	tagcgtgat	tttttgtgca	4740
ctggcgggct	tcaggcgtgc	gaatgcctcc	ggcgcgtgcc	ggattattct	gaggagggtca	4800
ctttcagggg	gaagctgtgg	ccagccggct	gtaattacgg	ttacgtgaca	gaatcatgcg	4860
ctccttcaca	cgacgtccca	cttcacgttt	taccgcctca	cgactggcag	tgaagcgccc	4920
ttccgagatt	tcacgcgtca	gttgccgttt	caccaggggtg	acaatatect	gacgtgcct	4980
gtccgcata	cgacgcgcac	gggcacgctt	cattccacgg	gacttaagct	ctgtctggta	5040
actgcggaaa	cgctcacgaa	caaaacgcca	ggcttttcgcc	atcagttcat	ccatacccag	5100
ggtatccagc	ccctgctttt	tgcgctgtct	gttttcccat	tccacacgac	tgcggcgcgc	5160
ggcgccact	gcctcctctg	atacatcgag	ggcagcaaac	agtgacgatg	tgaacgtgat	5220
atcggtcgga	atgtagcacc	cgataagcgg	gtcatattcc	gtctggtagg	taatcagtcc	5280
cagctctgca	aggaacgtca	gggcccggtg	ggcccggtg	atggagagtt	ttccggcagc	5340
agactccgtc	gccagtcgcg	actcaatggc	cagcgtgggtg	atggaacact	ggacgcgggt	5400
ggccagtggt	tcatagtgaa	aacacagccc	ctgcagcagc	gcatcaatgg	cccggcgcag	5460
tagtaccggt	ggcatgcgcc	gacgcagacc	aagcgaacgg	gcatgtgcca	catgaatggc	5520
gaaatcaaaa	cgggaggtga	agcccaccgc	cttttccatc	agtttccgcg	agaacggaag	5580
cgtccgggca	ccttcacggg	gtgtgaacac	cggattcggg	ttctttacct	ggcggttaata	5640

-continued

agtttgttga agatcagtc	caccatcctg cacttacaat	gcgagagaagg agcgagcaca	5700
gaaagaagtc ttgaactttt	ccgggcatat aactatactc	ccgcatagc ttgattgttg	5760
gctgtacggg ttaagtgggc	cccggtaatc tttctgactc	gccaaagtta aagaagatta	5820
tcgggggttt tgcttttctg	gctcctgtac atccacatca	gaaccagttc cctgccacct	5880
tatggcgtgg ccagccataa	aattccttaa acgacagta	atctagcatg ctacgccaca	5940
aagtaaagtc tttactttag	tatatccagt ctctgcagtt	catctttgat gattttctca	6000
acgaactgag cctgtgttat	ccccctcttc tcgcagtact	caaccatgag atctttcaga	6060
ggatttttga caaaaacttt	tatctctttg tgtgtaagac	gttttcttgc aacagcggcc	6120
agttgtttct cagagtcagt	cataggctta cctctgcgca	caaaccgctt ttgactcaat	6180
gaggaagtc	ctgcattttc tgtctgcgac	atctcgcctc ctcaatactc	aaacagggat 6240
cgtttcgcag	aggatactat tgcataaggg	cgaattctag ctcaactcatt	aggcacccca 6300
ggctttacac	tttatgcttc cggtcgtat	gttggtgtga attgtgagcg	gataacaatt 6360
tcacacagga	aacagctatg accatgatta	cgccaagctt aaggtgcacg	gcccacgtgg 6420
ccactagtac	ttctcgagct cacatgtgtt	taaactgtac atgtggcgcg	ccacatgtca 6480
tggcgggcca	gtgctaatac tttggtatct	ttcgtgctat tctcctttta	cgtaagtact 6540
taaggggaac	gtatttacta cttttaaaga	aagctccgtt actatctgaa	atgcaatgag 6600
ataataaacg	attactctcg gttagtatct	catgatcatt tacagacttt	gcataatcgc 6660
gaatactaca	taacagagtt ttaacattac	aggatgaata gtctctctta	tcaactccta 6720
ataaaagaag	gattatgggc agtcgttttt	gctgtcttat catatattat	ttagtgcagt 6780
tcctaacaaa	aacataacaa ttttctacaa	ttgatcctgt gactacataa	atatcttcaa 6840
caaaaatatt	tatatcagca aaagagttat	tatgaatgat tataaatcat	cagggtgttt 6900
ttcaacgacc	aactccagag cacaacgata	aatgtctttt ctaaccaggt	cacattcaga 6960
tgaaatcttt	ctgagtaaaa cattttattat	atccttattg ctaaggaaaag	atattgggtt 7020
caataacctca	ttagcaatgg agttaatgat	ggaagattct atttcaaatt	cagactgac 7080
aatacctatt	ttattacaga aattaactcc	catataaacac atctccttta	cttattaatg 7140
agtgttttca	acctgctcaa gccaaacaat	aagacgtagg acttcaccca	agtgttcaaa 7200
atcaacaaaa	ctatatatttg tatgtgtttt	atatagtttt ctagtctaatt	tcacatcacg 7260
cacagtcggt	ataccaactt catttgcata	ttttctgaca gccaaaggac	actggttagt 7320
ttcaatgaga	gaaataaaag gtgcaggcgc	tatttctgga ttaaaataaa	taccaattgc 7380
aatatgagtc	gggttcatta ccactaattt	tgaattacgt atacagatt	tagtctgctc 7440
tgaaagaatc	tcgatatgca actcagcct	tctcgacttt gtctcaaagt	gtccctcttg 7500
ctctatatat	tctcttttta tctcctgttt	atccatcatc atatctttca	tgtataaaat 7560
gaactcaatc	acaaagtcaa gaataataac	aagaatagaa aatgccaaaga	aaaataatat 7620
tatatcctta	aacagcctcc cccaataag	atataagcca tcaacactag	aaaacacctg 7680
agaaaaaatt	atttttcggg catatatcca	aaagaaatag gttgttaatg	ctagaataat 7740
aagaagcaga	atgcttttga aaaattcttt	tattgtcttt atactaaata	ttttttttaa 7800
ccctttaaca	gggtttaata ctgaaaaatc	aatcttgata gcttttagtcg	caagaacaaa 7860
ctttgtttga	accaatgttg gcaacacagc	cgaaagaaca cagaaaaaaa	gtgggaagcc 7920
aattatctta	aaaaagacaa taaccactgc	aaaaaagtat ttacctcat	taatttcgaa 7980
gtcattaatt	attacatatc tgtataagag	cattacatca cttaaggaaa	agaatgatat 8040

-continued

tatagtaaat gtcctacca gaataataac aacagtcgtt aaatccttaa atttaaatga	8100
ctgtcctttt ttgoggcat ccttttagtt tttaggtgtc ggcttttctg ttttatttgc	8160
cataatattt tatctaaca atagatttgt gaaaaattta tggtcaccga ggaaaaattg	8220
aacttttagaa aagtaaatag tagaagaaca gatgaaaata ataatatttg caagtaaaact	8280
tttaatagtt aatgatacgg aaaaagcatt catctgcgga gcaaatctcg ataatacacc	8340
aagtaatact tctgatagta acaatactat cataacagga ctggctaaaa taacagcctg	8400
acttgccaat aaagtcagaa atgttaagat atttgagaca cggaagaac attgagaaaa	8460
taacgggcat atattataag acaattgtat ggattctaaa ataaagacca taccaccact	8520
gtatagaaat acaactgcag aaaaaagatt gaaaaatttt gccaaactcag acgtatcaac	8580
accattggca ggatcaattg aactactaag cgttgccccg cgctggttgt caataatgct	8640
accaacagca tgaaatatcc aaaagggaag agacaagcaa aaagagagaa ggaggccaac	8700
aattattttc ttgaacataa ggaataaac atgttcaaaa acagaagaac ctatgtctac	8760
cttaccagaa gtaattaatc ctgatgcaac aagaaaaatc acagggtattc taatagatgg	8820
agaaattata cgtttattta aaaatggaag aaagaaaaat aatggagcca atctaaaaaa	8880
aacgccgttc aggagtatta aaaacacatg aatactttcg aaccagcttg aaatgtccat	8940
caaaccctac tcttaattaa aaacattatt tcatgacaaa aagacaataa aacctcacca	9000
taccatccag aaagaagtag caaagatatt gagacaccta taagctttat accaaaagga	9060
agtgtctgct cttggagttg agtcactgtc tgtaacaaac caatagataa tccaataacc	9120
gtagctatac ctactggcca taaggaaaag ataaggatta aataaagagc cttattaccc	9180
atataaacta tatcagacac attacctcca tcatctaagc aggaatattg atatattgtt	9240
caatcaaggc cttagacagg ataccctacc catcaagcgc aacaaataat acaagtttta	9300
ttggtactga tattgtgatt ggactcatca tcatcatacc aagtgcaggt aaaatgcttg	9360
aaataacaag atcaacaaca acaaatggca aatataaata aaaacctatt ttgaacgcat	9420
ctttgatttc acttagcgca tatgcaggca agagggaaaa caatgaataa tcattattct	9480
ctgcactctt taaatcagca ttttcttctt cagaccttg gaaaaaacgt gctaactcaa	9540
gatctgtatg ctttttcaag tattgcttat attccatcaa gccagaatca gaaaaacgga	9600
caatatcaga tattgtatca aatttttgag ggccatttag ataattttca taaccagctt	9660
caataatggg tttcatcacg aacaaagcca taatcaatgc aataaccattg agagtcatat	9720
ttgatggaac ctgctgcaaa ccaagtgcatt ttcgtacctat taaaaaaca atagaaaact	9780
ttatataaca ggtaccagca gcaactaaaa atggtaaatag cgtaaaaaaa gaaagtgtag	9840
cgatgagggg catgtcactc agcatgagat tactccttta ccatccaaga actaatctcg	9900
ataccataac catcctcaat agaaacaagt tccccatgcc caaccaaaga accatttact	9960
ttaatattta catgtttaac tatgtcatcg ggaaacttga ataattcgtt ttctacatac	10020
attttttagtt cattgattgt catatttttt tctaaaagaa taaagtccac ttttacattg	10080
atatcatcat aattgaataa tgccaaagaa acttctgtgt gttctgactc accattactt	10140
tctgacagat taatttttgc ctcatatta tcattcacia tataatcccc aataataact	10200
tgattacaca ataataaatt ataaagtctc tgaactatga ttatatcacc agtgcgaatt	10260
ctttttaaag tagccagcct tacattagat gaacctaaagc aaaactctaa aagatgatgc	10320
aatgagtcaa gagaaacttg attacgtaat accggcatct tgtcaaaata aataatacct	10380

-continued

aaagaagagg	aaaaacagca	taaatgactc	ccagtagttg	catactggat	ttcattcgtg	10440
atccggaaag	taatcttata	agacagttct	ggaatcgaaa	atacagatga	cgactctata	10500
aagacacatt	ttgataaatg	cagtaaagat	tcttgtgtag	aaaatcctga	aaatacctcg	10560
tatttttttt	tcaaaaaaga	taatgtatca	atacgaatta	cccaatcccc	acacacggaa	10620
gtaagagtga	ctaaactttc	actgcctgta	aatttatttt	tgaatgtctg	aatcgaaaaa	10680
cgctcgcaga	gttgctttgc	atataaaatc	tgtagttttt	cgttagcgtc	aaaatgttta	10740
attcttagca	ttcttcttca	ctctgttcat	tatactctgt	actattattt	ttattccctt	10800
tattatgttg	actgctatca	aatctgtaat	taccctcagc	atcttgtttc	aaaattaagt	10860
caagtttatt	tcctacaaat	tgatctgacg	gttttagaac	aaaagaacca	gactccactg	10920
aaatactaac	agaatgtcca	gccccccact	gtttaaagtg	gtaagtcatt	tcactatttt	10980
tttgaaaaac	atttttatca	ttttcaacta	ttgatttttc	ttgcaatgac	atggatgggg	11040
ttatgtctgc	agtattttta	ttcattacag	gcagctcaaa	tgatttagca	acacttccat	11100
ttagtaaaga	aatgatgtca	gcaatattct	tatcaataga	caagctatga	ttttttaaac	11160
caatttgacc	atttctatta	tcagattcta	gcagattatc	agggtttaca	gaagaagtat	11220
gacttctttc	gaagttcttg	tctgcatttc	cactcttatt	atatgaaaag	atccccctca	11280
tatcgccgat	atttattttt	tcccttgctt	tgatatcttg	atctggtatc	aaaatatttt	11340
ctttaatggc	cattgattct	ggaatatcag	aatatgtcgc	tcgttcgttt	ctaagttcga	11400
acgcattttt	aatgccacta	taaatttctc	ctttatcatg	cttatgttca	aagtgccttat	11460
caatagtaaa	atagtttagat	gcaattaaag	agtactcagg	aatttcaacc	ttcctcttct	11520
tcttcaaaaac	gagactatca	atattatcta	ttgaagatag	tttctcacat	ttttctatct	11580
gtttgtcact	actaaaattt	aggtttatat	tatctaattg	catacttcat	atcctcttgt	11640
gataactcat	caagtatcaa	ctgtttggca	aagttatttc	ttttaatcaa	gccaatccgt	11700
ttacatattt	tactctcttt	atgaagcagt	tttcttttag	aagcatttgc	ctcatcaata	11760
tcttttttgaa	tttcttcaat	tcttcccttt	agaagaattc	gttcagactg	atattggata	11820
atatgctttt	ttactatagc	gatcttacgt	cttgtattga	aaagctgact	aacactctga	11880
gcgcctacag	gttctaagtc	attctccagt	tcaggaaggt	atataatcct	tttatctaata	11940
gcctccacat	ctcgatacaa	agagtttttt	ttttagtgtt	aatatgtgtg	cttcaattaa	12000
tttaacagac	cgttgatatt	tatctttgat	ttttaaaacc	ttatctaatt	gtttcaccaa	12060
taagctccat	tgtctgtcga	taagtaaaac	cgagtcgata	atcttgcttc	agaaaactct	12120
caacaacaga	tattttgttg	tagattttat	cctgagaggg	attctcccc	ggtttgtatt	12180
caccaaagtc	aataattgtc	cttagctctt	ctatttcaga	aagtagctcc	cgaaatgcag	12240
cggccataat	acgatgtttt	tcatcaacaa	cctgtgtaaa	taccctgctt	atacttttta	12300
aggaatcaat	tgcaggaaat	tgtccttttt	gggctagatt	tctgctcaaa	tatatatgtc	12360
catctaaaat	ggatcttacc	tcttcagcta	atggatcagc	aaaatcatca	tcctccaaaa	12420
gaacagtgtg	aaatgcagta	atagagccac	ctgcctttta	ctttcctggc	ctttcaagaa	12480
gtctgggtaa	gctatcaaaa	accgaaaccg	gatagcctct	tctggcaggt	gattctccag	12540
cggctaaggc	cacatcacgt	aatgctctgg	cataccttgt	taatgaatca	ataaaaagcg	12600
ctactttatg	tccttcagtc	ctaaaaaatt	cggctatagc	agtggctata	tatgcagcat	12660
tacaacgata	aaccgaagag	taatccgaag	ttgcatatac	taaaacacac	ctgctttttt	12720
tctcagagtt	tttcaataaa	tcaaccgttt	cagtaacctc	tcgacctcgc	tcaccaatta	12780

-continued

```

acccaataac atatatatca gcaccactat gttcaatgag catattcatg agaaaagttt 12840
tgccacaacc agctgacgca aaaatcccca ttcgtgtgcc ttcaccacac gtgagtaaag 12900
aatcaataac cttaatacct gttaaaaaag gcttctcaat tgcagccctt tcactatata 12960
gctggaggagc attatctaca ggtcgataaa gaatttcact gttatctgta acagcaaact 13020
tatcagtaac ctcccctaaa ggattgacta ctgctcccaa taatccacgg ccgacttgcg 13080
tatgaagaaa ctgggcagtg ggcttaatca acgtttgccc tgaaagtcca cgagaatttc 13140
caatcaagct taatatgttt ttttcatcat gaaatcctac aacctgagct cttgcaacaa 13200
tttcattact ttcaattcca gctggaatgt taaaaatctc accaatcgaa acatcgctaa 13260
gacttgtttc caagattggc cccgagattc tattaggaaa agataattga gtgagcaatt 13320
ttgtatagct cattgattat aagaccccat ttaagatttc catcctctga taaaactcat 13380
gcagaatctc tgcaaaaaaca atcccatcgt gaacgtagtc atctttaata acaactctta 13440
actgtaagta ctcacagaa cgatgaagct ctaccagctc attaatgcta taactaaagt 13500
tcattaacat taaatttaat atgttataag cagatgactg aagttttaca tcacttggtg 13560
catcaaaatt agcccataac ataacctgct cattgacaag agctatatta atagcaggca 13620
tactatccaa agaaatggta atcgctgaat ggctgtctaa atctgtaata attgatggag 13680
gacaaccaat cgtgaaaaga ctatctctaa ctaattgaac taaattaatg ttactcatat 13740
ttaaacctca ctaaatagtc ttaatacat taatggata tgcttcatca atctcagcat 13800
acgatataac gagtatactc ttaaatctgt tatctataag tcttttaaca aaactacgta 13860
tatctactga taccaaaaga acgaaatttt tttttgcatt tctcaattct ctcaaagcat 13920
gtgctaaagt ttccattacc tcatccgaaa cctctatata catattcaag aaagagccac 13980
cagagggttg ccttatccct tttcttattg catcctcaat atatccggaa agcatcacia 14040
ctttaatctc accagaaaca gctattttac taaaaatata cctagaaagt gatgcacgga 14100
catgttcaac taatgttatg acatcttttt ctctgggagc ccaaagcgcc aaagactcca 14160
taataagttt taaattgcga acagaaatat tttctccaag caatctttgt aatacctcag 14220
aaattctctg tatagtcaca tgtcggaaga cttcctttta tagatcagga taccgatttt 14280
caaactgata taacatattt tttgtttctt gtataccaaa aatctcattt atgttggtta 14340
ataaagcttg tgacaactgg tgataaaatt catcttgagc acttttaatt ttagcatcaa 14400
tattagttag attttctgta tatgaaacgt ctaccaaga aataacacgt tcattatatg 14460
aagttgaaac aacaggaata cctagagcat ctatatctcc attttcatct gtaatacaca 14520
ctttgtcaaa atatataatta aaactatcgg cagcaccctc atttattaac aaaacaatat 14580
catcgacctt aagctcattg cttgttctat ataaaaatg tggaagcctg acaccataat 14640
caataaagaa ctgacttctt atcctttcaa taagacctc catatcattt gcatttatct 14700
tattttcggc aaataataaa attaaaggaa cggtttctga actaatagca tccagatttt 14760
ctatcattgc caaagatgag tcattggctat tatcaatata aaatgtacca gtatagccac 14820
tagaatcaga ctcagacaga cttttttctt tttctacgac ctttttataa tagaataaag 14880
ccgtcaaagt aactgctatc aggaaaaaaa caaaaaacgg aaacctggc aacatcccta 14940
tcgccaaggc aagagctgat gtaacaatga gaacaaaagg attcccaaat atctgggaca 15000
ttatgttacg toccatatta tcgctatcac cattaactct tgtaacgata aatcctgcac 15060
ttatagaaat taacagagca ggaatctgag agaccaatcc gtcaccaata gttagtatgg 15120

```


-continued

tataagtaga taatgcacca gatagggaca ttccatgttg actcatacca acagaaatac	15180
ctcctattaa attgacaaaa ataataatga tgccagcaat tgcatacct ttaatgaact	15240
tcattgtctc atcaaatgaa ccatacaatt gactctcacg ttccagaata ctgctgtttt	15300
ctttagctcc tgcggcatca ataattccag cctttaaatc cgcataatg ctcatgtgtt	15360
tccccggcat accgtcaaga gaaaagcgag cagctacctc cgcaaccctt tcagatcctt	15420
ttgttataac aatgaactgc acaacagtta ctattgaaaa aataacaaaa cctacagcca	15480
gactatcacc aatgacaaac tggccaaatg tagtaataat tttaaccgca tccgcatcaa	15540
ctaaaattag tcgactagta cttattgata aagcgagacg aaataatgtc gttatcaaca	15600
aaaccgaagg gaatgtagaa aaactcaata ttctttcaat ataaaatgat cccataaaaa	15660
cgagaatagc aagtacaata ttaagcccta ttaaaaaatc aacaaggtat gtaggtaatg	15720
gtatgatcaa catagcaatg atcataacca taagcactaa tataattaat tcaggcttag	15780
tacttacttg ctttaaaaa gactggatca cttttatctc ctgttatcta gaaagctctt	15840
tcttgtagtc acttgtagca agctctcgta gaattatgat cgcaatttct ctttcacttt	15900
cacttttaaa tataatcatga ggactttcat taacagaaga tagatatgtt tgatacacta	15960
cagaatcctc atgactggct cgtgatactt ctataacaga agtaagaatt tgctcacttt	16020
cacttgata ttgaaaaata tataataaag ttagcaatag ctgctctgca cttaaaactgc	16080
catcttttag caagcttgag gattccagtt tcttcataaa aattacgtct gaagttctaa	16140
tcactttaat tgcaattaat ttagataaca caaaacaaaa ctcataagca ttacagctcg	16200
gtttctcaga atccatgtct acaatcagtg actgctctac aaaattcaca attgtatgtc	16260
tgtgattaaa accaaaatta ccaagccatt ctatatactg atctgtgtgt gatatgttcc	16320
ccatgataaa accacgataa catgcacgca aaagctgtgg ttcaagtttc attttgttgc	16380
caaataattt agcctgtata gctgaattta ttccagcatt aacacttttt tcaactactg	16440
atattatctc attaataatt ttagttaata attcgacttg ttccagcagta agatcttttt	16500
cattcaaaag ttcccttaat gctaagactt gatcacttgg atctttaaaa tatctcttta	16560
gtctgtctat aaaatcccgat tctagaggaa ggttatcttt taacgttctc tttaaatcaa	16620
aaatcttgtg attaatattca tcttcttccc catctaaaa acgttccctga ctatcactat	16680
ttgttctctt cagtttctca aggtctcttc tgtttataaa tgacgaaaga gcggaagcca	16740
tctcatcaga tgagtcaata tacttgctat tagccaagcc agaataagc tctgcatcag	16800
cagtctcatc tccatcatct gaatttgtca ttgcattcaa cttatcaatg aatgcagagc	16860
taaaaactcc tgtattttta acatcaagca tacatcacct actttatagt tagtaattta	16920
agtatgaaac caatgatttt tcgtcttcca gcaaagtagt ttccagatggg attatttgtg	16980
tagtcttttg tatttctctt tcagatatga ggcttttata ttctgcagta ttataataac	17040
ttgactcctt tatctcacgg ggttgaatca gaaaaaccct tacaatatta ctgatattac	17100
ttgttttata cttgaacaca ttaccgatta caggaataga ggagagaaaa ggaatagata	17160
ttatttcggt agagttcggt tcatacggtat aaccacctat caacaaactt ttgccttgtg	17220
gaactctcgc tattgtactg atttttgtcc ttccaacttc tggaaggaca gatgtatttt	17280
cattattata attgtaattt gactgactat ttectgttcc atcctctatc gttaaagaca	17340
tctcaatttg tcccogagat gaaaagcgag gtataacatt tatcaatgtt ccatacgtaa	17400
catgctccaa cgaagaattg cgctcaccaa ctaatgacac ataaaaagtc ctgttattat	17460
caaatatagc cggaatatct tcttgtgtca aaatcaccgg ccttgaaact acatttgctt	17520

-continued

tctttttctg atttaatgcc ataactgaag caataaactt attgccgtcc aaagtgttta	17580
tcgatgtga tgagctcata ttaaatgacg caccaaacga atcaccaaat gaagccgttc	17640
cttgccaatt aacaccaagg ttatttaatt cagattttatc aatatcgatt atccataatg	17700
ataattctat atgtogtttg gctatatcta actgagtgat aatatcgca ataactgtga	17760
tttgtgatc attaccttg actaatatag aattgggttc tggatatgca ataagtata	17820
catcctctaa tatagaacta ttggtaactg aactaaatga aaagtcatta ctatcctcag	17880
atactttctg tgtaataatta aagggtggca tcggatcatt ttgggttgt ctgttactca	17940
gagccttacc attatttaaat aaacgttcta caacagtagc caccctggg atgacaatat	18000
cctccctct catattataa gttctgtcac taacaaatgt attttttaat ttataaccc	18060
caaaattaac tttatctgtc ccaatagaag aaacctgctt atcaagcaa gttgtgtat	18120
tagctaccag ttcaaccaat gccggaggac cagatatata gaatgttta tctaataatg	18180
ttcctcttat aggatatctg tgatcataaa ggtagcatc cttcagatc tgaataagat	18240
aattaagtga tatattctcc agcaaaataa ctttgcttat taattctcca gagtcatata	18300
tgtataatgc atttccatct ttataccata tgagtccac aagtaatgtc agttttcaa	18360
gcatttcttc tggattagat aaatcaaaact cccagatat tctcttttt gcagcttgt	18420
tgctaacaac aataggataa tttataacg cagaaaaacg ttcaaaaaa gatcctaccg	18480
tatcactttg agcaacatat ttcgtatgt cattttgttc taataaatga gagtcaatat	18540
tattagcatt aactatcaat ggtaacaata caatcaagag agtcaaagat ttaatattaa	18600
attttttcat ttattttttt cactaaaaa gtaattctc ataattctc tcgagaaaag	18660
cctagtcgtg ttttatctc atttgaaaaa tgggatgtg atgcataccc attatcatg	18720
gcagctgaag tgattgtttg attgtgcaaa aatacatcta ataaaccatt tactaaccgc	18780
caagtgttca gttgtcttt gacttttgcc cctaaagctt ttctacataa agatctaaaa	18840
taagcctccg acacaccata atgctctgtc agactcttta ttcttacctc cttttctttc	18900
tcaatagttc tcagaagata aagaaccaa tgatatgact cagtcattcg tatcatagcc	18960
agtgcatcatt ttacattctg gtcagatgaa aagatgaatc ttaacattaa cttatcaagc	19020
atcgattcgg tattacaact atcccgaaga tcattctcat gatagaaatg aatacatctt	19080
tcagcatggc tatgctctac tagagcacca ctgaaatagt ccaaaaaagc aagaaatctt	19140
gcacaactat taatgttaaa gcgataatga tagttcacat tatcactgaa ttttatattt	19200
tgatttcgga caaaaaaaaa ccttccatg ctggagccag ctcaattatg ttacgttctt	19260
tatcttttac aagctcaacg ttaactgggt catgtccact agagtaaag taaaacatat	19320
tacttgatt tagaccttta tatttactca caatacaatt atatccctcg aattaatttt	19380
aatttaactt ccccatcctt ctctataagg atggtatcat ttatcttatt accttttaa	19440
acaacccctt ctttcagcat agaaaatagt ttatctatag ctatcttttg tgcattcatg	19500
aagcaccat ctacttcctt aacattaaag ttgcccata atttacctgc attaaaaacta	19560
taaaccgctt taccactaat acatttagag taaacattta tacttttatt ggttgattta	19620
aacgaccata ataaatcatt tggaatagaa aaataatcct tacttacagg aacaatatgc	19680
cattcttttt cagaattaga tgatgacttt aaagcacacc cactcagtg taatagcaat	19740
atacttaaaa taaatatttt caacttatta ctacatgtc gaatcatctg cctctctccc	19800
tgcttcaaag aactaatta atgaatcgaa ggttgtagtc ttaatataga aatcaatatt	19860

-continued

gttatatatg gacaatgatg cataaaatcc ctcattgcaaa tgcacatatt gattttgcac	19920
tatgagttta ttaagaagtg aacttttttg tgcagctgta agataaggaa aacgtgaatt	19980
gaataattgt tcaatcttat tgtgcagttc atcattatct atggaattta attcactctg	20040
tgagaaatat tgcattatct catagattat atccctcatca aatgattgag aagatgataa	20100
tctctcattt aaattaacaa cttcttcact atttaattct tgcctgcagag cattgcttgc	20160
atttatattga ttaattatta acaggattct cttttaacaa gtcttctgca atttgtctca	20220
gtttcttatg tgctaactct ttgaaatatt ttttatcatt ttttgtaatt ttttcgcaa	20280
gttctatttt aacacctgat attacatcct gaggactaaa ctccactact tgattaccac	20340
tacagaatat ataattatcc ccttcagaat attttaaaat aatcttgaca tctgagcgat	20400
atttatggag atcaagttct aactttttcc tcagtgttaa attacacttg ggcaaaacca	20460
cttttaattc tgttactgta ttacgcaacg aactaagcca ctgctccaat aaagtgttaa	20520
tgatcctctc atcatgtaaa ttttctgaaa gcaaaactag tatatagtcc tcaatatttc	20580
ttctattttc attttgcttt aacttccatt cacaataata gtcaataata tgctcaaaag	20640
cagtctgtat accaatttga taaccatcac atacagcatc tatacgcagt gattcagctt	20700
ttttcgtagc ctcttttata acacaaattg ctttcttctt actctcagt tctctatctt	20760
ttatggtttt acataattcc ttttcaattc tctttataac aacaccatca taggcatgat	20820
gtctggatc cggtaaagtg cctttttgca tattgcatac cttcatttaa tcgggacaaa	20880
tttcttttca caggcgata atttccctcc aaagcaattg gattaaataa taagtttatt	20940
ctttcatata taccaacaga tatattttcc ttccacgcaa taagatgcca aaaaccactt	21000
cgaatagcgg cctccccttg cttactgatt ttgccccat aataactaat aacctctca	21060
ctatgattaa ttaaccgaga ctcttgacta ccgataaaat aagcaacagc tggaagttaa	21120
ttccagtttg caacaaacat agcggttacc ccgcttaaag gttcaatata gcaatttaga	21180
tcatatttat taattatgag attattaatc tcaattctga ttacgccatt ttctaacagt	21240
tcagaaggca aattcaaccg atttctattt atatagaacg ctggatcaaa aattattgaa	21300
agatattttt tataaattcc atccattctt atcatatttt gtttctcttg aaccacctg	21360
ttttgaatgc ccataccaaa agaattgacaa ctagtacagc catcccgaga aataaatata	21420
ttacttcatt tgttaaaaaat tccgatttaa cttccttaac aggttgtaaa tttgtataaa	21480
catattcttc tttcggagtt aatatgacag atatatcttc atacttaaca tcagaaaagg	21540
tgtttttcaa aaatcgctta atattactaa ctaatagttc agactcttcc ggtgagtcac	21600
atatagcgat tactgatata tgcacgggtt ttgaagatat attttttctt tcaagatcat	21660
agctaacatg tatttttgcc gaaataaacac caccaataga aactaaagac tgttccagcc	21720
gttgctcaat agcactatat aaacgggcct tttcagctct tggagaagac actaatgaat	21780
ctgtaggaaa catttgtag atataactc tctccgatt tggcaaatcg tacatgcgca	21840
tcaaatcaac tgcgatgca aatgtccct tttcgacttg taccgagatc ccctgtttac	21900
ctccatcaac ttttctagca gtaattatg ggcgttctag cacagatatt atttcatttg	21960
cctgtctttg agataaatta gaaattaact cttcacgttg ctcacatcca atcaacatca	22020
gcaacaagaa taaaaataaa cctttatacc taatcatata aattaagact ttaataaagt	22080
ttcaacagcc gaaactgttt ttcttgctaa cgtgccagct aatgatactc ctattgaata	22140
atttgagagc gtcgtctgta acttgccaa agattctggg tttaattgatt ctggattgct	22200
taccatctcc attatttggt atacttgaat atctgtatcc atcttaatat cactatattt	22260

-continued

```

agccgacacg acgtottcca gacttgatat ctcttgagat tgaaaatcac tggctttgat 22320
aatatcaacc tgattgactg gataaatgta attcatgaat gctccctatt atctgaagtt 22380
ttgaataatt gcagcatcaa catccttaat cactttcact gtattggatt gcgcgttcct 22440
atataatgta tattcagata atttactttg gtattcagcc agcaactgtg gattcgaagg 22500
atTTTTtagct aatttatcta gtgccaatgt tagttcacct tgtaatgttt gagttccatc 22560
atcaaaaagtt tcagataatg aactcaatgt ccaatcatca ttccgtactg taacactcat 22620
tttatcctca cttattttta tctaaaaaaa accagtgttt atcattcaac attacataac 22680
tgtctttgct gttaagatat gatttacctt taaaatcatc atctatcaaa agtacagaaa 22740
actcaatata ctgatctcca tatatatattt tatgttcaga aatcagtcct ttaatgagct 22800
caagcttttc atctgtcagt tcttctctga cagaatatgt aactttatta ttttcacaaa 22860
tctctttata ctggacattt acctttgtga aaataccttt agcatcccc cttgcattat 22920
gatcgctaag atagttgaat ttaatatattt tgatatacgg aaattcaaca atcagagcat 22980
ttttcaattt atcaagttcg gtatcttttg aactatttct tcttttactt aaaaacacca 23040
gtaattccgc tttatcactc actaaattca acacatacaa attgataaat gggaatcggt 23100
gattgatata ttgttgaaat ctatttatatt ctttgtttga caccagaata tatctgcatt 23160
tggtaaatcc ggttttatatt aaggetacag acgcccatac cgcacattt tgtgtttcgg 23220
caagaatata attgcactgt ccagtgttga ctattccata tctcttatca tctataattt 23280
cagcaatttc ttttacttca tctttcttta aagagaggga aaaaattata attaacacaa 23340
caataacagc aaagaaaaaa aacaccgagt ggttcttata cattccgttt agtatatggc 23400
ctctcgattt atcctccctc atatttttaa ggtgaaaaga aatcccgta tattcaaatg 23460
aattaaattg tgaattgata tcaatttgct cacttttttc atcactaaat tgacagatat 23520
tatctttgct tatggtcaaa taaattatac catccgattt actatctgtt acaggaataa 23580
tattttctgg gaaattgtcg tttccaaact caagtgttcc tatcgtccgt ccgagaacaa 23640
ttaagttatt ccataatat aacggaaact catccccgac accgtttgtg agtttcaacta 23700
ataacctgaa tgggtgcaaga ttgagttct ttgcctcaga cataaatcct cttatatcct 23760
tcgattatcc tgcctgctct ctgctgctat ccgcaaatac cgtctataga taccctctgc 23820
gtactttaat ctttcttttt tctttttcgg agacgttcct gcattatagg ctccaacagc 23880
ctcccacct cttccataca tcatcataaa ctcatttaat aactttgctg cataatgtac 23940
agaaagacat ggatgactaa ttagcatttc ctcagaatat cccatttctc taagtctttt 24000
agaatgaaag tcatttatatt gcattatacc ataactctta cttccattgt tattaacatt 24060
gactgcagat ttattaaatc cggattcttt ttccgcaatc gcttttaaca aacttgatgg 24120
aatattgtac ctttaccag ctttatccca acaatcagct ctccctaaat atgggataaa 24180
acacagcaaa ataaatacga aacgggacat taatacccc tcatctctcg cgcaaatca 24240
tccttcagcc aacgcacacg cgaatatata tattctgtat taacatatac atcctcgaat 24300
tcatttatcc cggtgaagtc taatcgcgca attagatttc cgccttcac atcagtagct 24360
aagcatatct tttctgagta atttaggttc aacgcatata taagatcatt tatattttca 24420
ggcaatgcac aaaaagggca taataaattt attgaatctc cttctttttc aatatatatt 24480
agcacatcat catcaagcat gattgcttga tcatcgatat caattaaagg gataccaag 24540
gctcgcaaaa taacatctgc taaatcttcc atatatcct cttatacaaa tgacgaatac 24600

```

-continued

cctttcacca	tattccatat	ttttgggtcc	cctattcttt	cagaataaga	tagctctaac	24660
gaagatagcg	gcaatttttt	catgacttta	tttccgggca	caccagtatt	catctcttgg	24720
atttccatat	taccactatt	cattagaata	gtagaaaata	atcttttctc	ttctgaggat	24780
aatttactat	ttaattgaga	aaactgacct	gtttcatgtt	ttcttattat	ttctctttta	24840
atttcagcat	cctgcatacc	tgttctgtcc	ttcccactct	tacaattcca	gcaaggatca	24900
gcacctatag	tataagccaa	tagagtcatt	ctttgtgaca	atttatacgg	ctctccatta	24960
tcattttttt	gtaacttttt	ggttacaatc	tcttttatct	ggttagccag	atatatcaca	25020
tcattcttac	atgggtgatt	tttttctatt	gcttctgcag	cccaacctcc	aatcacacca	25080
tttttgagga	aattatcccc	caacaagaa	catattgatt	catcattaag	ctgtcaaca	25140
ttctccagc	ctaagcccat	ttttagcgct	aattcattta	caccaaagtt	aaatgtcacc	25200
acttttagat	taacgcttac	ttctttcaga	agaccatcac	tattccgaat	taataatttc	25260
gttggtcccc	cccttttaga	gtttaaccct	tttaatgcgt	tcacctgate	ttttaacata	25320
ctctcctctc	ccccggttaa	actggtcggc	gtcagaagag	aagtcgaaac	aatctttaa	25380
tctactgttt	taccagacaa	agcctgtgat	aataactctg	gtctgtcata	taatgctgcy	25440
cttactaact	cctcagcttt	attacgagca	gcaacagctc	tttcagatga	gttttttttc	25500
aatccatagg	ctgatataac	tccatgtctg	ataccgaaa	aaatttcctt	cccctcatca	25560
tccaccactt	ttgaaagcca	catattcgcy	atatgatcac	tctctctcgt	agaagcacia	25620
caaattcctt	ttccattata	ctctttgaca	aatatatatt	tattccctat	cttcatatga	25680
gaggcaggag	ttaattgaaa	accataattt	ttcccatgat	gactaatatt	cttgtcgacc	25740
ggtccccaat	tcttattatt	taaataatca	gtaaatattt	gttttatttt	cgtttgcaat	25800
tcttttttat	ttttaatgcc	tagctgtgct	cttatctctt	cacttattac	tttattaagc	25860
tcattaatat	gtcttttatt	aacttcacca	agtttatagg	tatctaactg	ttccatcatg	25920
gctttaattt	ttgacgtatc	accacctgcc	aactttattg	cctcgcaaaa	aacttcagct	25980
ctgtcacgcy	caaacaaaag	gctttttcct	gttaagcttg	tttgggtctg	taaatataat	26040
cgattaagcy	cttcgatccc	cttatttctt	tctccatttt	taactctttg	atatgaaatc	26100
acgcttacac	ctttgtgctt	tccggtttct	tctgcgcctt	tataggaatc	tccgctttga	26160
aatgaaacct	gatgcaatcc	caaattagtt	atgtgcattt	aattatcctc	agtcataaat	26220
ggggcattga	ttgcagtttt	gtacctcgtt	ttacaacaaa	aaagatactt	attcaacagg	26280
gttctctctt	ttctataaac	actccgagcc	tgatacatca	ggcactttat	cttgtgggat	26340
ttcatgatga	aacgagcact	acatatgctc	acgagggtaca	tattatgtac	agcaaaagaa	26400
aatttgatta	tgtaaacagg	ctcaagtctc	ttaaatttatt	aattgaatat	attatatacc	26460
aaatacattt	tacatcccca	caatcaccaa	gtaattggaga	gttaataaag	tatgatcctc	26520
aaaattagca	atttcattga	cgcaagcaat	acaaaagggc	ctatacgcyt	tgaagataca	26580
gagcatggac	ctatattgat	agcacagaaa	ttcaacctaa	aggatctctt	tttcagaaca	26640
ttaagcacta	ttaatgcaaa	gataaattca	cagatattaa	atgaacaact	aaaaaattat	26700
cgactagaga	atcaaaaaatc	cttattgctt	ttcctcaata	cattagctag	cgaaaaatct	26760
gcagaaagtg	cgtttgccgc	ctatgaagca	gctaaaaact	caattcaaca	ctctttcaca	26820
ggtagagaca	tcaaattaat	gttgaatacc	gcagagcggt	ttcatggcat	cggtacagcc	26880
aaaaatcttg	aaagacattt	agtttttctg	tgctggggaa	atagaggcat	aacctattta	26940
gggcatacta	gtatcagtat	aaaaaacaat	ttactacagg	aaccaactca	tacctatctc	27000

-continued

```

agctgggtatc cggcggggaa tgttacaaaa gatacagaaa taaactacct ttttgaaaaa 27060
cgctcagggtt acagtgtcga tacctataag caagacaaac taaatatgat ttcagaccaa 27120
acagccgaaa gacttgatgc aggccaagag gttagaaact tattaatttc aaaacaagat 27180
caaaataaca acaaaaaaat attttttccc agagctaate aaaaaaaga cccctatggc 27240
tattggggggg tcagtgtcga taaagtatac attccattat cagggtgataa taaaacaaag 27300
gatggtaaaa tcagccataa cctttttggc cttgatgaaa caaacatgac aaaatttata 27360
tgcaaaaaga aagcagatgc tttcagacaa ttggcaaat ataaattaat cagtaaatct 27420
gaaaactgtg ctggtatggc acttaattgt cttaaagcag gaaattctga aatatacttt 27480
ccactccctg acgttaaatt ggttgctaca cctaacgatg tatacgcgta tgcaaaaaa 27540
gttaggcaac gcattgaaag tctcaatcag tcttataacg aaataatgaa gtatattgaa 27600
tctgattttg atctttccag attgactcaa ttacgacgca gctatcttaa aagtttcaat 27660
aaaattaacc ttatccacac accaaaaaca ttcaagcctt tatcaatttc actttacaaa 27720
catcctactg aaaatgtatc ttcagaagac tttgatgcgg tcatcaacgc ttgccactca 27780
tacttagtaa aatcagcacc cagcaatatg actcgtgtat taaacgaatt aaaaaccgaa 27840
gcaacagata aaaaagaaga aatcatcgaa aaatcaataa aaattattga ttattacaat 27900
tctcttaaat cgccagatct tggaacccaa cttttacatc acgacttatt acagatcaat 27960
aagcttttat tgaataactc tcattcta atatagaagg ccatagaaat gtgtcgcaaa 28020
ctatatgata aactttatga aataacaggt gcaaagcttg attttaacga taaaaatcaa 28080
gcctttatat tgctogaaga acaaatacct gtttgtataa cagataatga tgaatacatc 28140
tttttgaccg gacttcttaa tgaacatgaa ctttttacag aaaatattat taatccagaa 28200
cacatactca tccttaacta ctctttgagt agagattatg gttcttctat ctgtcttctc 28260
cctgatactc atcaatgtgt ttggaccaag aaacactaca aaaaatatct ttcacctgat 28320
gaactcattg aatctttgta tgaattttta ttttgtatta aactaacat tgcaaacatc 28380
acttcagaag tgaactaaca tatagggggt atcatgcaaa ttctaaacaa aatacttcca 28440
caggttgaat tcgcgattcc tcgccatca tttgattcgt taagtcgcaa taagcttgta 28500
aagaaaatat tatctgtttt taacttaaaa caacgcttc ctcagaagaa ttttgatgc 28560
cgggtgaata taaataaaat cagggacagc gtcatagaca aaataaaga tagtaattcg 28620
gggaatcaac tattttgttg gatgagccaa gagcgaacca cttatgtctc ttcaatgata 28680
aacagaagta tagatgaaat ggccatacac aatggagttg ttttgacctc agataacaaa 28740
aaaaacatat ttgcagctat tgaaaaaaaa ttcccgata taaaacttga tgaaaaatca 28800
gcacaaacat ctatcagtc tacagcacta aacgagattg cctcatctgg cctcagagca 28860
aagattttta aacgctatag cagtgcacat gatgtgtta acacacaaat gaaggatctt 28920
acaaatctag tatccagttc cgtctacgat aaaatattta atgaatcaac aaaagtttta 28980
caaatagaaa taagcgcaga ggttctaaag gctgtatacc gtcaaagcaa tacaattaa 29040
atcttttata ttaaatctta tacttattaa agtaaaagcag aaaaaagga gaccttatgt 29100
ctttaaatat caccgaaat gaaagcatct ctactgcagt aattgatgca attaatctg 29160
gcgtactact gaaagatatt aatgcaattc ctgatgatat gatggatgac atttattcat 29220
atgcttatga cttttacaac aaaggaagaa tagaggaagc tgaagtttcc ttcaggtttt 29280
tatgtatata cgacttttac aatgtagact acattatggg actcgcagct atttatcaga 29340

```

-continued

taaaagaaca	gttccaacaa	gcagcagacc	tttatgctgt	cgcttttgca	ttaggaaaaa	29400
atgactatac	accagtattc	catactggac	aatgccagct	tcggttgaaa	gcccccttaa	29460
aagctaaaga	gtgcttcgaa	ctcgtaattc	aacacagcaa	tgatgaaaaa	ttaaaaaata	29520
aagcacatc	atacttggac	gcaattcagg	atatcaagga	gtaattatta	tgcataatgt	29580
aagcaccaca	accactgggt	ttcctcttgc	caaaatattg	gcttcocactg	agcttggaga	29640
caatactatc	caagctgcaa	atgatgcagc	taacaaatta	ttttctctta	caattgctga	29700
tcttactgct	aacaaaaata	ttaatacaac	taatgcacac	tcaacttcaa	atatattaat	29760
ccctgaactt	aaagcaccaa	agtcattaaa	tgcaagttcc	caactaacgc	ttttaattgg	29820
aaaccttatt	caaatactcg	gtgaaaaatc	tttaactgca	ttaacaaata	aaattactgc	29880
ttggaagtcc	cagcaacagg	caagacagca	aaaaaaccta	gaattctccg	ataaaattaa	29940
cactcttcta	tctgaaactg	aaggactaac	cagagactat	gaaaaacaaa	ttaataaact	30000
aaaaaacgca	gattctaaaa	taaaagacct	agaaaaataa	attaacccaa	ttcaacaag	30060
attatccgaa	ctcgacccag	agtcaccaga	aaagaaaaaa	ttaagccggg	aagaaataca	30120
actcactatc	aaaaaagacg	cagcagttaa	agacaggaca	ttgattgagc	agaaaaccct	30180
gtcaattcat	agcaacttta	cagataaatc	aatgcaactc	gaaaaagaaa	tagactcttt	30240
ttctgcattt	tcaaacacag	catctgctga	acagctatca	accagcaga	aatcattaac	30300
cggacttgcc	agtgttactc	aattgatggc	aacctttatt	caactagtgt	gaaaaataa	30360
tgaagaatct	ttaaaaaatg	atctggctct	attccagtct	ctccaagaat	caagaaaaac	30420
tgaaatggag	agaaaatctg	atgagtatgc	tgctgaagta	cgtaaagcag	aagaactcaa	30480
cagagtaatg	ggttgtgttg	ggaaaatact	tggggcactt	ttaactatcg	ttagtgttgt	30540
tcgagcagct	ttttctggag	gagcctctct	agcactggca	gctgttggtt	tagctcttat	30600
ggttacggat	gctatagtac	aagcagcgac	cggcaattcc	ttcatggaac	aagccctgaa	30660
tccgatcatg	aaagcagtea	ttgaaccctt	aatcaaaactc	ctttcagatg	cattttacaaa	30720
aatgctcgaa	ggcttggggg	tcgactcgaa	aaaagccaaa	atgattgggt	ctattctggg	30780
ggcaatcgca	ggcgtctctg	tcctagtgtc	agcagtcgtt	ctcgtagcca	ctgttggtaa	30840
acaggcagca	gcaaaacttg	cagaaaatat	tggcaaaata	ataggtaaaa	ccctcacaga	30900
ccttatacca	aagtttctca	agaatttttc	ttctcaactg	gacgatttaa	tcactaatgc	30960
tgttgccaga	ttaataaat	ttcttggtgc	agcgggtgat	gaagtaatat	ccaaacaaat	31020
tatttccacc	catttaaac	aagcagtttt	attaggagaa	agtgttaact	ctgccacaca	31080
agcgggagga	agtgtcgctt	ctgctgtttt	ccagaacagc	gcgtcgacaa	atctagcaga	31140
cctgacatta	tcgaaatatc	aagttgaaca	actgtcaaaa	tatatcagtg	aagcaataga	31200
aaaattcggc	caattgcagg	aagtaattgc	agatctatta	gcctcaatgt	ccaactctca	31260
ggctaataga	actgatgttg	caaaagcaat	tttgcaacaa	actactgctt	gatacaaata	31320
aggagaatgt	tatggaaatt	caaaacacaa	aaccaaccga	gattttatat	acagatatat	31380
ccacaaaaca	aactcaaatg	tcttccgaaa	cacaaaaatc	acaaaattat	cagcagattg	31440
cagcgcata	tcacttaat	gtcggtaaaa	atcccgattt	aacaaccaca	ttaaatgatg	31500
atcaactttt	aaagttatca	gagcaggttc	agcatgattc	agaaatcatt	gctcgcctta	31560
ctgacaaaaa	gatgaaagat	ctttcagaga	tgagtcacac	ccttactoca	gagaacactc	31620
tggaattttc	cagtctttct	tctaattgctg	tttctttaat	tattagtgtg	gccgttctac	31680
ttctgctct	ccgactgca	gaaactaaat	tgggctctca	attgtcattg	attgcgttcg	31740

-continued

atgctacaaa atcagctgca gagaacattg ttcggcaagg cctggcagcc ctatcatcaa	31800
gcattactgg agcagtcaca caagtaggta taacgggtat cggtgccaaa aaaacgcatt	31860
cagggattag cgacccaaaa ggagccttaa gaaagaacct tgccactgct caatctcttg	31920
aaaaagagct tgcaggttct aaattagggt taaataaaca aatagatata aatatcacct	31980
caccacaaac taactctagc acaaaatddd taggtaaaaa taaactggcg ccagataata	32040
tatccctgtc aactgaacat aaaacttctc ttagttctcc cgatatttct ttgcaggata	32100
aaattgacac ccagagaaga acttacgagc tcaataccct ttctgcgag caaaaacaaa	32160
acattggcgg tgcaacaatg gaaacatcag ccgttgctgg taatatatcc acatcaggag	32220
ggcgttatgc atctgctctt gaagaagaag aacaactaat cagtcaggcc agcagtaaac	32280
aagcagagga agcatcccaa gtatctaaag aagcatccca agcgacaaat caattaatac	32340
aaaaattatt gaatataatt gacagcatca accaatcaaa gaattcgaca gccagtcaga	32400
ttgctggtaa cattcgagct taatatatcc aagagccata ataatatatg gctcttctctg	32460
taaggaaata accatgaata taacaactct gactaatagt atttccacct catcattcag	32520
tccaaacaat accaacgggt catcaaccga aacagttaat tctgatataa aaacaacgac	32580
cagttctcat cctgtaagtt cccttactat gctcaacgac acccttcata atatcagaac	32640
aacaatcag gcattaaaga aagagcttcc acaaaaaacg ttgactaaaa catcgctaga	32700
agaaatagca ttacattcat ctacagattag catggatgta aataaatccg ctcaactatt	32760
agatattctt tccaggcacg aatatccaat taataaagac gcaagagaat tattacattc	32820
agctccgaaa gaagccgagc ttgatggaga tcaaatgata tctcatagag aactgtgggc	32880
taaaattgca aactccatca atgatattaa tgaacagtat ctgaaagtat atgaacatgc	32940
cgttagtcca tatactcaaa tgtatcaaga ttttagcgct gttctttcca gtcttgccgg	33000
ctggatctct cccggaggta acgacggaaa ctccgtgaaa ttacaagtca actcgcttaa	33060
aaaggcattg gaagaactca aggaaaaata taaagataaa ccgctatatc cagcaataa	33120
tactgttagt caggaacaag caaataaatg gcttacagaa ttaggtggaa caatcggcaa	33180
ggtatctcaa aaaaacgggg gatatgttgt cagtataaac atgaccccaa tagacaatat	33240
gttaaaaagc ttagataatc taggtggaaa tggcgaggtt gtgctagata atgcaaaata	33300
tcaggcatgg aatgccgat tctctgccga agatgaaaca atgaaaaata atcttcaaac	33360
tttagttcaa aaatacagta atgccaatag tatttttgat aatttagtaa aggttttgag	33420
tagtacaata agctcatgta cagatacaga taaacttttt ctccatttct gaggtgcgcc	33480
atgcataatg taaataatac tcaagcgcca acattcttat ataaggcaac ttcaccatca	33540
tcaacagaat acagcgagtt aaaaagcaaa atatccgata tccatagtcc gcaaaacttct	33600
ctaaaaacac cagcatcagt gtctgaaaaa gaaaactttg caacgtcttt taatcagaaa	33660
tgtcttgatt ttttattttc ttctcaggg aaagaagatg tgtaagaag catattattcc	33720
aactcaatga atgcgtatgc caaaagcgag attctcgaat tttcaaatgt tttgtactcc	33780
ttagtacatc aaaatggtct taattttgaa aacgaaaagg gacttcaaaa aattgtcgca	33840
cagtattcgg aactaattat aaaagataaa ttatcccaag attctgcctt tggaccatgg	33900
tcggcaaaga ataagaaact ccataatta cgacaaaaca ttgagcacag acttgcaacta	33960
ttagcacaac aacacacatc tgggtgaagct ttatcattgg gacaaaaact cctcaatact	34020
gaagtatcat catttatcaa gaataatatt cttgctgaat taaagttaag taatgaaact	34080

-continued

gtttcatctc tcaaaactaga tgatttagtt gacgcacagg caaaacttgc ctttgatagt	34140
ttgcgcaatc aacgtaaaaa tactattgat agtaaaggat ttggtatagg taaactgtca	34200
agagacttaa atacagtagc cgtgtttcct gagctgttga gaaaagtctt taatgatatt	34260
ttagaagata taaaagattc gcatcctatc caagatggcc tccctacacc tcccgaagat	34320
atgccagatg gcggaccaac ccccgagacc aatgagaaaa catcccaacc tgtaattcac	34380
tatcatataa ataatgataa tagaacttac gataatagag tttttgaaa cagagtatat	34440
gacaatagct atcacgagaa ccagaaaaat gatgcacagt ctctacttc tcagacaaac	34500
gatctattat cccgtaacgg aaactcatta ctaaatccac aaagagcact agttcaaaaa	34560
gtaacttcgg ttctaccaca ctctatatca gatactgtcc agacatttgc aaataattca	34620
gcttttagaaa aggctttcaa ccatactcca gataattcgg atggaatagg ttcagacctg	34680
ttaactacga gtagtcaaga aagatctgca aataactctc tttctcgggg acacaggcct	34740
ctgaacatac agaactcttc aaccaccccc cctctccacc cggaaggagt gacaagcagt	34800
aatgataact catcagatac aactaaaagt agcgcttctc tttctcatag agtagcttcg	34860
caaatcaata aattcaactc aaactctgat tcaaaagtac ttcagactga ttttttatca	34920
agaaatggag acacatatct aacacgggaa acgatatttg aagcttcaaa aaaagtaaca	34980
aaactccctaa gtaatcttat atctctcatt ggaactaaat caggaacaca agaacgagag	35040
ttacaggaaa aatcaaagga cattacaaaa tccacaacag aacatagaat aaacaacaaa	35100
ttaaaagtta cagatgcaaa tataagaaac tacgtaacag aaaccaacgc agatacaatt	35160
gataaaaaac atgcgatcta tgaaaaggca aaagaagtat ctagcgccct cagcaaggta	35220
ttgtcaaaaa ttgacgatac ctctgcagaa ttacttacag atgatatatc tgatttaaaa	35280
aataacaatg atattacagc tgaaaacaat aatatatata aagcagcaaa agatgtaacc	35340
acttccctat caaaaagtatt aaagaatatc aataaggatt aaaatatgat aaaagaaaaa	35400
atattatcaa tagtggcctt ttgctatggt atagcgtaca gcaaaactatc tgaagaaact	35460
aaattttatg aagatcttct tgccgattct ctttctctga ttgaaatgct ggatatgatt	35520
agttttgagt tcaatctacg tatagatgaa tctacattag aacacattat tactattgga	35580
gatcttattt ctgtagtcaa aaatagtaca aaatcaatat agtaattgat aagcattttt	35640
tcatctatgg agctctcaca tcagagctcc acaagaatat tattctttta tccaataaag	35700
ataaattgca tcaatccagc tattaaaata gtaattttta agactaccgt tgactatcat	35760
caacagttac atacaatctt gctcacactg catttaactt ttgtcaatat aaatcaatat	35820
gaataaacag ggtgtgatat ggtggatttg tgcaacgact tgttaagtat aaaggaaggc	35880
caaaagaaaag agtttacct ccattctggt aataaagttt cttttatcaa agccaagatt	35940
cctcataaaa ggatccaaga ttttaacctc gtcaacccaa aaacgaatgt acgcatcaa	36000
gaatccctaa cagaagaatc attagccgat atcataaaaa ctataaagct acaacaattc	36060
ttccctgtaa taggaaggga gattgatggt agaattgaaa ttctggatgg cactcgtaga	36120
agagcatctg caatatatgc aggagcagat cttgaagttc tatattcaaa agaatatata	36180
tctactcttg atgccagaaa actagcaaac gatatacaaa cagcaaaaaga gcatagcatc	36240
cgagaacttg gtatttgtct taattttctg aaagtatcag ggatgtccta taaagacata	36300
gccaaaaaag agaactctgc tcgcgcaaaa gtcactcgtg cctttcagge agcaagcgtt	36360
ccacaggaaa taatatctct atttccaatc gcgtcagaac ttaactttaa tgactacaag	36420
atattattca attattataa aggacttgaa aaggctaag aatctcttag ttctacacta	36480

-continued

```

ccaatattaa aggaagaaat aaaagacctt gatacaaatt tgcccccgga catatataaa 36540
aaagaaattt taaacatcat aaagaaaagc aaaaacagaa agcaaaaccc ttcgctaaaa 36600
gttgactcgt tatttatttc taaagacaaa cggacttaca taaaagaaa agagaataaa 36660
acaaacagaa ctctaattatt tacattatct aaaataaaca aaacagtcca gagagaaata 36720
gatgaagcta ttcgggatat aatttctcgc catctatcgt cttcataaac atcgaacact 36780
gatgtaaact gcccaattca gataataaaa atcttcgttg aacttgactt tcagccgggtg 36840
gggtaattcc gtcggctatc tgttattcaa tccccacgca ctgtgtttcc ttaaaattac 36900
ggtgaaaact cataacctgc tgaccacgag aaaattatct ggtgaaatta gtaatattca 36960
tatcattgaa aatgaattgt gcaactaagg gatgcggcat gatggtcaaa attttatgtg 37020
aatatgagtg actggccgaa aatattcagc gtatttttaa tcgtttcaat aaaaacagcg 37080
aggattagca gtcaacgctt ttactactca atttttttta catggtgata aacaaaaaat 37140
ccaccattaa cagacttcgt tcacagtatc aaaaagcctt taagaatata ttagcagatt 37200
tctcaaatat aattactgtt ttattctcct aatcttccca taaaatgacc gacttacatt 37260
attttataat tataaacaag ttgatttta aaatcataatt aattattaac aagagattta 37320
tttttctaac cccaaaagt atgcttggtc aaagagtaac ttataatc catacccctc 37380
atagggctta cagtttaaat tgcgggtata aaatacatct attttcaaat aaaaataaagc 37440
atactccaat tacaacgagt gactaaaata aaaatatcat ccaagtcaat cactgtttgc 37500
tctgcaaaat atgtcagcct tctcaaacct accagcactt aaatatttaa gcacacaatc 37560
attccttgag ggagttcact tatttcaaga gaaacatttc gtcttcccac acaactaatc 37620
cgccaatat agctaaattt gtactcagta atatcatttg ttatgcccc caccceaaaa 37680
tattcaagct ccttcacaga taaaaaacga caatttaca agcctcatta gttataacta 37740
tggaataacc tattttagta taccctataa agttagaatt acttctctt gcacctaat 37800
ccatagttga aaaacaagaa taattttctc ctacagctgg atccatatag cttccatcag 37860
gtctacacaa caccagctgg aggccgacag gaaactccat aaaggataca gcaacagctt 37920
ccaaaactct ttggttagaa gataacacat ctcggtgatg tactatatta catcccctgc 37980
caattaataa gtcccttgca tcgggataaa taaaactaat tacttgcgaa aaatgttac 38040
tttcttcaac gacataagca ttcaatccaa gaagcctcgt agccatgact atcttatcag 38100
gcatagaata accggcatcc gcgatattat ctttgtgtat tctcggatta taattaccac 38160
tagttattaa gtataaatcc cgctcacac gattatccaa atcaagagaa ctcttccttg 38220
tcatttcgga catagaccct ttgtattgtg gtattttatc aaccctaat tcctttgcag 38280
cacagagtaa actggcagct ccacaagaat tgtcatacat ctgttcctc gccaaaaaac 38340
aaccttgctt gtttactcta ttctcattta ccttcacatc agtctgatga gatatcatat 38400
attctcttgg tatagcccc tgcaataaaa ccccatataa cataacacct gtatgaatac 38460
atccccgttt gcaaggcttc cgttgttccg acatcgacga gtaattgcag cacagtattg 38520
catccagcaa tgcaactac tatgagaatt agctgaccgg gtcccttatt ccggaacgaa 38580
ctttaaatcc aaaacctcat actcttttcc gtcgtcatca cgtccgtcac cataacaacc 38640
tttcagcatc caccctttta ttaagagtta atgtccatca gcatcagcaa ttaataccgc 38700
ctgatactga catatttttg tcgtatttac gtccataatc gggtattgtg atacaggcca 38760
caccagctta tttgctattg ttacaataac caccctgggt gagagctgtt ttttcaactt 38820

```

-continued

```

tggtatcaac tcccccaact ccttggtcag taaggtgcc gaccgagcac agtgaataga 38880
tagtatccgc aggtaccacc atgcttactt aatactcaga aggcgaacat atgaatatcc 38940
tccttagttc ctattccgaa gttcctattc tctagaaagt ataggaactt cgaagcagct 39000
ccagcctaca cgccccgggc ggccgctcag ctccagtctt cagttctcag gccgcccaca 39060
cgttcaaatt cccgggtggt attagtcaca ataatcagcc cccgactgcg ggcatgacct 39120
gcatcattt gatcaaatgg ccgcacaggc cgtccctgac gggcaagtgc tgcctttatc 39180
tgcccggtgt gtgtggctgc agcagcgtcg taatccagaa catcaatgcg ggaacaaat 39240
ccctcgatca cagcgagatt acgttcaggc atctggcttt tttctgcacc atatatcagc 39300
tccatcaggg tgaccgaact gatacacatt ttccctgggt tcagggttaa acgctccctg 39360
acgtggcggg gtttgttctt tctcgtaaaa atgcagatgt tggatcagag cataaacttc 39420
agcatcagaa tgactccctt tcttgcagtc cgggctgttc cctgttatcc ataaaatcgg 39480
tgctgacgct gtgtccgtcg aaccattcgt cccatgtctc tccggctggc gtaatgattc 39540
tggttcgtcc gacagcaatc acttcaacgc gttttacgtt ttccggcaat gcaaccgctt 39600
ttggcagctc gaccgcctgg ctgcggttgc tgagaaatc ggtggtttcc attatttgcc 39660
tccttatgtg gatagtctt gtgtatatct attgtgggat tgatacccta ggctatttaa 39720
tattccggag tatacgtagc cggttaacgt taacaaccgc gatcgacat gtacatgatc 39780
aacaacaggt acctaaaaat tttttatgat ataagtaaaa tttctttgga gttataccat 39840
aatattcatt aaatttcctt ataaaataag atgggcttga tttccgata agtcttgata 39900
catcattaat atagctttga ctattcaata aaagctttgc tgcagatgc atgcgaatat 39960
caagaaggat ttgttgaat gttaattttt cactctccaa tcgttttcta acagctattt 40020
ctgataaatt caagttattt gaaatatcag aaagacgcca tctcttctcg atgttttttt 40080
caacaatctt ccttatctga tcagaaaaac taagagaaga agctatcgat atcgaagtat 40140
ataaagcttc ctcatcagaa acagctgata aaaggcaagc taaactatag atctttcttt 40200
tgccgaaagg catctctttt atagatttga acaaatcgat agaaacctcc tctcagaga 40260
ggaggaatat tttttgttt aacccccctt tctcctcaga ataggagtgt tgaaatgaat 40320
aaattggttc cattattcta ataatgctta ataataaatt tctgtcaagg cttataatct 40380
caaatggatt aatgctatca gattttttta tggagacgtt tatttgata tttcgtcta 40440
taaaagcaat ttgcccctca tcgatatgca aagtctcatt gcctgagcta accgtcattg 40500
aacacctttt tgcatataaa ataatatagt tatgcaagcg aacctttata tctattttgt 40560
ttttatgtcc catatccatc ataaattcaa cagaatatac caaactatat ttagtgttta 40620
cagagtaact aagaaatttc tgtaacagtg attgtataca aaatctttcc gttattaaga 40680
tagattaagt aatttttcct atatttaact ttatttacct tttttgaaga aagagcttat 40740
gcagcttcta tttctctgtg caacaagcta agtatttgca ctgaaaggga ctataaccat 40800
attatggttt cacttttctt aggcgaatatc tgtgtaaatg ttcggaagg agtggagctt 40860
cgcaaaactc cagcatcata atactctata cggaaacgat aaagccccct cagactggag 40920
gtccacaaaa tccggttggt gttgatattg gtaagcgta accgaatgcc gcattgacga 40980
ttctcccgat gacagcggtt 40999

```

<210> SEQ ID NO 2

<211> LENGTH: 45646

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: Recombinant Plasmid

<400> SEQUENCE: 2

```

gctagcatgc gcaaatTTAA agcgcTgata tcgatcgCgc gccctgcagg tacCGttcgt      60
atagcataca ttatacgaag ttattgatcg gcacgtaaga ggTtccaact tTaccataa      120
tgaaataaga tCactaccgg cGgtattttt tgagttgtcg agattttcag gagctaagga      180
agctaaaatg gagaaaaaaa tCactggata taccaccgtt gatataTccc aatggcatcg      240
taaagaacat tttgaggcat ttcagtcagt tgctcaatgt acctataacc agaccgttca      300
gctggatatt acggcctttt taaagaccgt aaagaaaaat aagcacaagt tttatccggc      360
ctttattcac attcttgccc gcctgatgaa tgctcatccg gaattacgta tggcaatgaa      420
agacggtgag ctggtgatat gggatagtgt tcacccttgt tacaccgttt tccatgagca      480
aactgaaacg ttttcacgcg tctggagtga ataccacgac gatttccggc agtttctaca      540
catatattcg caagatgtgg cgtgttacgg tgaaaacctg gcctatttcc cTaaagggtt      600
tattgagaat atgtttttcg tctcagccaa tcccTgggtg agtttcacca gttttgattt      660
aaacgtggcc aatatggaca acttcttcgc ccccgttttc accatgggca aatattatac      720
gcaaggcgac aaggTgtgta tgcgcgtggc gattcaggtt catcatgccg tttgtgatgg      780
cttccatgtc ggcagatgct taatgaatac aacagtactg cgatgaataa cttcgataa      840
tgtatgctat acgaacggta cctgcagggt catgatgatc attgcaattg gatccaagca      900
ctacccccgc ctgttgcgac agtcacgtca tcgccgtccg ttaccggaac atgtttcccg      960
cgaaatacac catctggagc ctgaagaaag ctgttgTccg gagtgTggTg gtgaactgga      1020
ttattcgggg gaaatcagcg tctttctccc ggaaggcat actatcttac agtaatatTc      1080
attccgatga ttaataaatc agcaatgcgc gcatttttaa tataagcatg tacaatatgt      1140
acttgcacgc attaccctgc taatgcagtt ttttttTtaa taagttaaaa tccTacatag      1200
cacggagggg gaatgactag ttttgagcag ttaagtaagg ttgcgcagcg tgcagataaa      1260
atgttgctcg ccctaacaaa acagattcag gagcaaaagc aggagtttca ggctgatgtt      1320
ttttatcaag tctattcaaa gtctgctgta gctaagcttc caaaattaac gcgcgcaagc      1380
gtggatggTg cagtcggTga gatggaggct cagggtatc agtttgaaaa acgccagca      1440
ggaaccgcga caaaatatgc attaaccatc cagaacatta tcgatatcta tgcacatcgt      1500
ggcattccta aatatcgga tcgttatagt gaggcataTt ccatattcat cggtagtcta      1560
aaaggcggTg tgtctaaaac cgtgagtagt gtttccgtag ctacgccct gcgtgctcat      1620
cctcacctgc tgcagaaga tttacgtatt ctgctccttg atttagatcc ccagtcatca      1680
gccactatgt ttctgaacta cttgcacgcg gtggggctgg ttgatacaac cgcaccgcag      1740
gctatgtTgc agaacgtgag tcgtgaggaa ttactggaag attttatcgt tccgtcagtt      1800
atccctggcg tttatgttat gccggcttct attgacgacg cctttattgc atcaaatcgg      1860
gatactctgt gcgaggagca tctattagga caaaacaaac acgcgatatt gcgtgagaat      1920
attattgata agctgaagca tgatttcgac ttcacctcta ttgatacagg cccgcacTg      1980
gacgcatttt tgaaaaatgc cattgtgcc gcagatatca tgtttacacc ggtgcgcga      2040
gcacaagTtg atttccattc cacgcttaa ttttggtc gtttacctga acttgtacag      2100
attattgaac aggatggatg ttcatgccgt ttgcaggcaa atattggctt catgtcgaaa      2160
ctggcgaata agtcagatca taaatactgc catagcctga caaaagagat tttcggTggT      2220

```

-continued

gacatgcttg atgtttcaat gccacgtctt gatggttttg agagaagcgg cgagtcgttt	2280
gataccgtga tatctgcgaa tctgttact tatgttgca gtggtgaagc tttgaaaaat	2340
gcacgtatgg cagcagaaga tttcgctaaa gcagtttttg atcgattga atttataagg	2400
gccaaactact gatggaaaac agaaagcacc gaccgactat tggccgtacg ctgaatacaa	2460
atatcctgaa taatactgaa gaaatatcgg ctccagtcca tgtatttacc ctcaatactg	2520
gccgaaagggc aaaatttact gagatcaagg ttgatcatga caaagtggat actcaaaactt	2580
ttgtggttga agaggtgaat ggccgcgaac agacagccct gacgccggac tcactaaaag	2640
acatcaccag aactatacgt ttacaacaat tctatccatg tattgggatt cggactggcg	2700
atcttattga gattttggat ggttcccgtc gtcgggcagc agctttgcta tgcaagggtg	2760
gattacgtgt acttgtaacg gatgatgagc ttactgttcc tgaagctcag catctggcga	2820
aagatttgca gacgtcactt gaacataata ttcgtgaaat tgggttgctg ttagtacgtc	2880
tcaaagagggc cggatgaac cagaagcaaa ttgcggaaag agagggattg tcggcagcta	2940
aggtgactcg tgcctccaa gcagcgccg tacctaaaga tttgtgtct ctttttctg	3000
ttcagtctga gctgacatac gccgattaca ggcaactggc tgaattaagt gaacgactga	3060
ggctgggtga tatacttatt gatgaagtg tgaagaatat ctgccgagat attgagctta	3120
ttactgtga tgacaatctt tctgaagacg aggttaaaaa cagtattatg cggttataa	3180
ccaaggaaat gtcttcattg cttgattcag gagtaaggga taaagcgggt gtaactttgt	3240
tatggaaatt tgatagtaaa gataaatttg ctgaaaaacg agtaaaaggga cgtacgttct	3300
catatgaatt cggcagatta ccgttggaag ttcaggataa gttagatcgt atgattgctc	3360
ttgtactaaa agataatctg aattcgctgt gaaaattcaa gtttttcgcc agcaactcgc	3420
cagtttttaa ttggctgttt taaaagaaat tttgctttta ttagcgtttt ttttgaacag	3480
tatttcacgg tgaatcgcg tgtttcaccg tatcttcttt gcttgaaaaa gatagtttct	3540
cttagccgtt taacagtaac cgatgagata gtcttcccc tctttcacgg cagccttcaa	3600
ttgcgccatg atggataaac tttacaaca aaagtaagcg ggataccatg tctgaacaga	3660
aaatcaccgg cggatccata tatagggccc gggttataat tacctcaggt cgacgtccca	3720
tggccattcg aattcgccct tatgcaatga atcaatcaga gccatccagt ttaatcgtcg	3780
gttccgttca tatggagcaa aagtgggtt gaagaaacat ccgttttggtg gtgtttttt	3840
aatctttttg gggttttaat tcctatcgct caagtcttaa cgatatccct gaaacgacca	3900
cttaatatta ctacacttt tgcaaatgca ttaaatgctg cttgtattat tctgatgccg	3960
caccagcggc atggaaggcg gcacgctgtg gttacatgtg ataccggagt aaaacgccg	4020
aagcccgcg tcagccggta ctgattgaca gatttcacct taccatccc cagccctgcc	4080
aagccatacc cgttttcagc catgatagag cttctgtgcg cggtcggagt ggtcccgacg	4140
agggttttac ccgaagtcgg ggcgtatctc cgcgttagcg ggccgtgagg gccgcttacg	4200
agcgtgtatt cgatacgtcc agccagaaga ctgacagcga tgatgatgta gttacaacat	4260
tcacaattaa aagcgaactct gttcccgccc tttggggcgg ggccggcggc cttttcagtt	4320
atgagggggg cgttgtggtt tcagttttgc gttggacttg ggttttttag gggtttatgt	4380
ttgcatgttg taactaaacc ggcttcgggt gggggcccgc cgcttgcggt gggaggtgca	4440
tatttgtctg tccacaggac aggcagtgaa caggttttct ttttaaatga atgtaattaa	4500
gtagtttaaa ggagatataa acaggtgttt aaaagataca ttgcaccctg tagggttgac	4560
gtctggcgct ttatgacgtg aacggttgta accttatggt gaagtcctt gcagtttaat	4620

-continued

gtggataagc	aaaattcccc	gtctgtgagg	cgtgttttgt	atcaaaaaca	ggggggaccg	4680
gatgcacctg	aaggtggatg	atgaggttgt	tttttgtatg	tagcgtgat	tttttgtgca	4740
ctggcgggct	tcaggcgtgc	gaatgcctcc	ggcgcgtgcc	ggattattct	gaggaggta	4800
ctttcagggg	gaagctgtgg	ccagccggct	gtaattacgg	ttacgtgaca	gaatcatgcg	4860
ctccttcaca	cgacgtcca	cttcacgttt	taccgctca	cgactggcag	tgaagcgccc	4920
ttccgagatt	tcacgcgtca	gttgccgttt	caccaggggtg	acaatattct	gacgctgcct	4980
gtccgcatca	cgacgcgcac	gggcacgctt	cattccacgg	gaactaagct	ctgtctggta	5040
actgcggaaa	cgctcacgaa	caaaacgcca	ggctttcgcc	atcagttcat	ccatacccag	5100
ggtatccagc	ccctgctttt	tgcgtgtgtct	gttttcccat	tccacacgac	tgcggcgcg	5160
ggcggccact	gcctcctctg	atacatcgag	ggcagcaaac	agtgacgatg	tgaacgtgat	5220
atcggtcgga	atgtagcacc	cgataagcgg	gtcatattcc	gtctggtagg	taatcagtc	5280
cagctctgca	aggaacgtca	ggcccggggt	ggcccgggtg	atggagagtt	ttccggcagc	5340
agactccgtc	gccagtcgc	actcaatggc	cagcgtgggtg	atggaacact	ggacgcgggt	5400
ggccagtggg	tcatagtga	aacacagccc	ctgcagcagc	gcataaatgg	cccggcgacg	5460
tagtaccggt	ggcatgccc	gacgcagacc	aagcgaacgg	gcattgtcca	catgaatggc	5520
gaaatcaaaa	cgggaggtga	agcccaccgc	cttttccatc	agtttccgc	agaacggaag	5580
cgtccgggca	ccttcacggg	gtgtgaacac	cggattcggg	ttctttacct	ggcggtaata	5640
agtttggtga	agatcagtc	caccatcctg	cacttacaat	gcgcagaagg	agcagcacca	5700
gaaagaagtc	ttgaactttt	ccgggcatat	aactatactc	cccgcatagc	ttgattgttg	5760
gctgtacggt	ttaagtgggc	cccggtaatc	tttctgactc	gccaaagtta	aagaagatta	5820
tcgggggttt	tgttttctg	gtcctctgtac	atccacatca	gaaccagttc	cctgccacct	5880
tatggcgtgg	ccagccataa	aattccttaa	acgatcagta	atctagcatg	ctacgccaca	5940
aagtaaagtc	tttactttag	tatatccagt	ctctgcagtt	catctttgat	gattttctca	6000
acgaactgag	cctgtgttat	cccctctctc	tcgcagtact	caaccatgag	atctttcaga	6060
ggatttttga	caaaaacttt	tatctctttg	tgtgtaagac	gttttcttgc	aacagcggcc	6120
agttgtttct	cagagtcagt	cataggctta	cctctgcgca	caaaccgctt	ttgactcaat	6180
gaggaagtca	ctgcattttc	tgtctgcgac	atctcgcttc	ctcaatactc	aaacagggat	6240
cgtttcgcag	aggatactat	tgcataaggg	cgaattctag	ctcactcatt	aggcacccca	6300
ggctttacac	tttatgcttc	cggtcgtgat	gttgtgtgga	attgtgagcg	gataacaatt	6360
tcacacagga	aacagctatg	accatgatta	cgccaagctt	aaggtgcacg	gccacgtgg	6420
ccactagtac	ttctcgagct	cacatgtgtt	taaactgtac	atgtggcgcg	ccacatgtca	6480
tggccggccc	caatataata	ttggcttaaa	catcaggaga	tatgatggca	aatgatactt	6540
gacgactttc	tggagggata	gcattgcgaa	gcccttcact	gcttgggaagt	gttcacgcat	6600
gaacgagata	gcttgtttct	atttttattaa	taccaacacc	attacatgtt	atgtgcgatg	6660
gtaatatattga	tgaagtaact	tcgtaagccc	tccccccaga	ggcagagatt	gcagtgtcgt	6720
tttttagggac	aactgcgttg	atttttataa	tatgtccaca	attgttaata	actcgttgtt	6780
caggcatgaa	attaagcctg	ttttctatgt	tttcggacat	aatttgggca	tatatgtgtt	6840
ccacagagga	agacgtgatg	tgaagagatt	gccagaatgc	agatgttagt	gcagatataa	6900
atgcagggtc	ttgaagaggt	gaagtaaaaca	aaagtttttc	tgtttgaaaa	cataggtcct	6960

-continued

gttgggggaa gtttgaattg attattttat ttacttgggt gtaacttagc atttgaagct	7020
cttttttgct atcaatgaat tcattaattc ctaactcatt aattagatct gcaattaatt	7080
ttagtggtat ttcaatattg ttgaaatata tttttgtact attgctgtca gatgctcgaa	7140
tatctctgta aatatcatgc caaacactat taaaatcagt gtctttaacg tggaagtaat	7200
catcgtttg gattatcgat gtacagcaat taatggctgt ttcaataact tctggtgcag	7260
gcttgtagt tttaaaattc cctgataccg gagctgtaat agtatgtact tttcctaagt	7320
tgaattgttt gggggggagt gaacttaaca agttattata ttcttttgat gaaagttgag	7380
tggattgtat tttatctcca tcaagataaa aaaccgaaca tatgcctttg ttaaggcata	7440
gccgtattga aatgccatct atgtttgcgt gggcaaaact tccgtccatt ttatcttggt	7500
atagtgatgg tgagaaacta tccattagtt ttttttcgct atgtttttcg gctagagttt	7560
cgtgtggaga atatatgccg aaagtattta ttttcagtaa aatgtcaca aaactcaatt	7620
tattccaact tacttctgta gttgatttga cagttgacat ccatgatgag tcatttcttt	7680
catggttagt tatgtttgat gtctgcattt gatttctcct gatgtatttt cctattaatg	7740
ccaatatata ttaatatat taaattggga acataataaa atggaatgtt attctctct	7800
ttaagaaaac gatacatttc cactaaaaac aattatgcag ttattgcatt ggagagtctg	7860
gtaggaaagt tgtgatatga atacattcat gttattggct ttggggatta acatcaagg	7920
gcagatagaa ccattgggtg tatagaaaaa caaagtgggt aaactttggg caagcgtaca	7980
gaaagaactg aaaagttgct gtctgaagca gactatcata gtaaaattat taatttatta	8040
tcgaacatat agctttcccc ctcttttttt caaagcaaga cacaggtaaa tttctccgt	8100
tgcattgata tataacacag ctctcatggt ttgggtgagg ctttgtttaa tatgtttttg	8160
catatatcgt ccttttattc cggataaaac ggaatctttt caggggttta tcaaccactt	8220
actgataata tagtgcata atcaaattca caaatttttt tgtaatatga cccaatgttc	8280
acaggggggg gccggagaat tacctacggt aaaggaaaaa acatgcaaat tgtctttttc	8340
tccttttggt gttggtgcat cctgttgct cggggggcca atagcttttg ctactcctct	8400
ttcgggtact caagaacttc atttttcaga ggacaattat gaaaaattat taacacctgt	8460
tgatggactt tctcccttgg gagctgggtga agatggaatg gatgcgtggt atataacttc	8520
ttccaacccc tctcatgcat ctagaactaa gctacggatt aactctgata ttatgattag	8580
cgcaggtcat ggtggtgctg gtgataataa tgatggtaat agttgtggcg gtaatgggtg	8640
tgactctatt accgcatctg acttgtctat aatcaatcaa ggcattgatc ttggtggtag	8700
cggcggtagc ggtgctgacc ataacgggtga tgggtggtgag gctgttacag gagacaatct	8760
gtttataata aatggagaaa ttatttcagg tggacatggt ggcgatagtt atagtatag	8820
tgatgggggg aatggagggt atgccgtcac aggagtcaat ttaccataa tcaacaaagg	8880
gactatttcc ggtggtaatg gaggtaacaa ttatgggtgag ggtgatggcg gtaatggagg	8940
tgatgccatc acaggaagca gcctctctgt aatcaataag ggcacgttcg ctggaggcaa	9000
cggagggtgt gcttacggtt atgggtatga tggctacggt ggtaatgcta tcacaggaga	9060
taacctgtct gtaatcaaca atggagctat tttaggcggt aatgggtggac attgggggga	9120
tgctataaat ggtagcaata tgaccattgc taatagcggg tatataattt caggtaaaga	9180
agatgatgga acacaaaatg tagcaggtaa tgctatccac atcactggtg gaaacaattc	9240
attaatactc catgaagggt ctgtcattac tgggtgatga caggtaaca attcatccat	9300
tctgaaaatt atcaacaatg attacactgg gaccacacca actattgaag gtgatttatg	9360

-continued

tgctggtgat	tgtacaactg	tttcaactatc	aggtaacaaa	ttcaactgttt	cagggtgacgt	9420
ttcttttggg	gagaacagtt	ctttaaattt	agctggaatc	agtagtctgg	aagcttctgg	9480
aaatatgtca	tttggcaaca	atgtaaaagt	ggaggtctatt	ataaataact	gggcgagaa	9540
ggactataaa	ctgctaagtg	cagataaagg	gataaacaggt	ttcagtgttt	ctaataatc	9600
tatcatcaat	cgttactca	ctactggtgc	tattgactat	acaaaaagct	atatcagtga	9660
ccagaataaa	ttgatctacg	gtttgagctg	gaatgataca	gatggcgaca	gtcatggaga	9720
gttcaatctg	aaagaaaacg	ctgaacttac	tgtagtagt	attctggcag	ataatctcag	9780
ccatcataat	ataaatagct	gggacggaaa	atccctaaca	aaatcagggg	agggaactct	9840
cattttggcg	gaaaaaata	cctactctgg	tttcaccaac	atcaatgcag	gcattctaaa	9900
aatggggaca	gttgaagcta	tgacacgtac	cgtggtgtt	attgttaata	aaggtgtac	9960
cttgaatttt	tcaggcatga	accaaactgt	taacacttta	ttaaatagtg	ggactgtgct	10020
aatcaataat	attaatgcc	cttttttggc	tgaccccgtc	attgtcacag	gtaacatgac	10080
tctggagaaa	aacggtcatg	ttattctcaa	taatagttcg	tcaaatgtcg	gtcagaccta	10140
tgttcagaaa	ggtaattggc	atggaaaggg	cggaatatta	tctttgggcg	cggttctcgg	10200
caatgacaac	agtaaaactg	accggctgga	aattgcaggc	catgcgtctg	gtattaccta	10260
tgttgcagtg	acaaatgagg	gaggtctctg	agataaaaact	cttgaagggt	ttcaaattat	10320
ttcgacagat	tcttctgata	agaatgcttt	tattcagaaa	ggccgtattg	ttgctggtag	10380
ttatgactat	cgcctgaac	agggcactgt	atctggactg	aataccaata	agtggatatct	10440
aactagtctg	atggataatc	aagaatcaaa	acagatgagc	aatcaagagt	ctactcaaat	10500
gagtagtcgc	cgagctagtt	cacagcttgt	atcttcaact	aatttgggtg	aaggtagtat	10560
tcacacatgg	cgcctgaag	ctggcagtta	tattgctaac	ctgatagcaa	tgaacacgat	10620
gttttagtcct	tctctctatg	accgacacgg	tagcactatt	gttgatccta	ctacaggcca	10680
gctcagcgaa	accaccatgt	ggattcgtac	tggtgggtgga	cataatgagc	ataatttagc	10740
tgatagacaa	ttaaaaacca	cagctaacag	gatggtttat	cagattgggtg	gagatatattt	10800
gaagacaaa	ttcaactgatc	atgatggctt	gcatgtgggt	attatgggag	cttatggata	10860
tcaggatagc	aaaactcata	ataagtatac	tagttatagt	tcacgaggaa	ctgtgagcgg	10920
ttatactgcc	ggtttgtaca	gttcttggtt	tcaggatgaa	aaagaacgaa	cagggtctata	10980
tatggatgct	tggttgacgt	acagttggtt	taataataca	gtcaaaggag	atgggttaac	11040
tggtgagaaa	tattccagca	aaggaataac	aggagctttg	gaagctggct	atatctaccc	11100
aaccatacgc	tggactgctc	ataataatat	tgacaacgca	ttgtatctca	atccacaagt	11160
ccagataaact	aggcatgggg	taaaagcaaa	cgactatatt	gaacacaatg	gcactatggt	11220
cacatcctct	gggggcaata	atattcaagc	aaaattggga	ttgcgtacat	ccttaattag	11280
tcagagttgt	atcgataagg	agactcttcg	taagttcgaa	ccatttttgg	aagtgaattg	11340
gaaatggagc	tcaaagcaat	atgggtgtaat	tatgaatggc	atgtcaaate	accagatagg	11400
caaccgtaat	gtgattgaac	tcaaaactgg	tgtggggggg	cgtcttgacg	ataacctaag	11460
catctgggga	aacgtatctc	agcaattggg	taataacagt	tacagagaca	cccaagggtat	11520
tttgggtgtg	aaatatacct	tctgataata	aaaatgcagg	gcgtgttgat	gtcctgcatt	11580
tctcggctcg	accaggctac	atccagatga	aatctttctg	agtaaaacat	ttatttatatc	11640
cttattgcta	aggaaagata	ttgggttcaa	tacctcatta	gcaatggagt	taatgatgga	11700

-continued

```

agattctatt tcaaattcag actgatcaat acctatttta ttacagaaat taactcccat 11760
ataacacato tcccttactt attaatgagt gttttcaacc tgctcaagcc aaacaataag 11820
acgtaggact tcatccaagt gttcaaaatc aacaaaacta tattttgtat gtgttttata 11880
tagttttcta gctaatttca catcacgcac agtcggtata ccaacttcat ttgcattttt 11940
tctgacagcc aaggcacact ggtaggtttc aatgagagaa ataaaagggtg caggcgctat 12000
ttctggatta aaataaatac caattgcaat atgagtcggg ttctattacca ctaattttga 12060
attacgtata tcagatttag tctgctctga aagaatctcg atatgcaact cagccttctt 12120
cgactttgtc tcaaagtgtc cctcttgctc tatatattct ctttttatct cctgtttatc 12180
catcatcata tctttcatgt ataaaatgaa ctcaatcaca aagtcagaa taataacaag 12240
aatagaaaat gccaaagaaa ataattattat atccttaaac agcctcccc aaataagata 12300
taagccatca acactagaaa acacctgaga aaaaattatt ttctgggtcat taatccaaaa 12360
gaaatagggt gttaattgta gaataataag aagcagaatg cttttgaaaa attcttttat 12420
tgtctttata ctaaatattt ttttaaccc tttaacaggg tttaatactg aaaaatcaat 12480
cttgatagct ttagtcgcaa gaacaaactt tgtttgaacc aatgttgga acacagccga 12540
aagaacacag aaaaaaagtg ggaagccaat tatcttaaaa aagacaataa ccactgcaaa 12600
aaagtattta cctcatttaa ttctgaagtc attaattatt acatatctgt ataagagcat 12660
tacatcactt aaggaaaaga atgatattat agtaaatgtc cctaccagaa taataacaac 12720
agtcgttaaa tccttaaat taaatgactg tccttttttt gcggcatcct ttagtttttt 12780
agggtgctgc tttctgttt tatttgccat aatattttat ctaacaaata gatttgtgaa 12840
aaatttatgt tcaccgagga aaaattgaac tttagaaaag taaatagtag aagaacagat 12900
gaaaataata aatattgcaa gtaaaccttt aatagttaat gatacgga aagcattcat 12960
ctgaggagca aatctgata atacaccaag taatacttct gatagtaaca atactatcat 13020
aacaggactg gctaaaataa cagcctgact tgccaataaa gtcagaaatg ttaagatatt 13080
tgagacacgg aaagaacatt gagaaaataa cgggcatata ttataagaca attgtatgga 13140
ttctaaaata aagaccatac caccactgta tagaaataca actgcagaaa aaagattgaa 13200
aaattttgcc aactcagacg tatcaacacc attggcagga tcaattgaac tactaagcgt 13260
tgccccgcgc tggttgctca taatgtctacc aacagcatga aatatccaaa agggaagaga 13320
caagcaaaaa gagagaagga ggccaacaat tatttccttg aacataagga aataaacatg 13380
ttcaaaaaca gaagaaccta tgtctacctt accagaagta attaatcctg atgcaacaag 13440
aaaaatcaca ggtattctaa tagatggaga aattataccg ttatttaaaa atggaagaaa 13500
gaaaaataat ggagccaatc taaaaaaac gccgttcagg agtattaaaa acacatgaat 13560
actttcgaac cagcttgaaa tgtccatcaa accccactct taattaaaaa cattatttca 13620
tgacaaaaag acaataaaac ctaccatac catccagaaa gaagtagcaa agatattgag 13680
acacctataa gctttatacc aaaaggaagt gtctgctctt ggagttgagt cactgtctgt 13740
aacaaccaa tagataatcc aataaccgta gctataccta ctggccataa ggaaagata 13800
aggattaaat aaagagcctt attaccata taaactatat cagacacatt acctccatca 13860
tctaagcagg aatattgata tattgttcaa tcaaggcctt agacaggata ccccacccat 13920
caagcgcaac aaataatata agttttattg gtactgatat tgtgattgga ctcatcatca 13980
tcataccaag tgcaagtaaa atgcttgaaa taacaagatc aacaacaaca aatggcaaat 14040
ataaataaaa acctattttg aacgcactt tgatttcact tagcgcataat gcaggcaaga 14100

```

-continued

```

gggaaaacaa tgaataatca ttattctctg cactctttaa atcagcattt tttcttccag 14160
acctttggaa aaaacgtgct aactcaagat ctgtatgctt tttcaagtat tgcttatatt 14220
ccatcaagcc agaatcagaa aaacggacaa tatcagatat tgtatcaaat ttttgagggc 14280
catttagata attttcataa ccagcttcaa taatgggttt catcacgaac aaagccataa 14340
tcaatgcaat accattgaga gtcataattg atggaacctg ctgcaaacca agtgcatttc 14400
gtaccattac aaaaaaataa gaaaacttta tataacaggt accagcagca actaaaaatg 14460
gtaatagcgt aaaaaaagaa agtgtagcga tgagggacat gtcactcagc atgagattac 14520
tcctttacca tccaagaact aatctcgata ccataaccat cctcaataga aacaagttcc 14580
ccatgcccaa ccaagaacc atttacttta atatttacat gttaaactat gtcacgggga 14640
aacttgaata attcgttttc tacatacatt tttagttcat tgattgtcat atttttttct 14700
aaaagaataa agtccacttt tacattgata tcatacataat tgaataatgc caaagaaact 14760
tctgtgtgtt ctgactcacc attactttct gacagattaa tttttgcctc attattatca 14820
ttcacaatat aatcccaat aataacttga ttacacaata ataaattata aagtttctga 14880
actatgatta tatcaccagt gcgaattctt tttaaagtag ccagccttac attagatgaa 14940
cctaagcaaa actctaaaag atgatgcaat gagtcaagag aaacttgatt acgtaatacc 15000
ggcatcttgt caaaaataat aatacctaaa gaagaggaaa aacagcataa atgactccca 15060
gtagtgtcat actggatttc attcgtgatc cggaaagtaa tcttatcaga cagttctgga 15120
atcgaaaata cagatgacga ctctataaag acacattttg ataaatgcag taaagattct 15180
tgtgtagaaa atcctgaaaa tacctcgtat ttttttttca aaaaagataa tgtatcaata 15240
cgaattaccc aatcccaca cacggaagta agagtgacta aactttcact gcctgtaaat 15300
ttatttttga atgtctgaat cgaaaaacgc tcgcagagtt gctttgcata taaaatctgt 15360
agtttttctg tagcgtcaaa atgtttaatt cttagcattc ttcttcactc tgttcattat 15420
atcctgtact attattttta ttccccttat tatgttgact gctatcaaat ctgtaattac 15480
cctcagcatc ttgtttcaaa attaagtcaa gtttatttcc tacaattga tctgacgggt 15540
ttagaacaaa agaaccagac tccactgaaa tactaacaga atgtccagcc cccactgtt 15600
taaagtggta agtcatttca ctattttttt gaaaaacatt tttatcattt tcaactattg 15660
atttttcttg caatgacatg gatgggggta tgtctgcagt atttttatc attacaggca 15720
gctcaaatga tttagcaaca cttccattta gtaaagaaat gatgtcagca atattcttat 15780
caatagacaa gctatgattt tttaaacc aa ttgaccatt tctattatca gattctagca 15840
gattatcagg gtttacagaa gaagtatgac ttctttcgaa gttcttgtct gcatttccac 15900
tcttattata tgaaaagatc cccctcatat cgcgatatt tattttttcc cttgctttga 15960
tatcttgatc tggatcaaaa atattttctt taatggccat tgattctgga atatcagaat 16020
atgtcgctcg ttctgttcta agttcgaacg catttttaat gccactataa atttctcctt 16080
tatcatgctt atgttcaaa tgcttatcaa tagtaaaata gttagatgca attaaagagt 16140
actcaggaat ttcaaccttc ctcttcttct tcaaaacgag actatcaata ttatctattg 16200
aagatagttt ctacattttt tctatctgtt tgctcactact aaaatttagg tttatattat 16260
ctaagccat acttcatatc ctcttgatg aactcatcaa gtatcaactg tttggcaaag 16320
ttatttcttt taatcaagcc aatccgttta catattttac tctctttatg aagcagtttt 16380
cttttagaag catttgctc atcaatatct ttttgaattt cttcaattct tccctttaga 16440

```

-continued

agaattcggt	cagactgata	ttggataata	tgetttttta	ctatagcgat	cttacgtctt	16500
gtattgaaaa	gctgactaac	actctgagcg	cctacaggtt	ctaagtcatt	ctccagttgc	16560
aggaagtata	taatcctttt	atctaagtc	tccacatctc	gatacaaa	gttttttttt	16620
tagtgtaaat	atgtgtgctt	caattaattt	aacagaccgt	tgatatttat	ctttgatttt	16680
taaaacctta	tctaattggt	tcaccaataa	gctccattgt	ctgctcataa	gtaaaaccga	16740
gtcgataatc	ttgcttcaga	aaactctcaa	caacagatat	tttggtgtag	attttatcct	16800
gagaggcatt	ctccccgggt	ttgtattcac	caaagtcaat	aattgtcctt	agctcttcta	16860
tttcagaaa	tagctcccg	aatgcagcgg	ccataatacg	atgtttttca	tcaacaacct	16920
gtgtaaatc	cctgcttata	ctttttaagg	aatcaattgc	aggaaattgt	cctttttggg	16980
ctagatttct	gctcaaatat	atatgtccat	ctaaaatgga	tcttacctct	tcagctaattg	17040
gatcagcaaa	atcatcatcc	tccaaaagaa	cagtgtaaaa	tgcagtaata	gagccacctg	17100
cctttaactt	tcctggcctt	tcaagaagtc	tgggtaagct	atcaaaaacc	gaaacccgat	17160
agcctcttct	ggcaggtgat	tctccagcgg	ctaaggccac	atcacgtaat	gctctggcat	17220
accttggtta	tgaatcaata	aaaagcgcta	ctttatgtcc	ttcagtccta	aaaaattcgg	17280
ctatagcagt	ggctatatat	gcagcattac	aacgatcaac	cgaagagtaa	tccgaagttg	17340
catatactaa	aacacacctg	ctttttttct	cagagttttt	caaataatca	accgtttcag	17400
taacctctcg	acctcgctca	ccaattaacc	caataacata	tatatcagca	ccactatggt	17460
caatgagcat	attcatgaga	aaagttttgc	cacaaccagc	tgacgcaaaa	atccccattc	17520
gctgtccttc	accacacgtg	agtaaagaat	caataacctt	aatacctggt	aaaaaaggct	17580
tctcaattgc	agccctttca	ctatatagcg	gaggagcatt	atctacaggt	cgataaagaa	17640
tttcaactgt	atctgtaaca	gcaaaacttat	cagtaacctc	ccctaaagga	ttgactactg	17700
ctcccaataa	tccacggccg	acttgcgtat	gaagaaactg	ggcagtgggc	ttaatcaacg	17760
tttgccgtga	aagtccacga	gaattttcaa	tcaagcttaa	tattgttttt	tcacatgaa	17820
atcctacaac	ctgagctctt	gcaacaattt	cattactttc	aattccagcc	tgaatgttac	17880
aaatctcacc	aatcgaaaca	tcgctaagac	ttgtttccaa	gattggcccc	gagattctat	17940
taggaaaaga	taattgagtg	agcaattttg	tatagctcat	tgattataag	accccattta	18000
agatttccat	cctctgataa	aactcatgca	gaatctctgc	aaaaacaatc	ccatcgtgaa	18060
cgtagtcato	tttaataaca	actcttaact	gtaagtactc	atcagaacga	tgaagctcta	18120
ccagctcatt	aatgctataa	ctaaagtcca	ttaacattaa	atttaatatg	ttataagcag	18180
atgactgaag	ttttacatca	cttgggtgat	caaaattagc	ccataacata	acctgctcat	18240
tgacaagagc	tatattaata	gcaggcatac	tatccaaaga	aatggtaatc	gctgaatggc	18300
tgtctaaatc	tgtaataatt	gatggaggac	aaccaatcgt	gaaaagacta	tctctaacta	18360
attgaaacta	attaatgtta	ctcatattta	aacctcacta	aatagtcttt	aatacattaa	18420
tggtatatgc	ttcatcaatc	tcagcatatc	atataacgag	tatactctta	aatctgttat	18480
ctataagtct	tttaacaaac	ctacgtatat	ctactgatac	caaaagaacg	aaattttttt	18540
ttgcatttct	caattctctc	aaagcatgtg	ctaaagtttc	cattacctca	tccgaaacct	18600
ctatatccat	attcaagaaa	gagccaccag	agggttgctt	tatccctttt	cttattgcat	18660
cctcaatata	tccggaaagc	atcacaactt	taatctcacc	agaaacagct	attttactac	18720
aaatatacct	agaaagtgat	gcacggacat	gttcaactaa	tggtatgaca	tctttttctc	18780
tgggagccca	aagcgccaaa	gactccataa	taagttttaa	attgcgaaca	gaaatatattt	18840

-continued

ctccaagcaa	tctttgtaat	acctcagaaa	ttctctgtat	agtcacatgt	cggaagactt	18900
cctttaatag	atcaggatag	cgattttcaa	actgatctaa	catatttttt	gtttcttgta	18960
tacccaaaaat	ctcatttatg	ttgtttaata	aagcttggtga	caactgggtga	taaaattcat	19020
cttgagcact	tttaatttta	gcataaatat	tagtttagatt	ttctgtatat	gaaacgtcta	19080
cccaagaaat	aacacgttca	ttatatgaag	tgaaacaac	aggaataacct	agagcatcta	19140
tatctccatt	ttcatctgta	atacacactt	tgtcaaaaata	tataaaaaa	ctatcggcac	19200
gcacctcatt	tattaacaaa	acaatatcat	cgaccttaag	ctcattgctt	gttctatata	19260
aaatagttgg	aagcctgaca	ccataatcaa	taaagaactg	acttcttctc	ctttcaataa	19320
gaccttccat	atcatttgca	tttatcttat	tttcggcaaa	taataaaatt	aaaggaacgg	19380
tttctgaact	aatagcatcc	agattttcta	tcattgccaa	agatgagtca	tggtatttat	19440
caatatcaaa	tgtaccagta	tagccactag	aatcagactc	agacagactt	tttctttttt	19500
ctacgacctt	tttataatag	aataaagccg	tcaaagtaac	tgctatcagg	aaaaaaaaaa	19560
aaaacggaaa	acctggcaac	atccctatcg	ccaaggcaag	agctgatgta	acaatgagaa	19620
caaaaggatt	cccaaatatc	tgggacatta	tgttacgtcc	catattatcg	ctatcaccat	19680
taactcttgt	aacgataaat	cctgcactta	tagaaattaa	cagagcagga	atctgagaga	19740
ccaatccgtc	accaatagtt	agtatgggtat	aagtagataa	tgaccagat	agggacattc	19800
catgttgact	cataccaaca	gaaatacttc	ctattaaatt	gacaaaaata	ataatgatgc	19860
cagcaattgc	atcaccttta	atgaacttca	ttgctccatc	aatgaacca	tacaattgac	19920
tctcacgttc	cagaataactg	cgtctttctt	tagctcctgc	ggcatcaata	attccagcct	19980
ttaaatccgc	atcaatgctc	atttgtttcc	ccggcatacc	gtcaagagaa	aagcgagcag	20040
ctacctccgc	aaccttttca	gacctttttg	ttataacaat	gaactgcaca	acagttaacta	20100
ttgaaaaaat	aacaaaacct	acagccagac	tatcaccaat	gacaaaactg	ccaaatgtag	20160
taataatttt	acccgcatcc	gcatacaacta	aaattagtcg	actagtactt	attgataaag	20220
cgagacgaaa	taatgtcggt	atcaacaaaa	ccgaagggaa	tgtagaaaaa	ctcaatattc	20280
tttcaatata	aatgatcccc	ataaaaaacga	gaatagcaag	tacaatatta	agccctatta	20340
aaaaatcaac	aaggatatgta	ggtaatggta	tgatcaacat	agcaatgac	ataaccataa	20400
gcactaatat	aattaattca	ggcttagtac	ttacttgctt	aagaaaagac	tggtactactt	20460
ttatctctcg	ttatctagaa	agctctttct	tgtagtgcact	tgtagacaagc	tctcgtagaa	20520
tattgatcgc	aatttctctt	tcactttcac	ttttaaatat	atcatgagga	ctttcattaa	20580
cagaagatag	atatgtttga	tacactacag	aatcctcatg	actggctcgt	gatacttcta	20640
taacagaagt	aagaatttgc	tcactttcac	ttggatattg	aaaaatatat	aataaagtta	20700
gcaatagctg	ctctgcactt	aaactgccat	cttttagcaa	gcttgaggat	tccagtttct	20760
tcataaaaaat	tacgtctgaa	gttctaataca	tcttaattgc	aattaattta	gataacacaa	20820
aacaaaactc	ataagcatta	cagctcggtt	tctcagaatc	catgtctaca	atcagtgcact	20880
gctctacaaa	attcaccaatt	gtatgtctgt	gattaaaacc	aaaattacca	agccattcta	20940
tatactgac	tggtgttgat	atgttcccca	tgataaaacc	acgataacat	gcacgcaaaa	21000
gctgtgggtc	aagtttcatt	ttgttgccaa	ataatttagc	ctgtatagct	gaattttattc	21060
cagcattaac	acttttttca	ctacctgata	ttatctcatt	aataatttta	gttaataaatt	21120
cgacttggtc	agcagtaaga	tctttttcat	tcaaaagtcc	ccttaatgct	aagacttgat	21180

-continued

cacttggatc	tttaaaatat	ctcttttagtc	tgtctataaa	atccccgatct	agaggaaggt	21240
tatcttttaa	cgtctctctt	aaatcaaaaa	tcttgtagt	aatttcacat	tcttccccat	21300
ctaaaatacg	ttcctgacta	tcactatttg	ttcctttcag	tttctcaagg	tctctctgt	21360
ttataaatga	cgaaagagcg	gaagccatct	catcagatga	gtcaatatac	ttgctattag	21420
ccaagccaga	atcaagctct	gcacagcag	tctcatctcc	atcatctgaa	tttgtcattg	21480
cattcaactt	atcaatgaat	gcagagctaa	aaactcctgt	atttttaaca	tcaagcatac	21540
atcacctact	ttatagttag	taatttaagt	atgaaaccaa	tgatttttcg	tcttccagca	21600
aagtagtttc	agatgggatt	atttgtgtag	tcttttgtat	ttctctttca	gatatgaggc	21660
ttttatatc	tgcagtatta	taataacttg	actcctttat	ctcacggggt	tgaatcagaa	21720
aaacccttac	aatattactg	atattacttg	ttttatactt	gaacacatta	ccgattacag	21780
gaatagagga	gagaaaagga	atagatatga	tttcgtaga	gttcggttca	tgcgtataac	21840
cacctatcaa	caaaactttg	ccttgtagaa	ctctcgctat	tgtactgatt	ttggctcttc	21900
caacttctgg	aaggacagat	gtattttcat	tattataatt	gtaatttgac	tgactatttc	21960
ctgttccatc	ctctatcggt	aaagacatct	caatttgtcc	ccgagatgaa	aagcgaggta	22020
taacatttat	caatgttcca	tacgtaacat	gtccaacga	agaattgcgc	tcaccaacta	22080
atgacacata	aaaagtcctg	ttattatcaa	atatagccgg	aatattctct	tgtgtcaaaa	22140
tcaccggcct	tgaactaca	ttgctttct	ttttctgatt	taatgccata	actgaagcaa	22200
taaaacttatt	gccgtccaaa	gtgcttatcg	atgctgatga	gctcatatta	aatgacgcac	22260
caaacgaatc	accaaataaa	gccgttctct	gccaatgaac	accaaggtta	tttaattcag	22320
atttatcaat	atcgattatc	cataatgata	attctatatg	tcgtttggct	atatctaact	22380
gagtataaat	atcgogaata	atctgtattt	gctgatcatt	acctttgact	aatatagaat	22440
tggtttctgg	atatgcaata	agtatacat	cctctaata	agaactattg	gtaactgaac	22500
taaatgaaaa	gtcattacta	tctcagata	ctttctgtgt	aatattaaag	ggtaggcacg	22560
gatcattttg	ggcttgctg	ttactcagag	ccttaccatt	atttaataaa	cgttctacaa	22620
cagtagccac	ccctgggatg	acaatatcct	ccccctcat	attataagtt	ctgtcactaa	22680
caaatgtatt	ttttaatttt	ataaccccaa	aattaacttt	atctgtccca	atagaagaaa	22740
cctgcttacc	aagcaaagtt	gctgtattag	ctaccagttc	aaccaatgcc	ggaggaccag	22800
atatatagaa	tgttttatca	ctaagtgttc	ctcttatagg	atatctgtga	tcataaaggt	22860
tagcatcctt	cagatactga	ataagataat	taagtatat	attctccagc	aaaataactt	22920
tgcttattaa	ttctccagag	tcatatatgt	ataatgcatt	tccatcttta	taccatatga	22980
gtccacaag	taatgtcagt	ttttcaagca	tttcttctgg	attagataaa	tcaaactccc	23040
cagatattct	cttttttgca	gcttggttgc	taacaacaat	aggataattt	aataacgcag	23100
aaaaacgttc	aaaaaaagat	cctaccgtat	cactttgagc	aacatatttc	gctatgtcat	23160
tttgttctaa	taaatgagag	tcaatattat	tagcattaac	tatcaatggg	aacaatacaa	23220
tcaagagagt	caaagattta	atattaaatt	ttttcattta	tttttttcac	taaaaaagta	23280
atattcgata	attctcttgc	agaaaagcct	agtcgtgttt	ttatctcatt	tgaaaaatgg	23340
gatgttgatg	cataccatt	attcatggca	gctgaagtga	ttgtttgatt	gtgcaaaaat	23400
acatctaata	aaccatttac	taaccgccaa	gtgttcagtt	gttctttgac	ttttgccct	23460
aaagcttttc	tacataaaga	tctaaaataa	gcctccgaca	caccataatg	ctctgtcaga	23520
ctctttatc	ttacotcctt	ttctttctca	atagttctca	gaagataaag	aaccaaataga	23580

-continued

tatgactcag	tcattcgtat	catagccagt	gcattactta	cattctggtc	agatgaaaag	23640
atgaatctta	acattaactt	atcaagcatc	gattcgggat	tacaactatc	ccgaagatca	23700
ttctcatgat	agaaatgaat	acatttttca	gcatggctat	gctctactag	agcaccactg	23760
aaatagtccc	aaaaagcaag	aaattttgca	caactattaa	tggtaaagcg	ataatgatag	23820
ttcacattat	cactgaattt	tatatattga	tttcggacaa	aaaaaaaccc	ttccatgctg	23880
gagccagctc	aattatgtta	cgttctttat	cttttacaag	ctcaacgtta	actggttcat	23940
gtccactaga	gtaaatgtaa	aacatattac	ttgtatttag	acctttatat	ttactcacia	24000
tacaattata	tcctcgaat	taattttaat	ttaacttccc	catccttctc	tataaggatg	24060
gtatcattta	tcttattacc	ttttaaaaca	accccgctct	tcagcataga	aaatagttta	24120
tctatagcta	tcttttgtgc	atccatgaag	caccatctca	cttccttaac	attaaagttg	24180
cccatgaatt	tacctgcatt	aaaactataa	accgccttac	cactaataca	tttagagtaa	24240
acatttatac	ttttattggg	tgtattaaac	gaccataata	aatcatttgg	aatagaaaaa	24300
taatccttac	ttacaggaac	aatatgccat	tctttttcag	aattagatga	tgactttaaa	24360
gcacaccac	tcagtgttaa	tagcaatata	cttaaaataa	atattttcaa	cttattacta	24420
ccatgtcgaa	tcctctgect	ctctccctgc	ttcaaagaca	ctaattaatg	aatcgaaggt	24480
tgtagtctta	atatagaaat	caatattggt	atatatggac	aatgatgcac	aaaatccctc	24540
atgcaaatcg	acatattgat	ttgcacttat	gagtttatta	agaagtgaac	ttttttgtgc	24600
agctgtaaga	taaggaaaac	gtgaattgaa	taattgttca	atcttattgt	gcagttcatc	24660
attatctatg	gaatttaatt	cactctgtga	gaaatattgc	attatttcat	agatttatatc	24720
ctcatcaaat	gattgcgaag	atgataatct	ctcatttaaa	ttaacaactt	cttcactatt	24780
taatctttgc	tgcagagcat	tgcttgcat	tatttgatta	atcattaaca	ggattctctt	24840
ttaacaagtc	ttctgcaatt	tgtctcagtt	tcttatgtgc	taactctttg	aaatattttt	24900
tatcatTTTT	tggttaattt	tccgcaagtt	ctattttaac	acctgatatt	acatcctgag	24960
gactaaactc	cactacttga	ttaccactac	agaatatata	attattccct	tcagaatatt	25020
ttaaaataat	cttgacatct	gagcgatatt	tatggagatc	aagttctaac	tttttctca	25080
gtgctaaatt	acacttgggc	aaaaccactt	ttaattctgt	tactgtatta	cgcaacgaac	25140
taagccactg	ctccaataaa	gtgctaataa	tcctctcatc	atgtaaattt	tctgaaagca	25200
aaactagttat	atagtctctc	atatctcttc	tattttcatt	ttgctttaac	ttccattcac	25260
aaatatagtc	aataatatgc	tcaaaagcag	tctgtatacc	aatttgataa	ccatcacata	25320
cagcatctat	acgcagtgat	tcagcttttt	tcgtagcctc	ctttataaca	caaattgctt	25380
tcttcttact	ctcagtgtct	ctatctttta	tggttttaca	taattccttt	tcaattctct	25440
ttataacaac	accatcatag	gcattgatgc	tggtatccgg	taaagtgcct	ttttgcata	25500
tgcatacctt	catttaatcg	ggacaaaatt	ctttctacag	gcgtataatt	tccttccaaa	25560
gcaattggat	taaataataa	gtttattctt	tcatatatac	caacagatat	attttccttc	25620
cacgcaataa	gatgocaaaa	accacttcga	atagcggcct	ccccttgctt	actgattttg	25680
cccccataat	aactaataac	catctcacta	tgattaatta	accgagactc	ttgactaccg	25740
ataaaataag	caacagctgg	aagtaaatc	cagttggcaa	caaacatagc	ggttaccccg	25800
cttaaagggt	caatatcgca	atttagatca	tattttattaa	ttatgagatt	attaatctca	25860
cttctgatta	cgcatttttc	taacagttca	gaaggcaaat	tcaaccgatt	tctattttata	25920

-continued

tagaacgctg gatcaaaaat tattgaaaga tattttttat aaattccatc cattcttate	25980
atattttggt tctcttgaac caccctgttt tgaatgccca taccaaaaga atgacaacta	26040
gtacagccat cccgagaaat aaatatatta cttcatttgt taaaaattcc gatttaactt	26100
ccttaacagg ttgtacattt gtataaacat attcttcttt cggagttaat atgacagata	26160
tattttcata cttaacatca gaaaagggtg ttttcaaaaa tcgcttaata ttactaacta	26220
atagttcaga ctctttcggg gagtcataa tagcgattac tgatatatgc atcggttttg	26280
aagatatatt tttttcttca agatcatagc taacatgtat ttttgccgaa ataacaccac	26340
caatagaaac taaagactgt tccagccgtt gctcaatagc actatataaa cgggcctttt	26400
cagctcttgg agaagacact aatgaatctg taggaaacat ttgtgagata tcaactctct	26460
ccggatttgg caaatcgtac atgcgcatca aatcaactgc cgatgcaaat gtcccctttt	26520
cgacttgtag cgagatcccc tgtttacctc catcaacttt tctagcagta atattatggc	26580
gttctagcac agatattatt tcatttgccg gtctttgaga taaattagaa attaactctt	26640
cacgttgctc acatccaatc aacatcagca acaagaataa aataaaacct ttatacctaa	26700
tcataataat taagacttta ataaagtctc aacagccgaa actgtttttc ttgctaactg	26760
gccagctaag gatactccta ttgaataatt tgagagcgtc gtctgtaact tggccaaaga	26820
ttctgggttt aatgattctg gattgcttac catctccatt atttgtgata cttgaatate	26880
tgtatccatc ttaatatcac tatatttagc cgacacgacg tcttcagac ttgatatctc	26940
ttgagattga aaatcactgg ctttgataat atcaacctga ttgactggat aaatgtaatt	27000
catgaatgct ccctattatc tgaagttttg aataattgca gcatcaacat ccttaatcac	27060
tttactgta ttggattgag cgttcctata taatgtatat tcagataatt tacttttgta	27120
ttcagccagc aactgtggat tcgaaggatt tttagctaatt ttatctagtg ccaatgttag	27180
ttcacctgtt aatgtttgag ttccatcatc aaaagtttca gataatgaac tcaatgtcca	27240
atcatcatc ggtactgtaa cactcatttt atcctcactt atttttatct aaaaaaac	27300
agtgtttatc attcaacatt acataactgt ctttgctgtt aagatatgat ttacctttaa	27360
aatcatcatc tatcaaaagt acagaaaact caatatactg atctccatat atatttttat	27420
gttcagaaat cagctatata atgagctcaa gcttttcac ttgtcagttct tctctgacag	27480
aatatgtaac ttattatttt tcacaaatct ctttatactg gacatttacc tttgtgaaaa	27540
tacctttagc atccccctt gcattatgat cgctaagata gttgaattta atatttttga	27600
tatacggaat ttcaacaatc agagcatttt tcaattttatc aagttcggta tcttttgaac	27660
tatttctttc ttactttaa aacaccagta attccgcttt atcactcact aaattcaaca	27720
catacaaatt gataaatggg aatcgttgat tgatatattg ttgaattcta ttatttctt	27780
tgtttgacac cagaatatat ctgcatttgg taaatccggt tttatttaag gctacagacg	27840
cccataccgc atcattttgt gtttcggcaa gaatataatt gcactgtcca gtgttgacta	27900
ttccatatct cttatcatct ataatttcag caatttcttt tacttcatct ttctttaag	27960
agagggaaaa aattataatt aacacaacaa taacagcaaa gaaaaaaac accgagtggg	28020
tcttatacat tccgtttagt atatggctc tcgattttat ctcctcata tttttaaggt	28080
gaaaagaaat cccgtcatat tcaaatgaat taaattgtga attgatatca atttgctcac	28140
ctttttcatc actaaattga cagatattat ctttgcttat ggtcaaataa attataccat	28200
ccgatttact atctgttaca ggaataatat tttctgggaa attgtcgttt ccaaactcaa	28260
gtgtttctat cgtccgtccg agaacaatta agttattccc ataataaac ggaaactcat	28320

-continued

```

ccccgacacc gtttgtgagt ttcactaata acctgaatgg tgcaagattt gagttctttg 28380
cctcagacat aaatcctctt atatccttcg attattctgc ttgctctctg ctgctatccg 28440
caaatagcgt ctatagatat cctctgcgta ctttaactct tcttttttct ttttcggaga 28500
cgttcctgca ttataggtgc caacagcctc ccacctctt ccatacatca tcataaactc 28560
atttaataac tttgctgcat aatgtacaga aagacatgga tgactaatta gcatttcttc 28620
agaatatccc atttctctaa gtcttttaga atgaaagtca tttatttgca ttataccata 28680
atctttactt ccattgttat taacattgac tgcagattta ttaaatccgg attctttttc 28740
cgcaatcgct ttaacaaaac ttgatggaat attgtacctt tcaccagcct tatcccaaca 28800
atcagctctc cctaaatag ggataaaaca cagcaaaata aatacgaaac gggacattaa 28860
tacccttca tttctcgcgc aaattcatcc ttcagccaac gcacacgcga aatataatat 28920
tctgtattaa catatacatc ctgaattca tttatcccg taagatctaa tcgcgcaatt 28980
agatttccgc cttcatcatc agtagctaag catatctttt ctgagtaatt taggctcaac 29040
gcataataa gatcatttat attttcaggc aatgcacaaa aagggcataa taaatttatt 29100
gaatctcctt ctttttcaat atatattagc acatcatcat caagcatgat tgcttgatca 29160
tcgatatcaa ttaagggat acccaaggct cggcaaaata catctgctaa atcttcata 29220
tattcctctt atacaaatga cgaataacct ttcaccatat tccatatttt tgggtcccct 29280
attctttcag aataagatag ctctaacgaa gatagcggca attttttcat gactttattt 29340
cgggacacac cagtattcat ctcttggtt tccatattac cactattcat tagaatagta 29400
gaaaataatc ttttctcttc tgaggataat ttactattta attgagaaaa ctgacctgtt 29460
tcatgttttc ttattatttc tcttttaatt tcagcatcct gcatacctgt tctgtccttc 29520
ccactcttac aattccagca aggtacagca cctatagtat aagccaatag agtcattctt 29580
tgtgacaatt tatacggctc tccattatca tttttttgta actttttggt tacaatctct 29640
tttatctggt tagccagata tatcacatca ttcttacatg gtggattttt ttctattgct 29700
tctgcagccc aacctccaat cacaccattt ttgaggaaat tatccccc aaagaacat 29760
attgattcat cattaagctt gtcaacattc ctccagccta agcccatttt tagcgctaatt 29820
tcatttacac caaagttaaa tgtcaccact tttagattaa cgcttacttc ttccagaaga 29880
ccatcactat tccgaattaa taatttcgtt ggctccccc ttttagagtt taacctttt 29940
aatgcgttca cctgatcttt taacatactc tctctctccc cggttaaaact ggtcggcgtc 30000
agaagagaag tcgaaacaat ctttaaatct actgttttac cagacaaagc ctgtgataat 30060
aactctggtc tgctatataa tgcctgcgct actaactcct cagctttatt acgagcagca 30120
acagctcttt cagatgagtt ttttttcaat ccataggctg atataactcc atgtctgata 30180
ccgaaaaaaa tttccttccc ctcatcatcc accacttttg aaagccacat attcgcgata 30240
tgatcactct ctctcgtaga agcacaacaa attccttttc cattatactc ttgacaaat 30300
atatttttat tccctatctt catatgagag gcaggagtta attgaaaacc ataatttttc 30360
ccatgatgac taatattctt gtgcacgggt ccccaattct tattatttaa ataacagta 30420
aatatttggt ttattttcgt ttgcaattcc tttttatttt taatgcctag ctgtgctctt 30480
atctcttcac ttattacttt attaaagetca ttaatatgtc ttttattaac ttcaccaagt 30540
ttataggat ctaatcgttc catcatggct ttaatttttg acgtatcacc acctgccaac 30600
tttattgctt cgcaaaaaac ttcagctctg tcacgcgcaa acaaaaggct ttttctggt 30660

```


-continued

aagcttggtt	ggttctgtaa	atataatcga	ttaagcgctt	cgatccctt	atttcttct	30720
ccatttttaa	ctctttgata	tgaatcacg	cttacacctt	tgtgctttcc	ggttcttct	30780
gcgcctttat	aggaatctcc	gctttgaaat	gaaacctgat	gcaatcccaa	attagttatg	30840
tgcatttaat	tatcctcagt	cataaatggg	gcattgattg	cagttttgta	cctcgtttta	30900
caacaaaaaa	gatacttatt	caacagggtt	ctctcttttt	ctataacact	ccgagcctga	30960
tacatcagcg	actttatctt	gtgggatttc	atgatgaaac	gagcactaca	tatgctcacg	31020
aggtagatat	tatgtacagc	aaaagaaaat	ttgattatgt	taacaggctc	aagtttctaa	31080
atttattaat	tgaatatatt	atataccaaa	tacattttac	atccccaca	tcaccaagta	31140
atggagagtt	aataaagtat	gacctcaca	attagcaatt	tcattgacgc	aagcaatata	31200
aaagggccta	tacgcgttga	agatacagag	catggaccta	tattgatagc	acagaaattc	31260
aacctaaagg	atctcttttt	cagaacatta	agcactatta	atgcaaagat	aaattcacag	31320
atattaaatg	aacaactaaa	aaattatcga	ctagagaatc	aaaaatcctt	attgcttttc	31380
ctcaatacat	tagctagcga	aaaatctgca	gaaagtgcgt	ttgccgccta	tgaagcagct	31440
aaaaactcaa	ttcaacactc	tttcacaggt	agagacatca	aattaatggt	gaataccgca	31500
gagcgtttcc	atggcatcgg	tacagccaaa	aatcttgaaa	gacatttagt	ttttcggtgc	31560
tggggaaata	gaggcataac	ccatttaggg	catactagta	tcagtataaa	aaacaattta	31620
ctacaggaac	caactcctac	ctatctcagc	tggatcccg	gcgggaatgt	tacaaaagat	31680
acagaaataa	actacctttt	tgaaaaacgc	tcagggttaca	gtgtcgatac	ctataagcaa	31740
gacaaactaa	atatgatttc	agaccaaaac	gccgaaagac	ttgatgcagg	ccaagagggt	31800
agaaacttat	taaattcaaa	acaagatcaa	aataacaaca	aaaaaatatt	ttttcccgaa	31860
gctaatacaa	aaaaagaccc	ctatgggtat	tgggggggtca	gtgctgataa	agtatacatt	31920
ccattatcag	gtgataataa	aacaaaggat	ggtaaaatca	gccataacct	ttttggcctt	31980
gatgaaacaa	acatgtcaaa	atttatatgc	aaaaagaaag	cagatgcttt	cagacaattg	32040
gcaaattata	aattaatcag	taaatctgaa	aactgtgctg	gtatggcact	taatgttctt	32100
aaagcaggaa	attctgaaat	atactttcca	ctccctgacg	ttaaattggt	tgctacacct	32160
aacgatgtat	acgcgatgc	aaataaagtt	aggcaacgca	ttgaaagtct	caatcagctc	32220
tataacgaaa	taatgaagta	tattgaatct	gattttgatc	tttccagatt	gactcaatta	32280
cgacgcagct	atcttaaaag	tttcaataaa	attaacctta	tccacacacc	aaaaacattc	32340
aagcctttat	caatttctact	ttacaacat	cctactgaaa	atgtatcttc	agaagacttt	32400
gatgcgggtca	tcaacgcttg	ccactcctac	ttagtaaaat	cagcaccag	caatatgact	32460
cgtgtattaa	acgaattaaa	aaccgaagca	acagataaaa	aagaagaaat	catcgaaaaa	32520
tcaataaaaa	ttattgatta	ttacaattct	cttaaatcgc	cagatcttgg	aaccaaaactt	32580
tacattcacg	acttattaca	gatcaataag	cttttattga	ataactctca	ttctaataata	32640
tagaaggcca	tagaaatgtg	tcgcaaaacta	tatgataaac	tttatgaaat	aacagggtgca	32700
aagcttgatt	ttaacgataa	aaatcaagcc	tttatattgc	tcgaagaaca	aatacctggt	32760
tgtataacag	ataatgatga	atacatcttt	ttgaccggac	ttcttaatga	acatgaactt	32820
tttacagaaa	atattattaa	tccagaacac	atactcatcc	ttaactactc	tttgagtaga	32880
gattatggtt	cttctatctg	tcttctccct	gatactcatc	aatgtgtttt	gaccaagaaa	32940
cactacaaaa	aatatctttc	acctgatgaa	ctcattgaat	ctttgtatga	atttttattt	33000
tgtattaaac	taaccattgc	aaacatcact	tcagaagtga	actaacatat	aggggggtatc	33060

-continued

atgcaaattc	taaacaaaat	acttccacag	gttgaattcg	cgattcctcg	cccatcattt	33120
gattcgtaa	gtcgcaataa	gcttgtaaag	aaaatattat	ctgtttttta	cttaaaacaa	33180
cgctttccto	agaagaattt	tgatgcgccg	gtgaatataa	ataaaatcag	ggacagcgtc	33240
atagacaaaa	taaaagatag	taattcgggg	aatcaactat	tttggttgat	gagccaagag	33300
cgaaccactt	atgtctcttc	aatgataaac	agaagtatag	atgaaatggc	catacacaat	33360
ggagttgttt	tgacctcaga	taacaaaaaa	aacatatttg	cagctattga	aaaaaaattc	33420
ccggatataa	aacttgatga	aaaatcagca	caaacatcta	tcagtcatac	agcactaaac	33480
gagattgcct	catctggcct	cagagcaaag	attttaaaa	gctatagcag	tgacatggat	33540
ttgtttaaca	cacaaatgaa	ggatcttaca	aatctagtat	ccagttccgt	ctacgataaa	33600
atatttaatg	aatcaacaaa	agtttttaca	atagaaataa	gcgcagagggt	tctaaaggct	33660
gtataccgtc	aaagcaatac	aaattaaatc	ttttatatta	aatcttatac	ttattaaagt	33720
aaagcagaca	aaaaggagac	cttatgtctt	taaatatcac	cgaaaatgaa	agcatctcta	33780
ctgcagtaat	tgatgcaatt	aactctggcg	ctacactgaa	agatattaat	gcaattcctg	33840
atgatatgat	ggatgacatt	tattcatatg	cttatgactt	ttacaacaaa	ggaagaatag	33900
aggaagctga	agttttcttc	aggtttttat	gtatatacga	cttttacaat	gtagactaca	33960
ttatgggact	gcgagctatt	tatcagataa	aagaacagtt	ccaacaagca	gcagaccttt	34020
atgctgtcgc	ttttgcatta	ggaaaaaatg	actatacacc	agtattccat	actggacaat	34080
gccagcttcg	gttgaaagcc	cccttaaaag	ctaaagagtg	cttcgaactc	gtaattcaac	34140
acagcaatga	tgaaaaatta	aaaataaaag	cacaatcata	cttggacgca	attcaggata	34200
tcaaggagta	attattatgc	ataatgtaag	caccacaacc	actggttttc	ctcttgccaa	34260
aatattggct	tccactgagc	ttggagacaa	tactatccaa	gctgcaaatg	atgcagctaa	34320
caaatatttt	tctcttacaa	ttgctgatct	tactgctaac	caaaatatta	atacaactaa	34380
tgccactca	acttcaaata	tattaatccc	tgaacttaaa	gcaccaaagt	cattaaatgc	34440
aagttcccaa	ctaacgcttt	taattggaaa	ccttattcaa	atactcgggtg	aaaaatcttt	34500
aactgcatta	acaaaataaa	ttactgcttg	gaagtccag	caacaggcaa	gacagcaaaa	34560
aaacctagaa	ttctccgata	aaattaacac	tcttctatct	gaaactgaag	gactaaccag	34620
agactatgaa	aaacaaatta	ataaactaaa	aaacgcagat	tctaaaataa	aagacctaga	34680
aaataaaatt	aaccaaattc	aaacaagatt	atccgaactc	gacctcaggt	caccagaaaa	34740
gaaaaaatta	agccgggaag	aaatacaact	cactatcaaa	aaagacgcag	cagttaaaga	34800
caggacattg	attgagcaga	aaacctgtgc	aattcatagc	aaacttacag	ataaatcaat	34860
gcaactcgaa	aaagaaatag	actctttttc	tgcattttca	aacacagcat	ctgctgaaca	34920
gctatcaacc	cagcagaaat	cattaaccgg	acttgccagt	gttactcaat	tgatggcaac	34980
ctttattcaa	ctagtgtgaa	aaaataatga	agaatcttta	aaaaatgatc	tggtctctatt	35040
ccagtctctc	caagaatcaa	gaaaaactga	aatggagaga	aaatctgatg	agtatgctgc	35100
tgaagtacgt	aaagcagaag	aactcaacag	agtaatgggt	tgtgttggga	aaatacttgg	35160
ggcactttta	actatcgtaa	gtgttggtgc	agcagctttt	tctggaggag	cctctctagc	35220
actggcagct	gttggtttag	ctcttatggt	tacggatgct	atagtacaag	cagcgaccgg	35280
caattccttc	atggaacaag	cctggaatcc	gatcatgaaa	gcagtcattg	aacccttaat	35340
caaaactcct	tcagatgcat	ttacaaaaat	gctcgaaggc	ttgggcgtcg	actcgaaaaa	35400

-continued

```

agccaaaatg attggtctta ttctgggggc aatcgcaggc gctcttgctc tagttgcagc 35460
agtcgttctc gttagcactg ttggtaaaca ggcagcagca aaacttgtag aaaatattgg 35520
caaaataata ggtaaaaacc tcacagacct tataccaaag tttctcaaga atttttcttc 35580
tcaactggac gatttaatca ctaatgctgt tgccagatta aataaatttc ttggtgcagc 35640
gggtgatgaa gtaatatcca aacaaattat ttccacccat ttaaccaag cagttttatt 35700
aggagaaagt gttaactctg ccacacaagc gggaggaagt gtcgcttctg ctgttttcca 35760
gaacagcgcg tcgacaaatc tagcagacct gacattatcg aaatatcaag ttgaacaact 35820
gtcaaaatat atcagtgaag caatagaaaa attcggccaa ttgcaggaag taattgcaga 35880
tctattagcc tcaatgtcca actctcaggc taatagaact gatgttgcaa aagcaatttt 35940
gcaacaaact actgcttgat acaataaagg agaatgttat ggaaattcaa aacacaaaac 36000
caaccagat tttatatata gatatatcca caaaacaaac tcaaagttct tccgaaacac 36060
aaaaatcaca aaattatcag cagattgcag cgcattattc acttaattgc ggtaaaaatc 36120
ccgtattaac aaccacatta aatgatgatc aacttttaaa gttatcagag caggttcagc 36180
atgattcaga aatcattgct cgccttactg acaaaaagat gaaagatctt tcagagatga 36240
gtcacacctt tactccagag aacactctgg atatttcag tctttcttct aatgctgttt 36300
ctttaattat tagtgtagcc gttctacttt ctgctctccg cactgcagaa actaaattgg 36360
gctctcaatt gtcattgatt gcgttcgatg ctacaaaatc agctgcagag aacattgttc 36420
ggcaaggcct ggcagcccta tcataagca ttactggagc agtcacacaa gtaggtataa 36480
cgggtatcgg tgccaaaaaa acgcattcag ggattagcga ccaaaaagga gccttaagaa 36540
agaaccttgc cactgctcaa tctcttgaaa aagagcttgc aggttctaaa ttaggggttaa 36600
ataaacaaat agatacaaat atcacctcac cacaaactaa ctctagcaca aaatttttag 36660
gtaaaaaata actggcgcca gataatatat cctgttcaac tgaacataaa acttctctta 36720
gttctcccca tatttctttg caggataaaa ttgacacca gagaagaact tacgagctca 36780
atacccttct tgccagcaga aaacaaaaca ttggccgtgc aacaatggaa acatcagccg 36840
ttgctggtaa tatatccaca tcaggagggc gttatgcac tgctcttgaa gaagaagaac 36900
aactaatcag tcaggccagc agtaaacaaag cagaggaagc atcccaagta tctaaagaag 36960
catcccaagc gacaaatcaa ttaatacaaa aattattgaa tataattgac agcatcaacc 37020
aatcaaagaa ttcgacagcc agtcagattg ctggtaacat tcgagcttaa tatatccaag 37080
agccataata atatattggt ctctctgtaa ggaaataacc atgaatataa caactctgac 37140
taatagtatt tccacctcat cattcagtcc aaacaatacc aacgggtcat caaccgaaac 37200
agttaattct gatataaaaa caacgaccag ttctcatcct gtaagttccc ttactatgct 37260
caacgacacc cttcataata tcagaacaac aaatcaggca ttaagaaaag agctttcaca 37320
aaaaacgttg actaaaacat cgctagaaga aatagcatta cattcatctc agattagcat 37380
ggatgtaaat aaatccgctc aactattaga tattctttcc aggcacgaat atccaattaa 37440
taaagacgca agagaattat tacattcagc tccgaaagaa gccgagcttg atggagatca 37500
aatgatattc catagagaac tgtgggctaa aattgcaaac tccatcaatg atattaatga 37560
acagtattct aaagtatatg aacatgccgt tagttcatat actcaaatgt atcaagattt 37620
tagcgctgtt ctttcagtc ttgccgctg gatctctccc ggaggtaacg acggaaactc 37680
cgtgaaatta caagtcaact cgcttaaaaa ggcattggaa gaactcaagg aaaaataata 37740
agataaacgg ctatatccag caaataatac tgttagtcag gaacaagcaa ataaatggct 37800

```

-continued

tacagaatta	ggtggaacaa	tcggcaaggt	atctcaaaaa	aacgggggat	atgttgtcag	37860
tataaacatg	acccaatag	acaatatgtt	aaaaagctta	gataatctag	gtggaaatgg	37920
cgaggttgtg	ctagataatg	caaaatatca	ggcatggaat	gccggattct	ctgccgaaga	37980
tgaaacaatg	aaaaataatc	ttcaaaacttt	agttcaaaaa	tacagtaatg	ccaatagtat	38040
ttttgataat	ttagtaaagg	ttttgagtag	tacaataagc	tcatgtacag	atacagataa	38100
actttttctc	cattttctgag	gtgcgccatg	cataatgtaa	ataatactca	agcgccaaca	38160
ttcttatata	aggcaacttc	accatcatca	acagaataca	gcgagttaaa	aagcaaaata	38220
tccgatatcc	atagttcgca	aactttctta	aaaacaccag	catcagtgtc	tgaaaaagaa	38280
aactttgcaa	cgtcttttaa	tcagaaatgt	cttgattttt	tattttcttc	ctcagggaaa	38340
gaagatgtgt	taagaagcat	ttattccaac	tcaatgaatg	cgtatgcaa	aagcgagatt	38400
ctogaatttt	caaatgtttt	gtactcctta	gtacatcaaa	atggtcttaa	tttgaaaaac	38460
gaaaagggac	ttcaaaaaat	tgtcgcacag	tattcggaac	taattataaa	agataaatta	38520
tccaagatt	ctgcctttgg	accatggctg	gcaaagaata	agaaactcca	tcaattacga	38580
caaaacattg	agcacagact	tgactatta	gcacaacaac	acacatctgg	tgaagcttta	38640
tcattgggac	aaaaactcct	caatactgaa	gtatcatcat	ttatcaagaa	taatattctt	38700
gctgaattaa	agttaagtaa	tgaaactgtt	tcactcttca	aactagatga	tttagttgac	38760
gcacaggcaa	aacttgcctt	tgatagtttg	cgcaatcaac	gtaaaaatac	tattgatagt	38820
aaaggatttg	gtataggtaa	actgtcaaga	gacttaaata	cagtagccgt	gtttcctgag	38880
ctggttgagaa	aagtccttaa	tgatatttta	gaagatataa	aagattcgca	tcctatccaa	38940
gatggcctcc	ctacacctcc	cgaagatatg	ccagatggcg	gaccaacccc	cggagccaat	39000
gagaaaacat	cccaacctgt	aattcactat	catataaata	atgataatag	aacttacgat	39060
aatagagttt	ttgacaacag	agtatatgac	aatagctatc	acgagaaccc	agaaaatgat	39120
gcacagtctc	ctacttctca	gacaaacgat	ctattatccc	gtaacggaaa	ctcattacta	39180
aatccacaaa	gagcactagt	tcaaaaagta	acttcogtgc	taccacactc	tatatcagat	39240
actgtccaga	catttgcaaa	taattcagct	ttagaaaagg	ctttcaacca	tactccagat	39300
aattcggatg	gaataggttc	agacctgtta	actacgagta	gtcaagaaag	atctgcaaat	39360
aactctcttt	ctcggggaca	caggcctctg	aacatacaga	actcttcaac	cacccccctt	39420
ctccacccgg	aaggagtgc	aagcagtaat	gataaactcat	cagatacaac	taaaagtagc	39480
gcttctcttt	ctcatagagt	agcttcgcaa	atcaataaat	tcaactcaaa	caactgattca	39540
aaagtacttc	agactgattt	tttatcaaga	aatggagaca	catatttaac	acgggaaacg	39600
atatttgaag	cttcaaaaaa	agtaacaaac	tccttaagta	atcttatatc	tctcattgga	39660
actaaatcag	gaacacaaga	acgagagtta	caggaaaaat	caaaggacat	tacaaaatcc	39720
acaacagaa	atagaataaa	caacaaatta	aaagttaacg	atgcaaatat	aagaaactac	39780
gtaacagaaa	ccaacgcaga	tacaattgat	aaaaatcatg	cgatctatga	aaaggcaaaa	39840
gaagtatcta	gcgcctcag	caaggatttg	tcaaaaattg	acgatacctc	tgagaatta	39900
cttacagatg	atatatctga	tttaaaaaat	aacaatgata	ttacagctga	aaacaataat	39960
atatataaag	cagcaaaaga	tgtaaccact	tccttatcaa	aagtatttaa	gaatatcaat	40020
aaggattaaa	atatgataaa	agaaaaaata	ttatcaatag	tggccttttg	ctatggtata	40080
gcgtacagca	aactatctga	agaaactaaa	tttattgaag	atctttctgc	cgattctctt	40140

-continued

tctctgattg	aatgctgga	tatgattagt	tttgagttca	atctacgtat	agatgaatct	40200
acattagaac	acattattac	tattggagat	cttattttctg	tagtcaaaaa	tagtacaaaa	40260
tcaatatagt	aattgataag	catttttttca	tctatggagc	tctcacatca	gagctccaca	40320
agaatattat	tcttttatcc	aataaagata	aattgcatca	atccagctat	taaaatagta	40380
atttttaaga	ctaccgttga	ctatcatcaa	cagttacata	caatcttgct	cacactgcat	40440
ttaacttttg	tcaatataaa	tcaatatgaa	taaacagggg	gtgatatggg	ggatttgtgc	40500
aacgacttgt	taagtataaa	ggaaggccaa	aagaaagagt	ttacactcca	ttctggtaat	40560
aaagtttcct	ttatcaaaagc	caagattcct	cataaaagga	tccaagattt	aaccttcgct	40620
aacaaaaaaa	cgaatgtacg	cgatcaagaa	tcctaacag	aagaatcatt	agccgatatc	40680
ataaaaaacta	taaagctaca	acaattcttc	cctgtaatag	gaaggagat	tgatggtaga	40740
attgaaatc	tggatggcac	tcgtagaaga	gcatctgcaa	tatatgcagg	agcagatcct	40800
gaagttctat	attcaaaaga	atatatatct	actcttgatg	ccagaaaact	agcaaacgat	40860
atacaaacag	caaaagagca	tagcatccga	gaacttggtg	ttggtcttaa	ttttctgaaa	40920
gtatcaggga	tgtcctataa	agacatagcc	aaaaagaga	atctgtctcg	cgcgaaagtc	40980
actcgtgcct	ttcaggcagc	aagcgttcca	caggaaataa	tatctctatt	tccaatcgcg	41040
tcagaactta	actttaatga	ctacaagata	ttattcaatt	attataaagg	acttgaaaag	41100
gctaataaat	ctcttagttc	tacactacca	atattaaagg	aagaaataaa	agaccttgat	41160
acaaatttgc	cccggacat	atataaaaaa	gaaattttta	acatcataaa	gaaaagcaaa	41220
aacagaaagc	aaaacccttc	gctaaaagtt	gactcgttat	ttatttctaa	agacaaacgg	41280
acttacataa	aaagaaaaga	gaataaaaca	aacagaactc	taatatctac	attatctaaa	41340
ataaacaaaa	cagttcagag	agaaatagat	gaagctattc	gggatataat	ttctcgccat	41400
ctatcgtcct	cataaacatc	gaacactgat	gtaaactgcc	caattcagat	aataaaaaatc	41460
ttcgttgaac	ttgactttca	gccggtgggg	taattccgct	ggctatctgt	tattcaatcc	41520
ccacgcactg	tgtttcctta	aaattacggt	gaaaactcat	acctgctga	ccacgagaaa	41580
attatctggt	gaaattagta	atattcatat	cattgaaaat	gaattgtgca	actaagggat	41640
gcgcatgat	ggcaaaaatt	ttatgtgaat	atgagtgact	ggccgaaaat	attcagcgta	41700
tttttaaatcg	tttcaataaa	aacagcgagg	attagcagtc	aacgctttta	ctactcaatt	41760
ttttttacat	ggtgataaac	aaaaaatcca	ccattaacag	acttcgttca	cagtatcaaa	41820
aaagctttta	gaatatatta	gcagatttct	caaatataat	tactgtttta	ttctcctaatt	41880
cttcccataa	aatgaccgac	ttacattatt	ttataattat	aaacaagttt	gatttttaaaa	41940
tcatattaat	tattaacaag	agattttatt	ttctaacccc	aaaagttatg	cttgggtcaaa	42000
gagtaactta	taatatccat	acccttcata	gggcttacag	tttaaatgct	gggtataaaa	42060
tacatctatt	ttcaaatata	ataaagcata	ctccaattac	aacgagtgc	taaaataaaa	42120
atatcatcca	agtcaatcac	tgtttgctct	gcaaaatatg	tcagccttct	caaacttacc	42180
agcacttaaa	tatttaagca	cacaatcatt	ccttgaggga	gttcaacttat	ttcaagagaa	42240
acatttcgct	ttcccacaca	actaatccgc	ccaatatagc	taaatttgta	ctcagtaata	42300
tcatttggtta	tgccccccac	cccaaaatat	tcaagctcct	tcacagataa	aaaacgacaa	42360
tttcaaaagc	ctcattagtt	ataactatgg	aaatacctat	tttagtatat	cctataaagt	42420
tagaattact	tcttcttgca	cctaattcca	tagttgaaaa	acaagaataa	ttttctccta	42480
cagctggatc	catatagctt	ccatcaggtc	tacacaacac	ccagtggagg	ccgacaggaa	42540

-continued

ctccaataaa	ggatacagca	acagcttcca	aaactctttg	gttagaagat	aacacatctc	42600
gttgatgtac	tatattacat	cccatgccaa	ttaataagtc	ccttgcatcg	ggataaataa	42660
aactaattac	ttgcgaaaaa	atgttacttt	cttcaacgac	ataagcattc	aatccaagaa	42720
gcctcgtagc	catgactatc	ttatcaggca	tagaataacc	ggcatccgcg	atattatctt	42780
tgtgtattct	cggattataa	ttaccactag	ttattaagta	taaatccgcg	tcacaacgat	42840
tatccaaato	aagagaactc	ttccttgtca	tttcggacat	agaccctttg	tattgtggta	42900
ttttatcaac	ccctaattcc	tttgcagcac	agagtaaaact	ggcagctcca	caagaattgt	42960
catacatctg	cttcctcgcc	aaaaaacaac	cttggtcggt	tactctattc	tcatttacc	43020
tcacatcagt	ctgatgagat	atcatatatt	ctcttggtat	agccccctgc	aataaaaccc	43080
catataacat	aacacctgta	tgaatacatc	cccgtttgca	aggttccgtg	tgttccgaca	43140
tcgacgagta	attgcagcac	agtattgcat	ccagcaatgc	gaactactat	gagaattagc	43200
tgaccgggtc	cttattcccc	gaacgaactt	taaatccaaa	acctcatact	cttttccgtc	43260
gtcatcacgt	ccgtcaccat	aacaaccttt	cagcatccac	ccttttatta	agagttaatg	43320
tccatcagca	tcagcaatta	ataccgctcg	atactgacat	atttttgtcg	tatttaacgtc	43380
cataatcggg	tattgtgata	cagggtcacac	cagcttattt	gctattgtta	caataaccac	43440
cttggttgag	agctgttttt	tcaactttgt	tatcaactcc	cccaactcct	tggtcagtaa	43500
ggctgccgac	cgagcacagt	gaatagatag	tatccgcagg	taccaccatg	cttacttaat	43560
actcagaagg	cgaacatatt	aatatcctcc	ttagtcccta	ttccgaagtt	cctattctct	43620
agaaagtata	ggaacttcga	agcagctcca	gcctacacgg	cccgggcggc	cgctcagctc	43680
cagtcttcag	ttctcaggcc	gcccacacgt	tcaaattccc	gggtgttatt	agtcacaata	43740
atcagcccc	gactgcgggc	atgacctgcg	atcatttgat	caaatggccc	gacagggcgt	43800
ccctgacggg	caagtcttgc	tcttatctgg	ccggtgtgtg	tggtgcagc	agcgtcgtaa	43860
tccagaacat	caatgcggga	aacaaatccc	tcgatcacag	cgagattacg	ttcaggcatc	43920
tggtcttttt	ctgcaccata	tatcagctcc	atcagggtga	ccgaactgat	acacattttc	43980
ccctgggtta	gggttaaaacg	ctccctgacg	ctggcggggt	tggtctttat	cgtaaaaatg	44040
cagatgttgg	tatcgagcat	aaacttcagc	atcagaatga	ctccctttct	tgcatgccgg	44100
gctgttcctt	gttatccata	aaatcgggtg	tgacgctgtg	tcggtcgaac	cattcgctcc	44160
atgtctctcc	ggctggcgta	atgattctgg	ttcgtccgac	agcaatcact	tcaacgcgtt	44220
ttacgttttc	cggcaatgca	accgcttttg	gcagtctgac	cgctggctg	cggttgetga	44280
gaaatacggg	ggtttccatt	atttgctctc	ttatgtggat	atgtcttggt	tatatctatt	44340
gtgggattga	taccttaggc	ctattaatat	tccggagtat	acgtagccgg	ctaacgttaa	44400
caaccgcgat	cgcacatgta	catgatcaac	aacaggtaac	ttaaaatttt	ttatgatata	44460
agtaaaat	ctttggaggt	ataccataat	attcattaaa	tttccttata	aaataagatg	44520
ggcttgatat	tccgataagt	cttgatacat	cattaatata	gctttgacta	ttcaataaaa	44580
gctttgctgc	atgatgcatg	cgaatatcaa	gaaggatttg	ttgaaatgtt	aatttttcac	44640
tctccaatcg	ttttctaaca	gctattttctg	ataaattcaa	gttatttgaa	atatcagaaa	44700
gacgccatct	cttctcgatg	tttttttcaa	caatcttcct	tatctgatca	gaaaaactaa	44760
gagaagaagc	tatcgatata	gaagtatata	aagcttcctc	atcagaaaca	gctgataaaa	44820
ggcaagctaa	actatagatc	tttcttttgc	cgaaggcat	ctcttttata	gatttgaaca	44880

-continued

```

aatcgataga aacctcctcc tcagagagga ggaatatattt ttgtttaac ccccttttct 44940
cctcagaata ggagtgttga aatgaataaa ttggttccat tattctaata atgcttaata 45000
ataaatattt gtcaaggctt ataactctaa atggattaat gctatcagat ttttttatgg 45060
agacgtttat ttgtatatatt cgctctataa aagcaatttg cccctcatcg atagtcaaag 45120
tttcattgcc tgagctaacc gtcattgaac acctttttgc atataaaata atatagtatt 45180
gcaagcgaac ctttatatct attttgtttt tatgtcccat atccatcata aattcaacag 45240
aatataccaa actatatatta gtgtttacag agtaactaag aaattttctgt aacagtgatt 45300
gtatacaaaa tctttccggtt attaagatag attaagtaat ttttctata ttttaacttta 45360
tttacatttt ttgaagaaag agcttatgca gcttctattt ctctgtgcaa caagctaagt 45420
atttgcactg aaagggaacta taaccatatt atggtttcac ttttcttagg caatatctgt 45480
gtaaatgttc ggaaaggagt ggagcttcgc aaaactccag catcataata tcctatacgg 45540
aacgcataaa gcccttcag actggaggct ccacaaatcc gggtgggtgt gatattggta 45600
agcgtaacc gaatgccgca ttgacgattc tcccgatgac agcggtt 45646

```

What is claimed is:

1. A recombinant *Shigella* minimal invasion plasmid construct, PRISM, comprising SEQ ID NO: 1, wherein the PRISM optionally comprises a selection gene or recombinant *Shigella* minimal invasion plasmid construct, pRISM-G, comprising SEQ ID NO: 2, wherein the pRISM-G optionally comprises a selection gene.

2. A recombinant *Shigella* minimal invasion plasmid construct of claim 1, wherein the PRISM or pRISM-G comprises a *Shigella* cell-entry region from a *Shigella flexneri* bacterium.

3. The recombinant *Shigella* minimal invasion plasmid construct of claim 1, wherein the selection gene is an antibiotic resistance gene.

4. A *Shigella* or *E. coli* bacterium comprising a recombinant *Shigella* minimal invasion plasmid construct, PRISM, comprising SEQ ID NO: 1, or a recombinant *Shigella* minimal invasion plasmid construct, pRISM-G, comprising SEQ ID NO: 2.

5. A composition comprising the *Shigella* or *E. coli* bacterium of claim 4 and a pharmaceutically acceptable excipient.

6. The composition of claim 5, further comprising an adjuvant.

25 7. A method of inducing an immune response in a subject, the method comprising administering a composition comprising a recombinant *Shigella* minimal invasion plasmid construct, pRISM, comprising SEQ ID NO: 1, or a recombinant *Shigella* minimal invasion plasmid construct, pRISM-G, comprising SEQ ID NO: 2, or a *Shigella* or *E. coli* bacterium comprising either pRISM or pRISM-G, to the subject in an amount sufficient to induce the immune response.

30 8. The method of claim 7, wherein the immune response is against a pathogen.

35 9. The method of claim 8, wherein the pathogen is a virus, bacterium, fungus or parasite.

10. The method of claim 9, wherein the pathogen is *Shigella* or *E. coli*.

40 11. The method of claim 7, wherein the pRISM or pRISM-G comprise a nucleic acid encoding a bacterial, viral, fungal, parasitic, or mammalian protein of interest, or an immunogenic portion thereof.

45 12. The *Shigella* or *E. coli* bacterium of claim 4, wherein the pRISM or pRISM-G comprise a nucleic acid encoding a bacterial, viral, fungal, parasitic, or mammalian protein of interest, or an immunogenic portion thereof.

* * * * *